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STAT 112 — Past Questions & Materials  
**by Renance**

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Course: STAT 112  
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# Paper 1 of 1 — STAT 112

Level 100 • Second Semester • Handout

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## CHAPTER

Introduction to

Statistics 1

LEARNING OBJECTIVES 1.1 Descriptive and  
Inferential Statistics

After reading this chapter, you should be able to:

1.2 Statistics in Research

1.3 Scales of Measurement

1 Distinguish between descriptive and inferential 1.4 Types of Data  
statistics.

1.5 Research in Focus:

Types of Data and

2 Explain how samples and populations, as well as a Scales of Measurement  
sample statistic and population parameter, differ. 1.6 SPSS in Focus:

Entering and Defining

3 Describe three research methods commonly used in Variables  
behavioral science.

4 State the four scales of measurement and provide an  
example for each.

5 Distinguish between qualitative and quantitative data.

6 Determine whether a value is discrete or continuous.

7 Enter data into SPSS by placing each group in separate  
columns and each group in a single column (coding is  
required).

## 2 PART I: INTRODUCTION AND DESCRIPTIVE STATISTICS

### 1.1 DESCRIPTIVE AND INFERENTIAL STATISTICS

Why should you study statistics? The topic can be intimidating, and rarely does anyone tell you, “Oh, that’s an easy course . . . take statistics!” Statistics is a branch of mathematics used to summarize, analyze, and interpret what we observe—to make sense or meaning of our observations. A family counselor may use statistics to describe patient behavior and the effectiveness of a treatment program. A social psychologist may use statistics to summarize peer pressure among teenagers and interpret the causes. A college professor may give students a survey to summarize and interpret how much they like (or dislike) the course. In each case, the counselor, psychologist, and professor make use of statistics to do their job.

The reason it is important to study statistics can be described by the words of

Mark Twain: “There are lies, damned lies and statistics.” He meant that statistics can be deceiving—and so can interpreting them. Statistics are all around you—from your college grade point average (GPA) to a Newsweek poll predicting which political candidate is likely to win an election. In each case, statistics are used to inform you. The challenge as you move into your careers is to be able to identify statistics and to interpret what they mean. Statistics are part of your everyday life, and they are subject to interpretation. The interpreter, of course, is YOU.

Statistics is a branch of mathematics used to summarize, analyze, and  
**DEFINITION** interpret a group of numbers or observations.

We begin by introducing two general types of statistics:

- Descriptive statistics: statistics that summarize observations.
- Inferential statistics: statistics used to interpret the meaning of descriptive statistics.

This book describes how to apply and interpret both types of statistics in science and in practice to make you a more informed interpreter of the statistical information you encounter inside and outside of the classroom. Figure 1.1 is a schematic diagram of the chapter organization of this book, showing which chapters focus on descriptive statistics and which focus on inferential statistics.

#### DESCRIPTIVE STATISTICS

Researchers can measure many behavioral variables, such as love, anxiety, memory, and thought. Often, hundreds or thousands of measurements are made, and procedures were developed to organize, summarize, and make sense of these measures. These procedures, referred to as descriptive statistics, are specifically used to describe or summarize numeric observations, referred to as data. To illustrate, suppose we want to study anxiety among college students. We could describe anxiety, then, as a state or feeling of worry and nervousness. This certainly describes anxiety, but not numerically (or in a way that allows us to measure anxiety).

Instead, we could state that anxiety is the number of times students fidget during a class presentation. Now anxiety is defined as a number. We may observe 50, 100, or 1,000 students give a class presentation and record the number of times each

#### CHAPTER 1 : INTRODUCTION TO STATISTICS 3

Transition from  
 descriptive to  
 inferential statistics  
 (Chapters 6-7)

Descriptive Statistics Inferential Statistics  
 (Chapters 2-5) (Chapters 8-18)

FIGURE 1.1

A general overview of this book.

Statistics This book begins with an introduction to descriptive statistics (Chapters 2–5) and then uses descriptive statistics to transition (Chapters 6–7) to a discussion of inferential statistics (Chapters 8–18).

student fidgeted. Presenting a spreadsheet with the number for each individual student is not very clear. For this reason, researchers use descriptive statistics to summarize sets of individual measurements so they can be clearly presented and interpreted.

Descriptive statistics are procedures used to summarize, organize, and make sense of a set of scores or observations. **DEFINITION**

Descriptive statistics are typically presented graphically, in tabular form (in tables), or as summary statistics (single values).

Data (plural) are measurements or observations that are typically numeric.

A datum (singular) is a single measurement or observation, usually referred to as a score or raw score.

Data are generally presented in summary. Typically, this means that data are **NOTE: Descriptive statistics**

presented graphically, in tabular form (in tables), or as summary statistics (e.g., an summarize data to make sense

average). For example, the number of times each individual fidgeted is not all that or meaning of a list of numeric

meaningful, whereas the average (mean), middle (median), or most common values.

(mode) number of times among all individuals is more meaningful. Tables and graphs serve a similar purpose to summarize large and small sets of data.

Most often, researchers collect data from a portion of individuals in a group of interest. For example, the 50, 100, or 1,000 students in the anxiety example would not constitute all students in college. Hence, these researchers collected anxiety data from some students, not all. So researchers require statistical procedures that allow them to infer what the effects of anxiety are among all students of interest using only the portion of data they measured.

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##### INFERENTIAL STATISTICS

The problem described in the last paragraph is that most scientists have limited access to the phenomena they study, especially behavioral phenomena. As a result, researchers use procedures that allow them to interpret or infer the meaning of data. These procedures are called inferential statistics.

Inferential statistics are procedures used that allow researchers to infer or

**DEFINITION** generalize observations made with samples to the larger population from

which they were selected.

To illustrate, let's continue with the college student anxiety example. All students enrolled in college would constitute the population. This is the group that researchers want to learn more about. Specifically, they want to learn more about characteristics in this population, called population parameters.

The characteristics of interest are typically some descriptive statistic. In the anxiety example, the characteristic of interest is anxiety, specifically measured as the number of times students fidget during a class presentation.

Unfortunately, in behavioral research, scientists rarely know what these population parameters are since they rarely have access to an entire population. They simply do not have the time, money, or other resources to even consider studying all students enrolled in college.

A population is defined as the set of all individuals, items, or data of interest.

**DEFINITION** This is the group about which scientists will generalize.

A characteristic (usually numeric) that describes a population is referred to as a population parameter.

**NOTE:** Inferential The alternative is to select a portion or sample of individuals in the population. Statistics are used to help. Selecting a sample is more practical, and most scientific research you read

the researcher infer how comes from samples and not populations. Going back to our example, this means

well statistics in a that selecting a portion of students from the larger population of all students sample reflect enrolled in college would constitute a sample. A characteristic that describes a parameters in a sample is called a sample statistic—this is similar to a parameter, except it population. describes characteristics in a sample and not a population. Inferential statistics use the characteristics in a sample to infer what the unknown parameters are in a given population. In this way, as shown in Figure 1.2, a sample is selected from a population to learn more about the characteristics in the population of interest.

A sample is defined as a set of selected individuals, items, or data taken from

**DEFINITION** a population of interest.

A characteristic (usually numeric) that describes a sample is referred to as a sample statistic.

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Population: All students enrolled in college.

Draw conclusions about anxiety levels for the

Sample: 50 students 100 students 1,000 students entire population of

FIGURE 1.2

students (not just among those in each sample). Samples and populations. In this example, levels of anxiety were measured in a sample of 50, 100, or 1,000 college students.

Researchers will observe anxiety

Observe the number of times each in each sample. Then they will use student fidgets during a class inferential statistics to generalize presentation in each sample. their observations in each sample to the larger population, from which each sample was selected.

#### MAKING SENSE: Populations and Samples

A population is identified as any group of interest, whether that group is all students worldwide or all students in a professor's class. Think of any group you are interested in. Maybe you want to understand why college students join fraternities and sororities. So students who join fraternities and sororities is the group you're interested in. Hence, this group is now a population of interest, to you anyways. You identified a population of interest just as researchers identify populations they are interested in.

Remember that researchers collect samples only because they do not have access to all individuals in a population. Imagine having to identify every person who has fallen in love, experienced anxiety, been attracted to someone else, suffered with depression, or taken a college exam. It's ridiculous to consider that we can identify all individuals in such populations. So researchers use data gathered from samples (a portion of individuals from the population) to make inferences concerning a population.

To make sense of this, say you want to get an idea of how people in general feel about a new pair of shoes you just bought. To find out, you put your new shoes on and ask 20 people at random throughout the day whether or not they like the shoes. Now, do you really care about the opinion of only those 20 people you asked? Not really—you actually care more about the opinion of people in general. In other words, you only asked the 20 people (your sample) to get an idea of the opinions of people in general (the population of interest). Sampling from populations follows a similar logic.

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On the basis of the following example, we will identify the population, sample,

**E X A M P L E 1.1** population parameter, and sample statistic: Suppose you read an article in the local

college newspaper citing that the average college student plays 2 hours of video games per week. To test whether this is true for your school, you randomly approach 20 fellow students and ask them how long (in hours) they play video games per

week. You find that the average student, among those you asked, plays video games for 1 hour per week. Distinguish the population from the sample.

In this example, all college students at your school constitute the population of interest, and the 20 students you approached is the sample that was selected from this population of interest. Since it is purported that the average college student plays 2 hours of video games per week, this is the population parameter (2 hours). The average number of hours playing video games in the sample is the sample statistic (1 hour).

## LEARNING

CHECK 1 1. \_\_\_\_\_ are techniques used to summarize or describe numeric data.

2. \_\_\_\_\_ describe(s) how a population is characterized, whereas \_\_\_\_\_ describe(s) the characteristics of samples.

- a. Statistics; parameters
- b. Parameters; statistics
- c. Descriptive; inferential
- d. Inferential; descriptive

3. A psychologist wants to study a small population of 40 students in a local private school. If the researcher was interested in selecting the entire population of students for this study, then how many students must the psychologist include?

- a. None, since it is not possible to study an entire population in this case.
- b. At least half, since this would constitute the majority of the population.
- c. All 40 students, since all students constitute the population.

4. True or false: Inferential statistics are used to help the researcher infer whether observations made with samples are reflective of the population.

Answers: 1. Descriptive statistics; 2. B; 3. C; 4. True.

## 1.2 STATISTICS IN RESEARCH

This book will describe many ways of measuring and interpreting data. Yet, simply collecting data does not make you a scientist. To engage in science, you must follow specific procedures for collecting data. Think of this as playing a game. Without the rules and procedures for playing, the game itself would be lost. The same is true in science; without the rules and procedures for collecting data, the ability to draw scientific conclusions would be lost. Ultimately, statistics are used in the context of science, and so it is necessary to introduce you to the basic procedures of scientific inquiry.

## CHAPTER 1 : INTRODUCTION TO STATISTICS

To illustrate the basic premise of engaging in science, suppose you come across the following problem first noted by the famous psychologist Edward Thorndike in 1898:

Dogs get lost hundreds of times and no one ever notices it or sends an account of it to a scientific magazine, but let one find his way from

Brooklyn to Yonkers and the fact immediately becomes a circulating anecdote. Thousands of cats on thousands of occasions sit helplessly yowling, and no one takes thought of it or writes to his friend, the professor; but let one cat claw at the knob of a door supposedly as a signal to be let out, and straightway this cat becomes the representative of the cat-mind in all books. . . . In short, the anecdotes give really . . . supernormal psychology of animals. (pp. 4-5)

Science is the study of phenomena, such as behavior, through strict observation, evaluation, interpretation, and theoretical explanation. **DEFINITION**

Here the problem was to determine the animal mind. Edward Thorndike posed the question of whether animals were truly smart, based on the many observations he made. This is where the scientific process typically begins—with a question. To answer questions in a scientific manner, researchers need more than just statistics; they need a set of strict procedures for making the observations and measurements. In this section, we introduce three research methods commonly used in behavioral research: experimental, quasi-experimental, and correlational methods. Each research method involves examining the relationship between variables. Each method is introduced here since we will apply these methods throughout the book.

#### EXPERIMENTAL METHOD

Any study that demonstrates cause is called an experiment. To demonstrate cause, though, an experiment must follow strict procedures to ensure that the possibility of all other possible causes have been minimized or eliminated. So researchers must control the conditions under which observations are made to isolate cause-and-effect relationships between variables. Figure 1.3 shows the steps in a typical experiment based on a sample taken from a population. We will work through this example to describe the basic structure of an experiment.

To conduct an experiment, a researcher must specifically control the conditions under which observations are made to isolate cause-and-effect **DEFINITION** relationships between variables.

Three requirements must be satisfied for a study to be regarded as an experiment: randomization, manipulation, and comparison.

The experiment illustrated in Figure 1.3 was designed to determine the effect of distraction on student test scores. A sample of students was selected from a population of all undergraduates. In one group, the professor sat quietly while students took the exam (low-distraction group); in the other, the professor rattled papers, tapped her foot, and made other sounds during the exam (high-distraction group). Exam scores in both groups were measured and compared.

For this study to be called an experiment, researchers must satisfy three requirements. These requirements are regarded as the necessary steps to ensure enough

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Population

### FIGURE 1.3

The basic structure of an Sample experiment that meets each

requirement for demonstrating Manipulate one variable—

cause-and-effect: randomization, randomly assign

manipulation, and comparison. In participants to each level of

this example, a random sample of the manipulated variable. Low-distraction condition:

High-distraction condition:

students was selected from a Example: Randomly A professor sits quietly at A professor makes loud population of all undergraduates assign participants to two a desk while students take sounds (paper ruffling,

to study the effects of distraction levels of distraction. an exam. foot tapping) at a desk while on exam performance. To qualify students take an exam.

as an experiment, (1) students

were randomly assigned to Measure a second variable—

experience a low- or high- the same variable is measured

in each condition, and the

distraction condition while taking

difference between groups

an exam (randomization), (2) the

is compared.

researcher created each level of

distraction (manipulation), and (3) Example: Measure exam

Measure grades on Measure grades on

a comparison group was included performance (or grades)

exam (0–100 points). exam (0–100 points).

where distraction was minimal or in each condition.

absent (comparison).

control to allow researchers to draw cause-and-effect conclusions. These require-ments are the following:

1. Randomization (of assigning participants to conditions)
2. Manipulation (of variables that operate in an experiment)
3. Comparison (or a control group)

To meet the requirement of randomization, researchers must use random assignment (Requirement 1) to assign participants to groups. To do this, a researcher must be able to manipulate the levels of an independent variable (IV) (Requirement 2) to create the groups. Referring back to the test distraction example shown in Figure 1.3, the independent variable was distraction. The researchers first manipulated the levels of this variable (low, high), meaning that they created the conditions. They then assigned each student at random to experi-

ence one of the levels of distraction.

Random assignment is a random procedure used to ensure that participants in

DEFINITION a study have an equal chance of being assigned to a particular group or condition.

An independent variable (IV) is the variable that is manipulated in an

experiment. This variable remains unchanged (or “independent”) between

conditions being observed in an experiment. It is the “presumed cause.”

The specific conditions of an IV are referred to as the levels of the IV.

Random assignment and manipulation ensure that characteristics of partici-

pants in each group (such as their age, intelligence level, or study habits) vary

entirely by chance. Since participant characteristics between groups now occur at

## CHAPTER 1 : INTRODUCTION TO STATISTICS 9

random, we can assume that these characteristics are about the same between

groups. This makes it more likely that any differences observed between groups

were caused by the manipulation (low vs. high levels of distraction) and not partici-

pant characteristics.

Notice also that there are two groups in the experiment shown in Figure 1.3. So NOTE: An experiment is one

tests scores for students experiencing high levels of distraction were compared to where researchers satisfy three

those experiencing low levels of distraction. By comparing test scores between requirements to ensure enough

groups, we can determine whether high levels of distraction caused lower scores control to allow researchers to

(compared to scores in the low-distraction group). This satisfies the requirement of draw cause-and-effect

comparison (Requirement 3), which requires that at least two groups be observed in conclusions. These are

an experiment. This allows scores in one group to be compared to those in at least randomization, manipulation,

one other group. and comparison.

In this example, test scores were measured in each group. The measured varia-

ble in an experiment is referred to as the dependent variable (DV). Dependent

variables can often be measured in many ways, and therefore require an opera-

tional definition. This is where a dependent variable is defined in terms of how

it will be measured. For example, here we operationally defined exam performance as

a score between 0 and 100 on a test. So to summarize the experiment in Figure 1.3,

levels of distraction (IV) were presumed to cause an effect or difference in exam

grades (DV) between groups. This is an experiment since the researchers satisfied

the requirements of randomization, manipulation, and comparison, thereby allow-

ing them to draw cause-and-effect conclusions.

The dependent variable (DV) is the variable that is believed to change in the

presence of the independent variable. It is the “presumed effect.” DEFINITION

An operational definition is a description of some observable event in terms of the specific process or manner by which it was observed or measured.

#### QUASI-EXPERIMENTAL METHOD

A study that lacks randomization, manipulation, or comparison is called a quasi-experiment. This most often occurs in one of two ways:

1. The study includes a quasi-independent variable.
2. The study lacks a comparison group.

In a typical quasi-experiment, the variables being studied can't be manipulated, which makes random assignment impossible. This occurs when variables are preexisting or inherent to the participants themselves. These types of variables are called quasi-independent variables. Figure 1.4 shows an example of a quasi-experiment that measured differences in multitasking ability by gender. Since the levels of gender (male, female) can't be randomly assigned (it is a quasi-independent variable), this study is regarded as a quasi-experiment.

A quasi-independent variable is a variable whose levels are not randomly assigned to participants (nonrandom). This variable differentiates the groups or conditions being compared in a quasi-experiment.

NOTE: A quasi-experiment is

A study is also regarded as a quasi-experiment when only one group is observed. Since only one group is observed, there is no comparison group. So differences between two levels of an independent variable can't be compared. In this (2) lacks a comparison group.

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Gender is not randomly assigned. Men are assigned to the male condition; women to the female condition. A sample of men and women is selected from a population.

#### FIGURE 1.4

The basic structure of a quasi-experiment. In this example, researchers measured differences in multitasking behavior by gender. The grouping variable (gender) is preexisting. That is, participants were already male or female prior to the study. For this reason, researchers can't manipulate the variable or randomly assign. Dependent measure: as many tasks as possible in 5 minutes.

participants to each level of The number of tasks The number of tasks  
gender, so this study is regarded completed is recorded. completed is recorded.  
as a quasi-experiment.

way, failing to satisfy any of the three requirements for an experiment (randomiza-  
tion, manipulation, or comparison) makes the study a quasi-experiment.

#### CORRELATIONAL METHOD

Another method for examining the relationship between variables is to measure pairs  
of scores for each individual. This method can determine whether a relationship  
exists between variables, but it lacks the appropriate control needed to demonstrate  
cause and effect. To illustrate, suppose you test for a relationship between time spent  
using a computer and exercising per week. The data for such a study appear in tabular  
form and plotted as a graph in Figure 1.5. Using the correlational method, we can  
examine the extent to which two variables change in a related fashion. In the exam-  
ple shown in Figure 1.5, as computer use increases, time spent exercising decreases.  
This pattern suggests that computer use and time spent exercising are related.

NOTE: The correlational Notice that no variable is manipulated to create different conditions or  
groups

method involves measuring to which participants can be randomly assigned. Instead, two variables  
are measured

the relationship between pairs for each participant, and the extent to which those variables are  
related is measured.

of scores. This book will describe many statistical procedures used to analyze data using the  
correlational method (Chapters 15–17) and the experimental and quasi-experimen-  
tal methods (Chapters 9–14, 17–18).

A researcher conducts the following study: Participants are presented with a list of  
E X A M P L E 1. 2 words written on a white background on a PowerPoint slide. In one group, the  
words are written in red (Group Color); in a second group, the words are written in  
black (Group Black). Participants are allowed to study the words for 1 minute. After  
that time, the slide is removed and participants are allowed 1 minute to write down  
as many words as they can recall. The number of words correctly recalled will be  
recorded for each group. Explain how this study can be an experiment.

#### C H A P T E R 1 : I N T R O D U C T I O N T O S T A T I S T I C S 11

(a)

Computer use Exercise

Participant (Hours per week) (Minutes per week)

A 3 80

B 2 83

C 0 96

D 10 60

E 8 78

F 12 46

(b) 120

FIGURE 1.5

100

(Minutes per week)

An example of the correlational  
80 method. In this example,  
researchers measured the amount  
Exercise

60 of time students spent using the  
computer and exercising each  
week. (A) The table shows two  
40 sets of scores for each participant.

(B) The graph shows the pattern of  
20 the relationship between these  
scores. From the data, we can see  
0 that the two variables (computer  
0 2 4 6 8 10 12 14 use and exercise) change in a  
related fashion. That is, as

Computer Use (Hours per week) computer use increases, time  
spent exercising decreases.

To create an experiment, we must satisfy the three requirements for demonstrating  
cause and effect: randomization, manipulation, and comparison. To satisfy each  
requirement, the researcher can

1. Randomly assign participants to experience one of the conditions. This ensures that some participants read colored words and others read black words entirely by chance.
2. Create the two conditions. The researcher could write 20 words on a PowerPoint slide. On one slide, the words are written in red; on the second slide, the same words are written in black.
3. Include a comparison group. In this case, the number of colored words correctly recalled will be compared to the number of black words correctly recalled, so this study has a comparison group.

Remember that each requirement is necessary to demonstrate that the levels of an independent variable are causing changes in the value of a dependent variable. If any one of these requirements is not satisfied, then the study is not an experiment.

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### LE A R N I N G

C H E C K 2 1. \_\_\_\_\_ is the study of phenomena through strict observation, evaluation, interpretation, and theoretical explanation.

2. State whether each of the following describes an experiment, quasi-experiment, or correlational method.

- A researcher tests whether dosage level of some drug (low, high) causes significant differences in health.
- A researcher tests whether political affiliation (Republican, Democrat) is associated with different attitudes toward morality.
- A researcher measures the relationship between income and life satisfaction.

3. True or false: An experiment is the only method that can demonstrate cause-and-effect relationships between variables.

Answers: 1. Science; 2. A. Experiment. B. Quasi-experiment. C. Correlational method; 3. True.

### 1.3 SCALES OF MEASUREMENT

Many statistical tests introduced in this book will require that variables in a study be measured on a certain scale of measurement. In the early 1940s, Harvard psychologist S. S. Stevens coined the terms nominal, ordinal, interval, and ratio to classify the scales of measurement (Stevens, 1946). Scales of measurement are rules that describe the properties of numbers. These rules imply that a number is not just a number in science. Instead, the extent to which a number is informative depends on how it was used or measured. In this section, we discuss the extent to which data are informative. In all, scales of measurement are characterized by three properties: order, differences, and ratios. Each property can be described by answering the following questions:

- Order: Does a larger number indicate a greater value than a smaller number?
- Differences: Does subtracting two numbers represent some meaningful value?
- Ratio: Does dividing (or taking the ratio of) two numbers represent some meaningful value?

Scales of measurement refer to how the properties of numbers can change with different uses.

Table 1.1 summarizes the answers to the questions for each scale of measurement. You can think of each scale as a gradient of the informativeness of data. In this section, we begin with the least informative scale (nominal) and finish with the most informative scale (ratio).

#### NOMINAL SCALES

Numbers on a nominal scale identify something or someone; they provide no additional information. Common examples of nominal numbers include ZIP codes,

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TABLE 1.1 Scales of measurement. Different scales of measurement and the information they provide

concerning the order, difference, and ratio of numbers.

Scale of Measurement

Nominal Ordinal Interval Ratio

Property

Order No Yes Yes Yes

Difference No No Yes Yes

Ratio No No No Yes

license plate numbers, credit card numbers, country codes, telephone numbers, and Social Security numbers. These numbers simply identify locations, vehicles, or individuals and nothing more. One credit card number, for example, is not greater than another; it is simply different.

Nominal scales are measurements where a number is assigned to represent something or someone. **DEFINITION**

In science, nominal variables are typically categorical variables that have been NOTE: Nominal values

coded—converted to numeric values. Examples of nominal variables include a person's race, gender, nationality, sexual orientation, hair and eye color, season of someone. They often reflect

birth, marital status, or other demographic or personal information. A researcher coded data in behavioral

may code men as 1 and women as 2. They may code the seasons of birth as 1, 2, 3, science. and 4 for spring, summer, fall, and winter, respectively. These numbers are used to

identify gender or the seasons and nothing more. We often code words with

numeric values when entering them into statistical programs such as SPSS. Coding is largely done because it is often easier to compute data using statistical programs, such as SPSS, when data are entered as numbers, not words.

Coding refers to the procedure of converting a nominal value to a numeric value. **DEFINITION**

## ORDINAL SCALES

An ordinal scale of measurement is one that conveys order alone. This scale indi- NOTE: Ordinal values convey

cates that some value is greater or less than another value. Examples of ordinal order alone. scales include finishing order in a competition, education level, and rankings.

These scales only indicate that one value is greater or less than another, so differences between ranks do not have meaning. Consider, for example, the U.S. News & World Report rankings for the top psychology graduate school programs in the United States. Table 1.2 shows the rank, college, and actual score for the top 25 programs, including ties, in 2009. Based on ranks alone, can we say that the difference between the psychology graduate programs ranked 1 and 11 is the same as the difference between those ranked 13 and 23? In both cases, 10 ranks separate

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TABLE 1.2 Ordinal scale. A list of the top 25 ranked psychology graduate school programs in 2009, including ties (left column), and the actual points used to determine their rank (right column). The U.S.

News & World Report published these rankings.

Rank College Name Actual Score

|    |                                           |     |
|----|-------------------------------------------|-----|
| 1  | Stanford University                       | 4.7 |
| 1  | University of California, Berkeley        | 4.7 |
| 3  | Harvard University                        | 4.6 |
| 3  | University of California, Los Angeles     | 4.6 |
| 3  | University of Michigan, Ann Arbor         | 4.6 |
| 3  | Yale University                           | 4.6 |
| 7  | University of Illinois, Urbana-Champaign  | 4.5 |
| 8  | Princeton University                      | 4.4 |
| 8  | University of Minnesota, Twin Cities      | 4.4 |
| 8  | University of Wisconsin, Madison          | 4.4 |
| 11 | Massachusetts Institute of Technology     | 4.3 |
| 11 | University of Pennsylvania                | 4.3 |
| 13 | University of North Carolina, Chapel Hill | 4.2 |
| 13 | University of Texas, Austin               | 4.2 |
| 13 | University of Washington                  | 4.2 |
| 13 | Washington University in St. Louis        | 4.2 |
| 17 | Carnegie Mellon University                | 4.1 |
| 17 | Columbia University                       | 4.1 |
| 17 | Cornell University                        | 4.1 |
| 17 | Northwestern University                   | 4.1 |
| 17 | Ohio State University                     | 4.1 |
| 17 | University of California, San Diego       | 4.1 |
| 23 | Duke University                           | 4.0 |
| 23 | Indiana University, Bloomington           | 4.0 |
| 23 | John Hopkins University                   | 4.0 |

the schools. Yet, if you look at the actual scores for determining rank, you find that the difference between ranks 1 and 11 is 0.4 points, whereas the difference between ranks 13 and 23 is 0.2 points. So the difference in points is not the same. Ranks alone don't convey this difference. They simply indicate that one rank is greater or less than another rank.

Ordinal scales are measurements where values convey order or rank alone.

DEFINITION

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INTERVAL SCALES

An interval scale measurement, on the other hand, can be understood readily by

two defining principles: equidistant scales and no true zero. A common example for this in behavioral science is the rating scale. Rating scales are taught here as an interval scale since most researchers report these as interval data in published research. This type of scale is a numeric response scale used to indicate a participant's level of agreement or opinion with some statement. An example of a rating scale is given in Figure 1.6. Here we will look at each defining principle.

Interval scales are measurements where the values have no true zero and the distance between each value is equidistant. **DEFINITION**

An equidistant scale is a scale distributed in units that are equidistant from one another. Many behavioral scientists assume that scores on a rating scale are distributed in equal intervals. For example, if you are asked to rate your satisfaction with a spouse or job on a 7-point scale from 1 (completely unsatisfied) to 7 (completely satisfied), like in the scale shown in Figure 1.6, then you are using an interval scale. Since the distance between each point (1 to 7) is assumed to be the same or equal, it is appropriate to compute differences between scores on this scale. So a statement such as, "The difference in job satisfaction among men and women was 2 points," is appropriate with interval scale measurements.

Equidistant scales are those values whose intervals are distributed in equal units. **DEFINITION**

However, an interval scale does not have a true zero. A common example of NOTE: Interval values have

a scale without a true zero is temperature. A temperature equal to zero for most equidistant scales but no

measures of temperature does not mean that there is no temperature; it is just an true zero.

arbitrary zero point. Values on a rating scale also have no true zero. In the example shown in Figure 1.6, a 1 was used to indicate no satisfaction, not 0. Each value (including 0) is arbitrary. That is, we could use any number to represent none of something. Measurements of latitude and longitude also fit this criterion. The implication is that without a true zero, there is no value to indicate the absence of the phenomenon you are observing (so a zero proportion is not meaningful). For this reason, stating a ratio such as, "Satisfaction ratings were three times greater among men compared to women," is not appropriate with interval scale measurements.

A true zero describes values where the value 0 truly indicates nothing.

Values on an interval scale do not have a true zero. **DEFINITION**

Satisfaction Ratings

FIGURE 1.6

1  
2 3 4 5  
6  
7

Completely Completely An example of a 7-point rating  
Unsatisfied Satisfied scale for satisfaction used for  
scientific investigation.

## 16 PART I: INTRODUCTION AND DESCRIPTIVE STATISTICS

NOTE: Ratio values have **RATIO SCALES**

equidistant scales and a true

zero. This scale of Ratio scales are similar to interval scales in that scores are distributed in equal measurement does not limit units. Yet, unlike interval scales, a distribution of scores on a ratio scale has a true

the conclusions researchers zero. This is an ideal scale in behavioral research because any mathematical opera-

can state. tion can be performed on the values that are measured. Common examples of ratio scales include counts and measures of length, height, weight, and time. For scores on a ratio scale, order is informative. For example, a person who is 30 years old is older than another who is 20. Differences are also informative. For example, the difference between 70 and 60 seconds is the same as the difference between 30 and 20 seconds (the difference is 10 seconds). Ratios are also informative on this scale because a true zero is defined—it truly means nothing. Hence, it is meaningful to state that 60 pounds is twice as heavy as 30 pounds.

Ratio scales are measurements where a set of values has a true zero and are **DEFINITION** equidistant.

In science, researchers often go out of their way to measure variables on a ratio scale. For example, if they want to measure eating, they may choose to measure the amount of time between meals or the amount of food consumed (in ounces). If they measure memory, they may choose to measure the amount of time it takes to memorize some list or the number of errors made. If they measure depression, they may choose to measure the dosage (in milligrams) that produces the most beneficial treatment or the number of symptoms reported. In each case, the behaviors were measured using values on a ratio scale, thereby allowing researchers to draw conclusions in terms of order, differences, and ratios—there are no restrictions with ratio scale variables.

## LEARNING

**CHECK 3.1.** The \_\_\_\_\_ refer to how the properties of numbers can change with different uses.

2. In 2010, Fortune 500 magazine ranked Apple as the most admired company in the world. This ranking is on a(n) \_\_\_\_\_ scale of measurement.

3. What are two characteristics of rating scales that allow some researchers to use these values on an interval scale of measurement?

- Values on an interval scale have order and differences.
- Values on an interval scale have differences and a true zero.

- c. Values on an interval scale are equidistant and have a true zero.
  - d. Values on an interval scale are equidistant and do not have a true zero.
4. Which of the following is not an example of a ratio scale variable?
- a. Age (in days)
  - b. Speed (in seconds)
  - c. Height (in inches)
  - d. Movie ratings (1 to 4 stars)

Answers: 1. Scales of measurement; 2. Ordinal; 3. D; 4. D.

## CHAPTER 1 : INTRODUCTION TO STATISTICS 17

### TYPES OF DATA 1.4

The scales of measurement reflect the informativeness of data. With nominal scales, researchers can conclude little; with ratio scales, researchers can conclude just about anything in terms of order, differences, and ratios. Researchers also distinguish between the types of data they measure. The types of data researchers measure fall into two categories: (1) continuous or discrete and (2) quantitative or qualitative. Each category is discussed in this section. Many examples we use for these types of data are given in Table 1.3.

**CONTINUOUS AND DISCRETE VARIABLES NOTE:** Continuous variables are measured along a

Variables can be categorized as continuous or discrete. A continuous variable is continuum, whereas discrete

measured along a continuum. So continuous variables are measured at any place variables are measured in

beyond the decimal point. Consider, for example, that Olympic sprinters are timed whole units or categories.

to the nearest hundredths place (in seconds), but if the Olympic judges wanted to clock them to the nearest millionths place, they could.

A continuous variable is measured along a continuum at any place beyond the decimal point. Continuous variables can be measured in whole units or **DEFINITION** fractional units.

A discrete variable, on the other hand, is measured in whole units or categories. So discrete variables are not measured along a continuum. For example, the number of brothers and sisters you have and your family's socioeconomic class (working class, middle class, upper class) are examples of discrete variables. Refer to Table 1.3 for more examples of continuous and discrete variables.

A discrete variable is measured in whole units or categories that are not distributed along a continuum. **DEFINITION**

### QUANTITATIVE AND QUALITATIVE VARIABLES

Variables can be categorized as quantitative or qualitative. A quantitative variable varies by amount. The variables are measured in numeric units, and so both

continuous and discrete variables can be quantitative. For example, we can measure food intake in calories (a continuous variable) or we can count the number of pieces of food consumed (a discrete variable). In both cases, the variables are measured by amount (in numeric units).

A quantitative variable varies by amount. This variable is measured numerically and is often collected by measuring or counting. **DEFINITION**

A qualitative variable, on the other hand, varies by class. The variables are often labels for the behaviors we observe—so only discrete variables can fall into this category. For example, socioeconomic class (working class, middle class, upper class) is discrete and qualitative; so are many mental disorders such as depression (unipolar, bipolar) or drug use (none, experimental, abusive). Refer to Table 1.3 for more examples of quantitative and qualitative variables.

A qualitative variable varies by class. This variable is often represented as a label and describes nonnumeric aspects of phenomena. **DEFINITION**

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TABLE 1.3 A list of 20 variables showing how they fit into the three categories that describe them.

Continuous vs. Qualitative vs. Scale of

Variables Discrete Quantitative Measurement

Gender (male, female) Discrete Qualitative Nominal

Seasons (spring, summer, fall, winter) Discrete Qualitative Nominal

Number of dreams recalled Discrete Quantitative Ratio

Number of errors Discrete Quantitative Ratio

Duration of drug abuse (in years) Continuous Quantitative Ratio

Ranking of favorite foods Discrete Quantitative Ordinal

Ratings of satisfaction (1 to 7) Discrete Quantitative Interval

Body type (slim, average, heavy) Discrete Qualitative Nominal

Score (from 0 to 100%) on an exam Continuous Quantitative Ratio

Number of students in your class Discrete Quantitative Ratio

Temperature (degrees Fahrenheit) Continuous Quantitative Interval

Time (in seconds) to memorize a list Continuous Quantitative Ratio

The size of a reward (in grams) Continuous Quantitative Ratio

Position standing in line Discrete Quantitative Ordinal

Political Affiliation (Republican, Democrat) Discrete Qualitative Nominal

Type of distraction (auditory, visual) Discrete Qualitative Nominal

A letter grade (A, B, C, D, F) Discrete Qualitative Ordinal

Weight (in pounds) of an infant Continuous Quantitative Ratio

A college students' SAT score Discrete Quantitative Interval

Number of lever presses per minute Discrete Quantitative Ratio

For each of the following examples, we will (1) name the variable being measured,

EXAMPLE 1.3 (2) state whether the variable is continuous or discrete, and (3) state whether the

variable is quantitative or qualitative.

- a. A researcher records the month of birth among patients with schizophrenia. The month of birth (the variable) is discrete and qualitative.
- b. A professor records the number of students absent during a final exam. The number of absent students (the variable) is discrete and quantitative.
- c. A researcher asks children to choose which type of cereal they prefer (one with a toy inside or one without). He records the choice of cereal for each child. The choice of cereal (the variable) is discrete and qualitative.
- d. A therapist measures the time (in hours) that clients continue a recommended program of counseling. The time in hours (the variable) is continuous and quantitative.

## CHAPTER 1 : INTRODUCTION TO STATISTICS 19

### LEARNING

1. True or false: The types of data researchers measure fall into two categories: (1) CHECK 4 continuous or discrete and (2) quantitative or qualitative.
  2. State whether each of the following are continuous or discrete.
    - a. Delay (in seconds) it takes drivers to make a left-hand turn when a light turns green
    - b. The number of questions that participants ask during a research study
    - c. Type of drug use (none, infrequent, moderate, or frequent)
    - d. Season of birth (spring, summer, fall, or winter)
  3. State whether the variables listed in Question 2 are quantitative or qualitative.
  4. True or false: Qualitative variables can be continuous or discrete.
  5. A researcher is interested in the effects of stuttering on social behavior with children. He records the number of peers a child speaks to during a typical school day. In this example, would the data be qualitative or quantitative?
- C. Qualitative. D. Qualitative; 4. False; 5. Quantitative.

Answers: 1. True; 2. A. Continuous. B. Discrete. C. Discrete. D. Discrete; 3. A. Quantitative. B. Quantitative.

### RESEARCH IN FOCUS: TYPES OF DATA AND SCALES OF MEASUREMENT 1.5

While qualitative variables are often measured in behavioral research, this book will focus largely on the latter. The reason is twofold: (1) Quantitative measures are more common in behavioral research, and (2) most statistical tests taught in this book are adapted for quantitative measures. Indeed, many researchers who measure qualitative variables will also measure those that are quantitative in the same study. For example, Jones and colleagues (2010) explored the costs and benefits of social networking among college students. The researchers used a qualitative method to interview each student in their sample. In the interview, students could respond openly to questions asked during the interview. These researchers then summarized responses into categories related to learning, studying, and social life.

For example, the following student response was categorized as an example of independent learning experience for employability: “I think it [social software] can be beneficial . . . in the real working environment” (p. 780).

The limitation for this analysis is that categories are on a nominal scale (the least informative scale). So many researchers who record qualitative data also use some quantitative measures. For example, researchers in this study also asked students to rate their usage of a variety of social software technologies, such as PowerPoint and personal websites, on a scale from 1 (never) to 4 (always). A fifth choice (not applicable) was also included on this rating scale. These ratings are on an interval scale, which allowed the researchers to also discuss individual differences related to how much students use social software technologies.

Inevitably, the conclusions we can draw with qualitative data are rather limited since these data are typically on a nominal scale. On the other hand, most statistics introduced in this book require that variables are measured on the more informative scales. For this reason, this book will mainly describe statistical procedures for quantitative variables measured on an ordinal, interval, or ratio scale.

## 20 PART I: INTRODUCTION AND DESCRIPTIVE STATISTICS

### 1.6 SPSS IN FOCUS: ENTERING AND DEFINING VARIABLES

Throughout this book, we will present instructions for using the statistical software program SPSS by showing you how this software can make all the work you do by hand as simple as point and click. Before you read this SPSS section, please take the time to read the section titled, “How To Use SPSS With This Book” at the beginning of this book. This section provides an overview of the different views and features in SPSS. This software is an innovative statistical computer program that can compute almost any statistic taught in this book.

In this chapter, we discussed how variables are defined, coded, and measured.

Let’s see how SPSS makes this simple. Keep in mind that the variable view is used to define the variables you measure, and the data view is used to enter the scores you measured. When entering data, make sure that all values or scores are entered in each cell of the data view spreadsheet. The biggest challenge is making sure you enter the data correctly. Entering even a single value incorrectly can alter the data analyses that SPSS computes. For this reason, always double-check the data to make sure the correct values have been entered.

Let’s use a simple example. Suppose you record the average GPA of students in one of three statistics classes. You record the following GPA scores for each class, given in Table 1.4.

TABLE 1.4 GPA scores in three statistics classes.

| Class 1 | Class 2 | Class 3 |
|---------|---------|---------|
|---------|---------|---------|

|     |     |     |
|-----|-----|-----|
| 3.3 | 3.9 | 2.7 |
|-----|-----|-----|

|     |     |     |
|-----|-----|-----|
| 2.9 | 4.0 | 2.3 |
|-----|-----|-----|

3.5 2.4 2.2

3.6 3.1 3.0

3.1 3.0 2.8

There are two ways you can enter these data: by column or by row. To enter data by column:

1. Open the variable view tab (see Figure 1.7). In the name column, enter your variable names as class1, class2, and class3 (note that spaces are not allowed) in each row. Three rows should be active.
2. Since our data are to the tenths place, go to the decimal column and reduce that value to 1 in each row.

FIGURE 1.7

SPSS variable view.

## CHAPTER 1 : INTRODUCTION TO STATISTICS 21

1. Open the data view tab. You will see that the first three columns are now labeled with the group names (see Figure 1.8). Enter the data, given in Table 1.4, for each class in the appropriate column. The data for each group are now listed down each column.

FIGURE 1.8

Data entry in data view for SPSS.

There is another way to enter these data in SPSS: You can enter data by row. This requires coding the data.

1. Open the variable view tab (see Figure 1.9). Enter classes in the first row in the name column. Enter GPA in the second row in the name column.
2. Go to the decimal column and reduce that value to 0 for the first row. You will see why we did this in the next step. Reduce the decimal column to 1 in the second row since we will enter GPA scores for this variable.

FIGURE 1.9

SPSS variable view.

1. Go to the values column and click on the small gray box with three dots. In the dialog box, enter 1 in the value cell and class 1 in the label cell, and then select add. Repeat these steps by entering 2 for class 2 and 3 for class 3, then select OK. When you go back to the data view tab, SPSS will now recognize 1 as class 1, 2 as class 2, and so on in the column you labeled classes.
2. Open the data view tab. In the first column, enter 1 five times, 2 five times, and 3 five times (see Figure 1.10). This tells SPSS that there are five students in each class. In the second column, enter the GPA scores for each class by row. The data for each class are now listed across the rows next to the corresponding codes (1, 2, and 3) for each class.

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## FIGURE 1.10

Data entry in data view for SPSS.

The data for all the variables are labeled, coded, and entered. So long as you do this correctly, SPSS will make summarizing, computing, and analyzing any statistic taught in this book fast and simple.

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## CHAPTER SUMMARY ORGANIZED BY LEARNING OBJECTIVE

LO 1-2: Distinguish between descriptive and inferential statistics. They are characterized by three

types of statistics; explain how samples and populations, properties: order, differences, and ratios. There

are four scales of measurement: nominal, ordinal, interval, and ratio. Nominal values are typically coded (e.g., seasons, months,

- Statistics is a branch of mathematics used to summarize, analyze, and interpret a group of numbers or observations. Descriptive statistics are procedures used to make sense of rating scale values, temperature), and ratio values are also equidistant but have a true zero (e.g., weight, height, calories).

that allow researchers to infer whether obser-

vations made with samples can be generalized to the population. LO 5-6: Distinguish between qualitative and quantitative data; determine whether a value is discrete or

- A population is a set of all individuals, items, or data of interest. A characteristic that

describes a population is referred to as a population parameter. A sample is a set of

- A continuous variable is measured along a continuum, whereas a discrete variable is measured in whole units or categories. So continuous variables are measured at any place beyond the decimal point, whereas discrete variables are not measured along a continuum. A quantitative variable varies by

LO 3: Describe three research methods commonly

amount, whereas a qualitative variable  
used in behavioral science.  
varies by class.

- The experimental design uses randomization, manipulation, and comparison to control SPSS LO 7: Enter data into SPSS by placing each variables to demonstrate cause-and-effect rela- group in separate columns and by placing each group

tionships. The quasi-experimental design is in a single column (coding is required).  
structured similar to an experiment but lacks  
randomization or a comparison group. •• SPSS can be used to enter and define variables.

- The correlational method is used to measure All variables are defined in the variable  
pairs of scores for each individual and exam- view tab. The values recorded for each vari-  
ine the relationship between the variables. able are listed in the data view tab. Data can  
be entered by column or by row in the data

LO 4: State the four scales of measurement and pro- view tab. Listing data by row requires coding  
vide an example for each. the independent variable. Independent  
variables are coded in the variable view tab in

- The scales of measurement refer to how the values column (for more details, see  
the properties of numbers can change with Section 1.6).

## 24 PART I: INTRODUCTION AND DESCRIPTIVE STATISTICS

### KEY TERMS

coding interval scale ratio scale  
continuous variable levels of the independent variable raw score  
data nominal scale sample  
datum operational definition sample statistic  
dependent variable (DV) ordinal scale scale of measurement  
descriptive statistics population scales of measurement  
discrete datavariable population parameter science  
equidistant scale qualitative variable score  
experiment quantitative variable statistics  
independent variable (IV) quasi-independent variable true zero  
inferential statistics random assignment

### END-OF- CHAPTER PROBLEMS

Factual Problems a. Describe

b. Infer

1. What is the difference between descriptive and

c. Summarize

inferential statistics?

12. State whether each of the following is true or false.

2. Distinguish between data and a raw score.

a. Graphs, tables, and summary statistics all illus-

3. By definition, how is a sample related to a population? b. Inferential statistics are procedures used to make inferences about a population, given
4. State three commonly used research methods in only a limited amount of data. behavioral science.
- c. Descriptive statistics can be used to describe
5. In an experiment, researchers measure two populations and samples of data. types of variables: independent and dependent variables. 13. A researcher measured behavior among all individuals in some small population. Are inferential
- a. Which variable is manipulated to create the statistics necessary to draw conclusions concerning this population? Explain.
- b. Which variable is measured in each group?
14. Appropriately use the terms sample and population to describe the following statement: A statistics class has 25 students enrolled, but only 23 students attended class.
6. State the four scales of measurement. Which scale of measurement is the most informative? 7. Can a nominal variable be numeric? Explain.
15. On occasion, samples can be larger than the population from which it was selected. Explain why on an interval and ratio scale of measurement? this can't be true.
8. What is the main distinction between qualitative and quantitative variables? 9. A quantitative variable varies by \_\_\_\_\_; a qualitative variable varies by \_\_\_\_\_. 16. A researcher demonstrates that eating breakfast in the morning causes increased alertness throughout the day. What research design must the researcher have used in this example? Explain.
10. What are the two types of data that are collected and measured quantitatively?
17. A researcher measures the height and income of participants and finds that taller men tend to earn greater incomes than shorter men. What
11. State whether each of the following words best describes descriptive statistics or inferential statistics. this example? Explain.
- CHAPTER 1 : INTRODUCTION TO STATISTICS 25
18. State whether each of the following variables are examples of an independent variable or quasi-independent variable. Only answer quasi-
22. Rank the scales of measurement in order from least informative to most informative.

independent for variables that can't be randomized. 23. What is the main disadvantage of measuring qual-

itative data? In your answer, explain why quanti-

a. Marital status

tative research is most often applied in the

b. Political affiliation behavioral sciences.

c. Time of delay prior to recall

24. State whether each of the following describes a

d. Environment of research setting

study measuring qualitative or quantitative

e. Years of education data.

f. Type of feedback (negative, positive) a. A researcher distributed open-ended questions

19. To determine whether a new sleeping pill was to participants asking how they feel when effective, adult insomniacs received a pill (either they are in love.

real or fake), and their sleeping times were subse- b. A researcher records the blood pressure of quently measured (in minutes) during an over- participants during a task meant to induce night observation period. stress.

a. Identify the independent variable in this study. c. A psychologist interested in drug addiction

b. Identify the dependent variable in this study. injects rats with an attention-inducing drug and then measures the rate of lever pressing.

20. A researcher tests whether cocaine use increases d. A witness to a crime gives a description of the

impulsive behavior in a sample of cocaine-depen- suspect to the police.

dent and cocaine-inexperienced mice.

a. Identify the independent variable in this study. 25. State whether each of the following are continu-

ous or discrete data.

b. Identify the dependent variable in this study.

a. Time in seconds to memorize a list of words

21. Researchers are interested in studying whether b. Number of students in a statistics class personality is related to the month in which

c. The weight in pounds of newborn infants someone was born.

d. The SAT score among college students

a. What scale of measurement is the month of

birth? 26. Fill in the table below to identify the characteris-

b. Is it appropriate to code the data? Explain. tics of each variable.

Type of Data (qualitative Type of Number Scale of

Variable vs. quantitative) (continuous vs. discrete) Measurement

Gender

Seasons

Time of day

Rating scale score

Movie ratings (one to four stars)

Number of students in your class

Temperature (degrees Fahrenheit)

Time (in minutes) to prepare dinner

Position standing in line

## 26 PART I: INTRODUCTION AND DESCRIPTIVE STATISTICS

Problems in Research in terms of randomization, manipulation, and comparison.

27. Grading the public school system. In

December 2009, a CBS News poll asked 1,048 29. Gender and mental toughness. Nicholls and

Americans to grade the U.S. public school system. colleagues (2009) measured mental toughness

In the poll, they asked, "How would you grade (among other measures) in a sample of athletes

the U.S. on the quality of the public schools in ranging in age from 15 to 58 years. They reported

this country?" The results are described in the that "males scored significantly higher than females

table below, with A being the best grade and F on total mental toughness" (p. 74). Is it possible that

being the worst grade: an experimental method was used to show this effect? Explain.

30. Facial distinctiveness. Wickham, Morris, and

Grade % Fritz (2000) wrote, "Although there has been

extensive interest in the influence of distinctive-

A 5 ness on memory for faces, measuring the dis-

B 23 tinctiveness of a face is problematic because

distinctiveness is difficult to define" (pp. 99-100).

C 32 What type of definition will researchers need to

make for the term distinctiveness?

D 26

31. Height and educational attainment.

F 12

Szklarska and colleagues (2004) hypothesized

that taller young men are more likely to move up

the scale of educational attainment compared

Source: Reported at <http://www.pollingreport.com/ed.htm> with shorter individuals from the same social

a. What was the sample size for this poll? background. They recruited 91,373 nineteen-

b. Based on the percentages given in the table, year-old men to participate in the study. Do these

how well was the U.S. public school system participants most likely represent a sample or

graded in general? population? Explain.

28. Racial attitudes in college. Shook and Fazio 32. Describing the scales of measurement.

(2008) randomly assigned White freshman col- Harwell and Gatti (2001) stated, "If the test score

lege students to live with a White (same-race variable possesses an interval scale, then the dif-

group) or Black (different-race group) roommate difference in proficiency reflected in scores of 10 and in a college dormitory. After a few months, 15 is exactly the same as the difference in proficiency researchers measured racial attitudes and proficiency reflected in scores of 15 and 20" (p. 105). Compared differences between groups. Is this study an example of an experiment? Explain your answer researchers to draw this conclusion?

Answers for even numbers are in Appendix C.

## APPENDIX C

### Chapter Solutions

#### for Even-Numbered

#### End-of-Chapter Problems

### CHAPTER 1

Factual Problems 16. An experimental research method since the researcher claims to have demonstrated cause.

2. Data describe a set of measurements (made up of raw scores); a raw score describes individual mea-

18.

surements.

a. Quasi-independent variable

4. Experimental, quasi-experimental, and correlational research methods.

c. Independent variable

6. The four scales of measurement are nominal, ordinal, interval, and ratio. Ratio scale measurements are the most informative.

f. Independent variable

8. Interval variables DO NOT have a true zero, and ratio variables DO have a true zero. 20.

10. Continuous and discrete numbers. a. Cocaine use (dependent vs. inexperienced)

b. Impulsive behavior

#### Concepts and Applications Problems

22. Nominal, ordinal, interval, and ratio

12.

a. False 24.

b. True a. Qualitative

c. True b. Quantitative

14. The statistics class has a population of 25 students enrolled, but a sample of only 23 students attended. c. Quantitative

27

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26.

Type of Data (qualitative Type of Number Scale of  
Variable vs. quantitative) (continuous vs. discrete) Measurement

Gender Qualitative Discrete Nominal

Seasons Qualitative Discrete Nominal

Time of day Quantitative Continuous Ratio

Rating scale score Quantitative Discrete Interval

Movie ratings (one to four stars) Quantitative Discrete Ordinal

Number of students in your class Quantitative Discrete Ratio

Temperature (degrees Fahrenheit) Quantitative Continuous Interval

Time (in minutes) to prepare dinner Quantitative Continuous Ratio

Position standing in line Quantitative Discrete Ordinal

Problems in Research 30. An operational definition.

28. Yes, this is an experiment. The race conditions 32. Equidistant scales. Variables on an interval scale

were created by the researchers (manipulation), have equidistant scales, meaning that differences students were randomly assigned to each condi- along this scale are informative.  
tion (randomization), and racial attitudes in two groups were compared (comparison).

STA-PROB-WORK BOOK

FOR

PHYSICAL SCIENCE

AND

ENGINEERING

STA 213/MTH 214 STATISTICS FOR PHYSICAL SCIENCE AND ENGINEERING: 3 Units

Course Outline

Scope for statistical methods in physical sciences and engineering. Measures of location, partition and

dispersion. Elements of probability. Probability distribution: binomial, Poisson, geometric, hypergeometric, negative-binomial, normal, Weibull, Gompertz etc. Estimation (Point and internal) and

tests of hypotheses concerning population means proportions and variances. Regression and correlation.

Non-parametric tests. Contingency table analysis. Introduction to design of experiments. Analysis of variance.

STA 211: Probability II (3 Units C: LH 45)

Further permutation and combination. probability laws. Conditional probability, independence. Bayes'

theorem. probability distribution of discrete and continuous random variables: binomial, Poisson,

geometric, hypergeometric, rectangular (uniform), negative exponential, binomial. Expectations and moments of random variables. Chebyshev's inequality. joint marginal and conditional distributions and moments. Limiting distributions. Discrete, and continuous random variables, standard distributions, moments and moment-generating functions. Laws of large numbers and the central limit theorem.

NAME OF STUDENT:

.....

MATRIC NO.:

.....

DEPARTMENT:

.....

COLLEGE:

.....

SESSION:

.....

2

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## Chapter One

## 1.1 Introduction

The subject of Statistics is an old discipline; as old as human society itself. It has been used since the

existence of man on earth. Though, the sphere of its utility was very much restricted. However, Statistics

was regarded as a universal language of sciences derived from the Latin word *status* or the Italian word

*statista* or the German word *statistik* or the French word *statistique* with each meaning a political

state. In the ancient times, the scope of the subject of Statistics was initially limited to the collection of

data by the governments for framing military and fiscal policies;

(i) Age and sex-wise population of the country;

(ii) Property and wealth of the country.

However, the subject of Statistics has now adopted in several places like the academia, economics, medical records, marketing, hydrology, philosophy, health and life sciences, psychology, sociology, education, medicine, business, nursing etc. In some cases, it has been adopted as a universal language of

sciences. Thus, the understanding and careful use of the statistical methods has enable us to accurately

describe the outcomes or finding of several scientific research, make qualitative and quantitative decisions, and possibly make useful estimations.

## 1.2. Definition of Statistics.

Statistics is the science that deals with gathering (collecting), classifying, analysing, presenting, and

interpreting data. Statistics helps us turn data into information to see the relationship between variables—

or the "big picture." This course is an introduction to statistics with an emphasis on modern engineering

applications. Students explore concepts of probability theory, discrete and continuous random variables,

bivariate probability distributions, categorical data analysis, and model building. Statistics can also be

defined as a subject or science that classifies facts representing the conditions of the people in a stat...

specially those facts which can be stated in number or in tables of numbers or in any tabular or classified

arrangement. In some contests, it's a science of the measurement of social organism, regarded as a whole

in all its manifestations. It can also be defined science of estimates and probabilities.

## 1.3 Importance and Scope of Statistical Methods in Physical Sciences and Engineering.

In the ancient times, the word Statistics was regarded only as the science of state-craft, and was used to

collect information relating to crime, military strength, population, wealth, etc., for devising military and

fiscal policies. Some of importance include:

- (i) Statistics in Planning
- (ii) Statistics in state
- (iii) Statistics in Mathematics
- (iv) Statistics in Economics
- (v) Statistics in Business and Management
- (vi) Statistics in Accountancy and Auditing
- (vii) Statistics in Industry
- (viii) Statistics in Insurance
- (ix) Statistics in Astronomy
- (x) Statistics in Physical Sciences
- (xi) Statistics in Social Sciences
- (xii) Statistics in Biology and Medical Sciences

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(xiii) Statistics in Psychology and Education

(xiv) Statistics in War

#### 1.4 Limitations of Statistics.

Though Statistics is indispensable to almost all sciences or social sciences, physical and natural sciences,

however, it has several limitations that restrict its scope. These include the following:

- (i) Statistics does not study qualitative phenomenon.
- (ii) Statistics does not study individuals.
- (iii) Statistical laws are not exact.
- (iv) Statistics is liable to be misused.

Thus, the use of Statistics by experts who are well experienced, and skilled in the analysis, and interpretation of statistical data for drawing correct, and valid inferences very much reduces the chances

of mass popularity of this important sciences.

#### 1.5 Definition of Basic terms in Statistics

a. Descriptive statistics: This refers to the methods of organizing, summarizing and presenting data in an informative way or manner.

b. Inferential statistics or statistical inference or inductive statistics: This involves finding information about a population based on the knowledge gather from the study of a representative sample of the population.

c. Variable: A variable is a symbol or anything that can assume any of a prescribed set of values. A variable that can theoretically assume any value between two given values is called a continuous variable; otherwise it is called a discrete variable. A variable is a characteristic under study that

assumes different values for different elements. In contrast to a variable, the value of a constant is fixed.

d. Population or Universe: It is the entire collection, or set of individuals or objects whose properties are to be analysed.

e. Sample: A sample is a subset of a population. It is a subset containing the characteristics of a larger population. It is a representative part of a population.

f. Data: Data are raw facts or unprocessed information.

g. Statistic: A statistic is a quantity whose numerical values can be obtained from data. Also, a fact or piece of data obtained from a study of a large quantity of numerical data. "the statistics show that the crime rate has increased". This is a characteristic of a sample.

h. Experiment: An ordered procedure which is performed with the objective of verifying, and determining the validity of the hypothesis.

i. Parameter: This is a characteristic of a population. Examples include population mean, and population standard deviation.

j. Sampling frame: It's a list of the items or people forming a population from which a sample is taken.

k. Sampling Units: Sample units are the members of the population from which measurements are taken during sampling. Sample units are distinct and non-overlapping entities, such as quadrats or transect, individual plants, branches within a plant, etc.

l. Census: the complete enumeration of a population or group at a point in time with respect to well-defined characteristics.

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m. Element or Member: An element or member of a sample or population is a specific subject or object (for example, a person, firm, item, state, or country) about which the information is collected.

Assignment 1.1:

1. Define Statistics

2. Define and explain the following terms in your understanding:

a. Population

b. Sample

c. Statistics

d. Data

e. Variable

f. Experiment

SOLUTION

8

SOLUTION

9

SOLUTION

10

## SOLUTION

11

## SOLUTION

12

## Chapter Two

## 2.1 Average or Measures of Location or Central Tendency.

In our daily endeavour, we summarize data to make decision in one way or the other. For example, in a

market survey, we report an average (mean) price of a particular commodity as a representative of the

price of the commodity in the market. Sometimes, we also use the most common price (mode) or the price

that tends to fall in the middle of all other prices (median) of a particular commodity. When we do all

these, we are making use of the knowledge of measures of location or central tendency.

## 2.2 Arithmetic Mean or Mean.

The arithmetic mean of a given set of observations is their sum divided by the number of observations.

Thus, it is given by

$$\frac{1}{n} \sum_{i=1}^n x_i$$

$$x_1, x_2, \dots, x_n$$

$$x_i$$

$$x_i, i = 1$$

$$n$$

The above formula is used when data are given in array.

Example 2.1: Find the mean of 5, 8, 10, 15, 24 and 28.

$$\frac{1}{n} \sum_{i=1}^n x_i$$

$$x_1, x_2, \dots, x_n$$

$$x_i$$

$$5, 8, 10, 15, 24, 28$$

$$x_i, i = 1$$

$$n = 6$$

$$\frac{1}{n} \sum_{i=1}^n x_i$$

$$n$$

$$\sum_{i=1}^n f_i x_i$$

When the sampled data are from a frequency distribution for ungrouped form, we adopt  $\sum_{i=1}^n x_i f_i$

$$\sum_{i=1}^n f_i$$

$$i = 1$$

$$i$$

Example 2.2: Find the mean or expectation

$$x = 1, 6, 11, 16, 21$$

F 13 10 20 5 30

x 1 6 11 16 21

F 13 10 20 5 30 n

$\sum f$

$\sum_{i=1}^n$

i

$=13+10+20+5+30=78$

fx 13 60 220 80 360 n

$\sum fx$

$\sum_{i=1}^n$

i i

$=13+60+220+80+630=1005$

13

n

$\sum fx i i$

1005

$x \sum_{i=1}^n i n = 12.88$

$\sum f$

78

i

$\sum_{i=1}^n$

The expectation can also be obtained using the probability as

n

$x \sum_{i=1}^n x i p i$  where p is the probability of occurrence.

$\sum_{i=1}^n$

Example 2.3: Obtain the expectation of the example 2.2

$s/n \sum x f P(x) \sum x p(x)$

$1 \cdot 13 \cdot 13/78 = 0.17 \cdot 1 \cdot 0.17 = 0.17$

$2 \cdot 6 \cdot 10 \cdot 10/78 = 0.13 \cdot 6 \cdot 0.13 = 0.78$

$3 \cdot 11 \cdot 20 \cdot 20/78 = 0.26 \cdot 11 \cdot 0.26 = 2.86$

$4 \cdot 16 \cdot 5 \cdot 5/78 = 0.06 \cdot 16 \cdot 0.06 = 0.96$

$5 \cdot 21 \cdot 30 \cdot 30/78 = 0.38 \cdot 21 \cdot 0.38 = 7.98$

Total n  $0.17+0.78+2.86+0.96+7.98=$

$\sum p \sum x \sum 1$

$\sum_{i=1}^n$

$= 12.75$

n

$x \sum_{i=1}^n x i p i = 12.75$

i = 1

### 2.2.1 Basic terms associated with Grouped data.

Let the following be given

Class interval 1-5 6 - 10 11 - 15 16 - 20 21 - 25

F 13 10 20 5 30

(i) Class interval and class limits: A symbol defining a class such as 1 – 5 in the Table above is called a class interval. The end numbers, 1 and 5, are called class limits; the smaller number (1) is the lower class limit (LCL), and the larger number (5) is the upper class limit (UCL).

(ii) Class boundaries: The size of the gap between classes is the difference between the upper class limit of one class and the lower class limit of the next class. Thus,

Lower Class Boundary (LCB) =  $LCL - D/2$

Upper Class Boundary (UCB) =  $UCL + D/2$

where D is the difference between the lower class limit of any class interval and the upper class limit of the class just before that class. In the example above, using class interval 6 – 10,  $D = 1$ . Thus, the class boundary for the class interval 1 – 5 is given as

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$LCB = 1 - \frac{1}{2} = 0.5$  and  $UCB = 5 + \frac{1}{2} = 5.5$

Class interval 1-5 6 - 10 11 - 15 16 - 20 21 - 25

Class Boundary 0.5 – 5.5 5.5 – 10.5 10.5 – 15.5 15.5 – 20.5 20.5 – 25.5

(iii) Width or Size of Class: This is difference between the upper class boundary and the lower class boundary of any class. Class Size (C) =  $UCB - LCB$ . Example:  $10.5 - 5.5 = 5$

(iv) Class Mark (Class Mid-Point): This is given by  $\frac{1}{2}(LCL + UCL)$ . Example:  $\frac{1}{2}(1+5) = 6/2 = 3$ .

Class interval 1-5 6 - 10 11 - 15 16 - 20 21 - 25

Class Mid-Point 3 8 13 18 23

Thus, we can obtain the mean of a grouped data as

n

$\sum f x_i$

$\times \frac{1}{n}$

$\sum f$

i = 1

i

where  $x_i$  is the class mid-point.

Example 2.4: Compute the mean given the following data set

Class interval 1-5 6 - 10 11 - 15 16 - 20 21 - 25

F 13 10 20 5 30

Solution

Class 1-5 6 - 10 11 - 15 16 - 20 21 - 25 Total

interval

F 13 10 20 5 30 78

Class Mid- 3 8 13 18 23

Point (x)

fx 39 80 260 90 690 1159

Class size = 25.5 - 20.5 = 5

15

n

$\sum f x_i$

1159

$\bar{x} = \frac{\sum f x_i}{n} = \frac{1159}{78} = 14.86$

$\sum f$

78

i

$i = 1$

### 2.2.2 The method of assumed Mean

The mean of statistical data can be calculated by using the method of assumed mean. This method is

applicable if the individual number or class mid-point represented by X is very large. It alleviates the

stress of working with large numbers by reducing the individual number by a constant number (A) called

assumed mean. The mean using assumed mean is given by the following formula

N

$\sum f d_i$

$\bar{x} = A + \frac{\sum f d_i}{N}$

N

where  $d_i = X_i - A$

Example 2.5: Calculate the mean of the distribution table

Class 13 - 17 18 - 22 23 - 27 28 - 32 33 - 37 38 - 42 43 - 47 48 - 52 53 - 57

Interval

Frequency 2 22 10 14 3 4 6 1 1

Determine the class mid-point of each class, calculate the deviation of each class mid-point from the

assumed mean  $A = 35$  and then calculate the product as shown in the Table below.

Class 13 - 18 - 23 - 28 - 33 - 38 - 43 - 48 - 53 - Total

Interval 17 22 27 32 37 42 47 52 57

Frequency 2 22 10 14 3 4 6 1 1 63

15 20 25 30 35 40 45 50 55

Mid-point

$d = X - A = -20 -15 -10 -5 0 5 10 15 20$

$fd = -40 -330 -100 -70 0 20 60 15 20 -425$

N

$\sum fd = 0$

$\sum 425$

$\bar{x} = A + \frac{\sum fd}{N} = 10 + \frac{0}{63} = 10$

N 63

2.3 Median.

The median is the value that divides a data set that has been ranked in increasing order in two equal

halves. If the data set has an odd number of values, the median is given by the value of the middle term in

the ranked data set. If the data set has an even number of values, the median is given by the average of the

two middle values in the ranked data set.

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As is obvious from the definition of the median, it divides a ranked data set into two equal parts. The

calculation of the median consists of the following two steps:

- Rank the given data set in increasing order.
- Find the value that divides the ranked data set in two equal parts. This value gives the Median.

Note that if the number of observations in a data set is odd, then the median is given by the value of the

middle term in the ranked data. However, if the number of observations is even, then the median is given

by the average of the values of the two middle terms.

The depth (number of positions from either end), or position of the median is determined by the formula.

$n = 1$

Depth of median =

2

For example, find the median of these numbers 3, 4, 5, 7, 9. In the above example,  $n = 5$ , thus, the depth of

the median is given as

$\frac{n+1}{2} = \frac{5+1}{2} = 3$

Depth of median = 3

2

This implies that the median is the third number from either end in the ranked data, that is, median is 5.

For frequency distribution data,

x 1 6 11 16 21

F 13 10 21 5 30

We can obtain the median of the distribution as

x 1 6 11 16 21

F 13 10 21 5 30

Cumulative 13 23 44 49 79

$79 \div 2 = 39.5$

Thus, the depth of median = 39.5 . Thus, the median is 11.

2

Now, even data set, we have

x 1 6 11 16 21

F 13 10 20 5 30

Cumulative 13 23 43 48 78

$78 \div 2 = 39$

Thus, the depth of median = 39 . Thus, the median is 11.

2

2.3.1 The median of grouped data set.

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The median of a grouped data is given by the following formula

$\frac{N}{2}$

$\frac{N}{2} - C_f$

Median =  $L_1 + \frac{\frac{N}{2} - C_f}{f_m} \times C$

$f_m$

$C$

where,

$L_1$  is the lower class boundary of the median class,

$n$

$N = \sum f_i$  total frequency of the distribution,

$i = 1$

$C_f$  is the cumulative frequency of the class just before or preceding the median class

$f_m$  is the frequency of the median class,

$C$  is the class size.

Example 2.6: Compute the median of the following data set

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 5 12 2 15 13 20 7 5 1

The median class is 50 - 59.

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 5 12 2 15 13 20 7 5 1

Cumulative 5 17 19 34 47 67 74 79 80

$L1 = 49.5$

$n$

$N \sum f_i = 80$

$i \leq 1$

$Cf_b = 34$

$f_m = 13$

$C = 10$

$\frac{N}{2}$

$\frac{80}{2} = 40$

Median  $\frac{L1}{C} = \frac{49.5}{10} = 4.95$

$f_m = 13$

$\frac{4.95}{13}$

Thus, the median give as 54.1.

### 2.3.2 The graphical method of obtaining the Median

a. Presenting the data of given series on a graph in the form of less than ogive or more than ogive.

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b. Presenting the data of given series on a graph, simultaneously in the form of both less than ogive and more than ogive. The point where these two ogives meet is the median value of the given series.

### 2.4 Mode.

The mode is the value that occurs most frequently in a set of observations and around which the other

items of the set cluster densely. In other words, the mode of a distribution is the value at the point around

which the items tend to be most heavily concentrated.

Example 2.7: Find the mode of the following data set. 2, 2, 3, 5, 6, 6, 9, 10, 3, 4, 3, 9, 3.

The mode in this set of data is 3 because it is the most frequent number.

The model of random variable is said to be bimodal if the mode has 2 variables.

Example 2.8: Compute the model of the following data 2, 2, 3, 5, 6, 4, 4, 6, 9, 10, 3, 4, 3, 9, 4, 3.

The mode in this case is 3 and 4. Thus, it is a bimodal.

Also, in some cases, the data might not have mode. For example, for the data 6, 7, 13, 5, 4, 11, 2, 18 and

9. This data set has no mode.

However, for data that are grouped, the mode is obtained mathematically as

$\frac{D1}{D2}$

Mode  $\frac{L1}{C} = \frac{49.5}{10} = 4.95$

$\frac{D1}{D2}$

where,

$L_1$  is the lower class boundary of the modal class,

$f_0$  is the frequency of class before or preceding the modal class,

$f_1$  is the frequency of the modal class,

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$f_2$  is the frequency of the class after or succeeding the modal class,

$C$  is the class size,

$$D_1 = f_1 - f_0$$

$$D_2 = f_1 - f_2$$

Example 2.9: Obtain the mode of the following data

x 1 6 11 16 21

F 13 10 20 5 30

In the above case, the data are not grouped. Thus, the mode is obtained as 21. This implies that 21 appeared 30 times compared to other random values.

In some cases, the mode can be applied to data that are grouped.

Example 2.10: Calculate the mode of the distribution table

Class 13 - 17 18 - 22 23 - 27 28 - 32 33 - 37 38 - 42 43 - 47 48 - 52 53 - 57

Interval

Frequency 2 22 10 14 3 4 6 1 1

In the above example, the modal class is 18 - 22. Hence,

$$L_1 = 17.5$$

$$f_0 = 2$$

$$f_1 = 22$$

$$f_2 = 10$$

$$C = 5$$

$$D_1 = f_1 - f_0 = 22 - 2 = 20$$

$$D_2 = f_1 - f_2 = 22 - 10 = 12$$

$$\therefore D_1 > D_2$$

$$\text{Mode} = L_1 + \frac{D_1}{D_1 + D_2} \times C = 17.5 + \frac{20}{20 + 12} \times 5 = 17.5 + 3.125 = 20.625$$

$$\therefore D_1 > D_2 \therefore 20 > 12$$

#### 2.4.1 Graphical method of obtaining the mode with a given set of data

##### Calculation of Mode by Graphical Method

1. Mode by Graphical Method.

2. Draw a histogram of the given set of series.

3. The rectangle of the histogram with the greatest height will be the modal class of the series.

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Figure 2.1: Calculation of Mode by Graphical Method

#### 2.5 Empirical Relationship between Mean, Median and Mode.

This section describes the relationships among the mean, median, and mode for three such histograms and

frequency distribution curves. Knowing the values of the mean, median, and mode can give us some idea

about the shape of a frequency distribution curve.

(i) For a symmetric histogram and frequency distribution curve with one peak (see Figure 2.2), the values of the mean, median, and mode are identical, and they lie at the center of the distribution.

Figure 2.2: Mean, median, and mode for a symmetric histogram and frequency distribution curve.

(ii) For a histogram and a frequency distribution curve skewed to the right (see Figure 2.3), the value of the mean is the largest, that of the mode is the smallest, and the value of the median lies between these two. (Notice that the mode always occurs at the peak point.) The value of the mean is the largest in this case because it is sensitive to outliers that occur in the right tail. These outliers pull the mean to the right.

Figure 2.3: Mean, median, and mode for a histogram and frequency distribution curve skewed to the right.

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(iii) If a histogram and a frequency distribution curve are skewed to the left (see Figure 2.4), the value of the mean is the smallest and that of the mode is the largest, with the value of the median lying between these two. In this case, the outliers in the left tail pull the mean to the left.

Figure 2.4: Mean, median, and mode for a histogram and frequency distribution curve skewed to the left.

The cases (ii) and (iii) can be empirically and mathematically expressed as

$$\text{Mode} = \text{Mean} - 3(\text{Mean} - \text{Median})$$

## 2.6 Geometric Mean.

The geometric mean (G) of a set of N positive numbers  $X_1, X_2, X_3, \dots, X_N$  is the Nth root of the product of the number.

$$G = \sqrt[N]{x_1 \times x_2 \times x_3 \dots \times x_N}$$

For a grouped data, we have

$$G = \sqrt[N]{f_1 x_1 \times f_2 x_2 \times f_3 x_3 \dots \times f_N x_N}$$

Example 2.11: Find the geometric mean of the numbers 2, 3, 5

The geometric mean is obtain as  $G = \sqrt[3]{2 \times 3 \times 5} = 3.11$ .

## 2.7 Harmonic Mean.

The harmonic mean H of a set of N numbers  $x_1, x_2, x_3, \dots, x_N$  is the reciprocal of the arithmetic mean of

the reciprocals of the numbers. It is given as

$$H = \frac{1}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_N}}$$

$$H = \frac{N}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_N}}$$

$$N = x_1 \times x_2 \times x_3 \dots \times x_N$$

## 2.8 Relation between Arithmetic Mean, Geometric Mean and Harmonic Mean.

The geometric mean of a set of positive numbers  $X_1, X_2, X_3 \dots X_N$  is less than or equal to their arithmetic

mean but is greater than or equal to their harmonic mean.

$H \leq G \leq X$ .

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Assignment 2.1: Compute the median

x 1 6 11 16 21

F 13 10 21 5 30

SOLUTION

23

Assignment 2.2: Calculate the mode, mean, median of the distribution table

Class 13 - 17 18 - 22 23 - 27 28 - 32 33 - 37 38 - 42 43 - 47 48 - 52 53 - 57

Interval

Frequency 20 12 11 15 13 14 6 9 8

SOLUTION

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Chapter Three

3.1 Measure of Dispersion.

Average or measure of central tendency gives us an idea of the concentration of the observations about

the central part of the distribution. Thus, when decisions are made, it is not enough to report a number that

describes the centre of the sample. For instance, why do we send two marketers for a market survey, and

we prefer the reported by one of them. This may be due to consistency and the closeness of the reported

price to average price in the market. A measure of the closeness of the average values is what statisticians

refer to as measures of dispersion or variation. Thus, an estimate with lower dispersion is better than the

other with higher dispersion. There are several methods for finding dispersion in a set of data.

1. Range

2. Mean deviation

3. Variance

4. Standard Deviation

5. Coefficient of variation

3.2 Objectives and Significance of the Measures of Dispersion.

The main objectives of studying measure of dispersion include the following:

a. To find out the reliability of an average.

b. To control the variation of the data from the central value.

- c. To compare two or more sets of data regarding their variability.
- d. To obtain other statistical measures for further analysis of data.

### 3.2.1 Characteristics of an ideal measure of dispersion

- a. It should be rigidly defined.
- b. It should be easily understood and calculated.
- c. It should be based on all observations.
- d. It should be flexible for further mathematical treatment.
- e. It should be affected as little as possible by fluctuations of sampling.
- f. It should not be affected much by extreme observations

### 3.3 Range.

The range is the simplest of all the measures of dispersion. It is defined as the difference between the

extreme observations of the distribution. Thus, the range is the difference between the greatest (maximum) and the smallest (minimum) observation of the distribution.

Thus,

$$\text{Range} = X_{\max} - X_{\min} = \text{Largest} - \text{Smallest}$$

Examples 3.1: Find the range of the following prices of goods purchased by a vendor 6, 4, 1, 6, 10, 3

The highest observation is 10 and the smallest is 1.

Thus, the Range is

$$\text{Range} = 10 - 1 = 9.$$

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In the above case, the data are not grouped or ungrouped but in array.

Example 3.2: Compute the range of the frequency distribution

Now, supposed the data were given in a frequency distribution form. ie. Ungrouped

X 10 20 30 40 50 60 70 80 90 100

Frequency 5 12 2 15 13 20 7 5 3 8

The largest value is 120 and the smallest value is 10. Thus the range is given as

$$\text{Range} = \text{Largest} - \text{Smallest} = 100 - 10 = 90.$$

Now, suppose for a group data.

Example 3.3: Obtain the Range of the following data.

X 10-19 20-29 30-39 40-49 50-59 60-69 70-79 80-89 90-99

Frequency 5 12 2 15 13 20 7 5 1

The range is obtain as

X 10-19 20-29 30-39 40-49 50-59 60-69 70-79 80-89 90-99

Frequency 5 12 2 15 13 20 7 5 1

Mid-Point 14.5 24.5 34.5 44.5 54.5 64.5 74.5 84.5 94.5

The largest value is 94.5 and the smallest is 14.5.

$$\text{Thus, the Range} = 94.5 - 14.5 = 80.0$$

Note: In case of a frequency distribution, the frequencies of the various values of the variable or variables

or classes are IMMATERIAL since range depends only on the two extreme observations.

Another important measure is the Coefficient of range. The Coefficient of range is used to compare the

variability of the distributions of different units of measurement.

Thus,

$$X_{\max} - X_{\min}$$

$$\text{Coefficient of Range} = \frac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}$$

$$X_{\max} - X_{\min}$$

Example 3.4: Calculate the coefficient of range of monthly earnings for a year in naira

S/N 1 2 3 4 5 6 7 8 9 10 11 12

Monthly 139 150 151 151 157 158 160 161 162 162 173 175

Earnings

26

$$X_{\max} - X_{\min} = 175 - 139 = 36$$

$$\text{Coefficient of Range} = \frac{36}{175 + 139} = 0.115$$

$$X_{\max} - X_{\min} = 175 - 139 = 36$$

Merits and Demerits of Range

1. Range is not based on the entire set of data.
2. Range is very much affected by fluctuations of sampling
3. Range cannot be used if we are dealing with open end classes
4. Range is not suitable for mathematical treatment
5. Range is very sensitive to the size of the sample
6. Range is too indefinite to be used as a practical measure of dispersion.

Uses of Range

1. Range is used in the industry for statistical quality control of the manufactured product by the construction of R-chart. i.e. The control chart for range
2. Range is by far the most widely used measure of variability in our day-to-day life.
3. Range is used as a very convenient measure by meteorological department for weather forecast. This is because; it is primarily interested to know the limits within which the temperature is likely to vary on a particular day.
4. Range is used in the stock market fluctuations, variations in money rates and rate of exchange.

### 3.4 Mean Deviation or Average Deviation.

The mean deviation is the mean absolute deviations from the mean. Thus, is given as

N

$$\sum X_i$$

$$MD = \frac{\sum |e_i|}{N}$$

N

Note that |....| implies the answer must be positive.

Example 3.5: Compute the mean deviation

S/N 1 2 3 4 5 6 7 8 9 10

X 2 12 15 17 20 24 27 30 35 40

The solution is given as

$n$

$\sum X_i$

222

$X_i - \bar{X}$

$\bar{X} = 22.2$

$n = 10$

S/N 1 2 3 4 5 6 7 8 9 10 Total

X 2 12 15 17 20 24 27 30 35 40 222

27

$X_i - \bar{X} = 2 - 12 = -10, 12 - 12 = 0, 15 - 12 = 3, 17 - 12 = 5, 20 - 12 = 8, 24 - 12 = 12, 27 - 12 = 15, 30 - 12 = 18, 35 - 12 = 23, 40 - 12 = 28$

$22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0, 22.2 - 22.2 = 0$

$= = = 7.2 = 5.2 = 2.2 = 1.8 = 4.8 = 7.8 = =$

20.2 10.2 12.8 17.8

$n$

$\sum X_i$

90

$MD = \sum |e_i|$

$\bar{X} = 9$

$n = 10$

For data in frequency distribution, the mean deviation is given as

$N$

$\sum f X_i$

$i$

$MD = \sum |e_i|$

$n$

$\sum f_i$

$i$

Example 3.6: Suppose we have the following to obtain the mean deviation

x 12 20 24 32 50 62 70 75 86 90

frequency 4 6 8 2 3 7 10 11 9 10

Solution

The first thing to do is to calculate the mean

$n$

$\sum f X_i$

4207

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 60.1$$

 $\sum f_i$ 

70

i

i = 1

$$\sum_{i=1}^n X_i = \sum_{i=1}^n f_i X_i = 3616.6$$

$$1 \cdot 12 \cdot 4 = 48 \quad 12 - 60.1 = -48.1 \quad 192.4$$

$$2 \cdot 20 \cdot 6 = 120 \quad 20 - 60.1 = -40.1 \quad 240.6$$

$$3 \cdot 24 \cdot 8 = 192 \quad 24 - 60.1 = -36.1 \quad 288.8$$

$$4 \cdot 32 \cdot 2 = 64 \quad 32 - 60.1 = -28.1 \quad 56.2$$

$$5 \cdot 50 \cdot 3 = 150 \quad 50 - 60.1 = -10.1 \quad 30.3$$

$$6 \cdot 62 \cdot 7 = 434 \quad 62 - 60.1 = 1.9 \quad 13.3$$

$$7 \cdot 70 \cdot 10 = 700 \quad 70 - 60.1 = 9.9 \quad 99$$

$$8 \cdot 75 \cdot 11 = 825 \quad 75 - 60.1 = 14.9 \quad 163.9$$

28

$$9 \cdot 86 \cdot 9 = 774 \quad 86 - 60.1 = 25.9 \quad 233.1$$

$$10 \cdot 90 \cdot 10 = 900 \quad 90 - 60.1 = 29.9 \quad 299$$

Total 70 4207 1616.6

N

$$\sum f_i X_i = 3616.6$$

$$MD = \frac{\sum f_i X_i}{n} = 23.09$$

 $\sum f_i$ 

70

i = 1

Example 3.7: Obtain the mean deviation for a group data set

Class 13-17 18-22 23-27 28-32 33-37 38-42 43-47 48-52 53-57

Interval

Frequency 2 22 10 14 3 4 6 1 1

Solution

The first thing to do here is to calculate the mean.

n

$$\sum f_i X_i$$

1780

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = 28.25$$

 $\sum f_i$ 

63

i

i = 1

N

$$\sum f_i X_i = 481.25$$

$$MD = \frac{\sum f_i |X_i - n|}{n} = 7.64$$

$$\sum f_i$$

$$63$$

$$i = 1$$

$$S/N \text{ Class Interval Frequency} = \sum f_i X_i - \frac{(\sum f_i X_i)^2}{\sum f_i}$$

$$1 \quad 13-17 \quad 2 \quad 15 \quad 15 \times 2 = 30 \quad 15 - 28.25 = 13.25 \quad 26.5$$

$$2 \quad 18-22 \quad 22 \quad 20 \quad 22 \times 20 = 440 \quad 20 - 28.25 = 8.25 \quad 181.5$$

$$3 \quad 23-27 \quad 10 \quad 25 \quad 10 \times 25 = 250 \quad 25 - 28.25 = 3.25 \quad 32.5$$

$$4 \quad 28-32 \quad 14 \quad 30 \quad 14 \times 30 = 420 \quad 30 - 28.25 = 1.75 \quad 24.5$$

$$5 \quad 33-37 \quad 3 \quad 35 \quad 3 \times 35 = 105 \quad 35 - 28.25 = 6.75 \quad 20.25$$

$$6 \quad 38-42 \quad 4 \quad 40 \quad 4 \times 40 = 160 \quad 40 - 28.25 = 11.75 \quad 47$$

$$7 \quad 43-47 \quad 6 \quad 45 \quad 6 \times 45 = 270 \quad 45 - 28.25 = 16.75 \quad 100.5$$

$$29$$

$$8 \quad 48-52 \quad 1 \quad 50 \quad 1 \times 50 = 50 \quad 50 - 28.25 = 21.75 \quad 21.75$$

$$9 \quad 53-57 \quad 1 \quad 55 \quad 1 \times 55 = 55 \quad 55 - 28.25 = 26.75 \quad 26.75$$

$$\text{Total } 63 \quad 1780 \quad 481.25$$

Example 3.8: Obtain the mean deviation

Class interval 1-5 6 - 10 11 - 15 16 - 20 21 - 25

$$F \quad 5 \quad 4 \quad 2 \quad 7 \quad 2$$

Solution

$$S/N \quad \sum f_i X_i \quad \sum f_i \quad \sum f_i X_i^2$$

$$X \quad F$$

$$1 \quad 3 \quad 5 \quad 15 \quad 9.25 \quad 46.25$$

$$2 \quad 8 \quad 4 \quad 32 \quad 4.25 \quad 17$$

$$3 \quad 13 \quad 2 \quad 26 \quad 0.75 \quad 1.5$$

$$4 \quad 18 \quad 7 \quad 126 \quad 5.75 \quad 40.25$$

$$5 \quad 23 \quad 2 \quad 46 \quad 10.75 \quad 21.5$$

$$\text{Total } 20 \quad 245 \quad 126.5$$

$$n$$

$$\sum f_i X_i$$

$$245$$

$$X = \frac{\sum f_i X_i}{n} = 12.25$$

$$\sum f_i$$

$$20$$

$$i$$

$$i = 1$$

$$N$$

$$\sum f X_i$$

$$i$$

126.5

$$MD = \frac{\sum f_i |X_i - \bar{X}|}{n}$$

$$= \frac{6.325}{1}$$

$$= 6.325$$

20

i

i = 1

Merit of Mean Deviation

1. It is rigidly defined and easily understood and calculated.
2. It is based on all the observations. Thus, it is a better measure of dispersion than the range and quartile deviation.

30

3. The averaging of the absolute deviations from an average iron out the irregularities in the distribution and thus mean deviation provides an accurate and true measure of dispersion.

4. It is less affected by extreme observation.

5. It provides a better measure for comparison about the formation of different distributions.

Demerit of mean Deviation

1. The computation ignores the signs of the deviation.
2. It is not a satisfactory measure of dispersion when compare to the mode or while dealing with a fairly skewed distribution.
3. The steps involved are mathematically unsound and illogical.
4. It is rarely used in sociological studies.
5. It cannot be computed for distributions with open end classes.
6. Mean deviation tends to increase with the increase in size of the sample. Though, not proportionately and not so rapidly as range.

Uses

Despite its mathematical drawback, the mean deviation has found it favour in economics, and business

statisticians. This is because of its simplicity, and accuracy. It is majorly use in computing personal wealth in a community or nation, business cycle, and the National Bureau of Economic Research.

Assignment 3.1: Obtain the mean deviation for a group data set

Class 13-17 18-22 23-27 28-32 33-37 38-42 43-47 48-52 53-57

Interval

Frequency 12 20 10 14 13 14 16 12 15

SOLUTION

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3.5 Variance.

Variance is a useful very important measure of the spread of the original values about the mean.

However, for a population, the variance is denoted in terms of a Greek letter  $\sigma^2$ . Thus, for sample the

variance is represented by  $s^2$ . The variance is the mean of squared deviations from the mean.

The Variance of the distribution is obtained as the squared of the standard deviations from the mean.

Thus, for a population, the variance is given as

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

n

2

i

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2$$

N

Now, when data are given in frequency distribution, the variance is given as

$$\frac{1}{n} \sum_{i=1}^n f_i (X_i - \bar{X})^2$$

2

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

n N

2 2

i i i

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

2

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

$$\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2$$

. The variance for a sample of a set of numbers N for variable  $X_1, X_2, \dots, X_N$  is given as

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

n

2

i

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$n \sum_{i=1}^n (X_i - \bar{X})^2$$

Example 3.9: Obtain the variance for a group data set

Class 13-17 18-22 23-27 28-32 33-37 38-42 43-47 48-52 53-57

Interval

Frequency 2 22 10 14 3 4 6 1 1

Solution

The first thing to do here is to calculate the mean.

$n$

$\sum fX_i$

1780

$X_i$

$n$

$\frac{1780}{63}$

$\sum f$

63

$i$

$i = 1$

32

S/N Class Interval Frequency X FX X<sup>2</sup> FX<sup>2</sup>

1 13 -17 2 15 15x2 = 30 225 450

2 18-22 22 20 22x20 = 440 400 8800

3 23 -27 10 25 10x25 = 250 625 6250

4 28 -32 14 30 14x30 = 420 900 12600

5 33 - 37 3 35 3x35 = 105 1225 3675

6 38 - 42 4 40 4x40 = 160 1600 6400

7 43 - 47 6 45 6x45 = 270 2025 12150

8 48 - 52 1 50 1x50 = 50 2500 2500

9 53 -57 1 55 1x55= 55 3025 3025

Total 63 1780 55850

$\sum fX^2$

$\frac{55850}{63}$

$i = 2$

$\frac{1780}{63}$

$\frac{55850}{63} = 88.45$

$\sum f$

$n$

$\sum f$

63 63

$i$

$i = 1$

### 3.6 Standard Deviation.

This is the square root of the mean of square deviations from the mean. The standard deviation is a far

more useful measure of spread or variability in a set of data. This can be defined as the square root of the

variance. The standard deviation for a population of a set of numbers  $N$  for variable  $X_1, X_2, \dots, X_N$  is given as

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}$$

Now, when data are given in frequency distribution, the standard deviation is given as

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n f_i X_i^2 - \left( \frac{1}{n} \sum_{i=1}^n f_i X_i \right)^2}$$

The standard deviation for a sample of a set of numbers  $N$  for variable  $X_1, X_2, \dots, X_N$  is given as

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

$n$

2

$i$

$$\sum_{i=1}^n$$

$$n-1$$

The standard deviation has a special feature in that it measured in the same units as the original data. That

is, if the original data set were measured in weights, then the mean and standard deviation are measured in

weights. Also, we noted that the large the population standard deviation, the greater the spread or variability among those numbers. The smaller the value of the population standard deviation, the smaller

the amount of variability in the data set.

Example 3.10: The arithmetic mean and standard deviation of series of 20 items were calculated by a

student as 20cm and 5cm respectively. But while calculating them an item 13 was misread as 30.

Find the

correct arithmetic mean and standard deviation.

Solution

$n = 20$ , mean = 20cm, standard deviation = 5cm; wrong value used = 30; correct value = 13

$n$

$$\sum_{i=1}^n X_i = n \bar{X} = 20 \times 20 = 400, \text{ and } \sum_{i=1}^n X_i^2 = n s^2 + n \bar{X}^2$$

$$= 20 \times 25 + 400 = 8500$$

$n$

$$\sum_{i=1}^n$$

$$= 20 \times 25 + 400 = 8500$$

$$\sum_{i=1}^n X_i^2 = 8500$$

If the wrong observation 30 is replaced by the correct value 13, then the number of observations remains

same viz., 20 and

$n$

$$\sum_{i=1}^n X_i = 400 - 30 + 13 = 383, \text{ and corrected } \sum_{i=1}^n X_i^2 = 8500 - 30^2 + 13^2 = 7769$$

$$= 7769$$

Corrected

$$\sum_{i=1}^n X_i = 383$$

$$\sum_{i=1}^n X_i^2 = 7769$$

$$\text{Corrected mean} = \frac{\sum_{i=1}^n X_i}{n} = \frac{383}{20} = 19.15$$

$$\text{Corrected mean} = \frac{\sum_{i=1}^n X_i}{n} = \frac{383}{20} = 19.15$$

$$n = 20$$

Corrected

$\sum n$

Corrected  $\sum X^2$

$\sum 2 \sum i \sum 1 \sum \text{Corrected mean} \sum 2 \sum 7769 \sum 19.5 \sum 388.45 \sum 366.72 \sum 21.73$

$n = 20$

Corrected Standard Deviation =  $21.73 \div 4.6615$ .

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Assignment 3.2: Obtain the Standard deviation for a group data set

Class 13-17 18-22 23-27 28-32 33-37 38-42 43-47 48-52 53-57

Interval

Frequency 12 20 10 14 13 14 16 12 15

SOLUTION

35

3.7 Interpreting Standard Deviation.

The standard deviation is one of the most useful measures of dispersion or variability of the data. Thus,

for a homogeneous set of data. That is, for the observations which are close to each other,  $\sigma$  will be small

and for a heterogeneous set of data. That is, for the observations which are widely spread,  $\sigma$  will be large. Hence, we can use the values of standard deviations for the relative or comparative study of two or

more groups.

3.8 Empirical Rule.

Empirical rule is an important rule of thumb used for interpreting the value of the standard deviation. The

empirical rule is classified into three categories: The results are shown in Figure 3.1.

Figure 3.1: Empirical Rule Graph

a. Approximate 68% of the data values lie within 1 standard deviation of the mean. i.e. in the interval  $\bar{X} - \sigma$ ,  $\bar{X} + \sigma$ .

b. Approximate 95% of the data values lie within 2 standard deviation of the mean. i.e. in the interval  $\bar{X} - 2\sigma$ ,  $\bar{X} + 2\sigma$ .

c. Approximate 99.7% of the data values lie within 3 standard deviation of the mean. i.e. in the interval  $\bar{X} - 3\sigma$ ,  $\bar{X} + 3\sigma$ .

Remarks

1. If the variable under study has normal distribution, then the results (a) to (c) are exactly true. However, these results apply very well to data with the frequency distributions which are approximately symmetric and bell-shaped.

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2. Range Rule of the thumb: The empirical rule can be used to estimate the value of the range of the

given data.

a. For normal distribution, 95.4% of the observation lies within the limit  $X \pm 2\sigma$ ,  $X \pm 2\sigma$ .

Hence, Range =  $X \pm 2\sigma - X \pm 2\sigma = 4\sigma$ , covers 95.4% of the data observations.

The estimate of the standard deviation can be obtained as

Range

4

which will be true in 95.4% of the cases.

b. For normal distribution, 99.73% of the observation lies within the limit  $X \pm 3\sigma$ ,  $X \pm 3\sigma$ .

Hence, Range =  $X \pm 3\sigma - X \pm 3\sigma = 6\sigma$ , covers 99.73% of the data observations.

The estimate of the standard deviation can be obtained as

Range

6

which will be true in 99.73% of the cases.

Example 3.11: The test scores of a sample of 100 students have a symmetric mound distribution with a

mean score of 570 and standard deviation of 70. Approximately, what percent of the scores are between

(a) 430 and 710, (b) 360 and 780.

The mean of  $X = 570$  and its standard deviation = 70.

a. We have  $X \pm 2\sigma = 570 \pm 140 = 710$  and  $X \pm 2\sigma = 570 \pm 140 = 430$ . Hence, 430 and 710 are exactly 2 standard deviation away from the mean. The percentage of scores between 430 and 710 is equal the percentage scores lying between  $X \pm 2\sigma$  and  $X \pm 2\sigma$ , i.e. with 2 standard deviation of the mean = 95% (approximately). Hence, approximately 95% of the scores are between 430 and 710.

b. We have  $X \pm 3\sigma = 570 \pm 210 = 780$  and  $X \pm 3\sigma = 570 \pm 210 = 360$ . Hence, 360 and 780 are exactly 3 standard deviation away from the mean. The percentage of scores between 360 and 780 is equal the percentage scores lying between  $X \pm 3\sigma$  and  $X \pm 3\sigma$ , i.e. with 3 standard deviation of the mean = 99% (approximately). Hence, approximately 99% of the scores are between 360 and 780.

Assignment 3.3: The test scores of a sample of 1000 students have a symmetric mound distribution

with a mean score of 870 and standard deviation of 10. Approximately, what percent of the scores are

between (a) 640 and 910, (b) 460 and 980.

SOLUTION

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SOLUTION

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Example 3.12: The following are the cholesterol levels of a group of 20 middle aged men.

190 230 295 310 260 245 270 220 240 240

250 275 235 180 250 250 202 215 210 320

Using the range rule of the thumb, obtain a rough estimate of the data standard deviation.

In the above example, we observed the largest observation ( $L$ ) = 320 and the smallest observation ( $S$ ) =

180.

Thus, Range =  $L - S = 320 - 180 = 140$ . Hence, by the range rule of thumb, an estimate of the standard

deviation of the data is given as

140

$\square \square$  Range

4

$\square \square$  35

4

The actual value of the standard deviation for the above data is 37.88 the estimated value of the standard

deviation obtained above is reasonably close to the exact value.

### 3.9 Chebyshev's Theorem.

The Chebyshev's Theorem provides us with a rule for interpreting the value of standard deviation. The

rule was proposed by a Russian mathematician called Chebyshev in 1853. According to Chebyshev's

theorem, the proportion of statistical variable within  $K$  standard deviations is at least one minus one divided by  $K$  squared. Here,  $K$  is any positive number greater than one. For  $K$  equal to two, at least 75

percent of the statistical variable is within two standard deviations of the mean. Mathematically,

$\square 1 \square$

For any  $k \geq 1$ , at least  $100 \square 1 \square \square\%$  of the data values lie within  $k$  standard deviations of the mean of

$\square k^2 \square$

the data values. i.e. within the limits  $X \square k \square$  and  $X \square k \square$ .

Taking  $k = 2$ , thus we have  $1 \square k^2 \square 1 \square 14 \square 34 \square 0.75 \square 75\%$ . Hence, by using the Chebyshev's rule we

conclude that at least 75% of the data values will fall within the limits  $X \square 2 \square$  and  $X \square 2 \square$ .

Taking  $k = 3$ , we have  $1 \square k^2 \square 1 \square 19 \square 89 \square 0.89 \square 89\%$ . Hence, using the Chebyshev's rule, we conclude

that at least 89% of the data values will fall within the limits  $X \square 3 \square$  and  $X \square 3 \square$ .

The empirical rule holds reasonably well in the case of data with approximately symmetrical mound

frequency distribution and exactly for the normal distribution. However, the beauty of Chebyshev's rule is

that it holds for all sorts of data irrespective of the nature of the distribution represented by it. See Figure

3.2a. to Figure 3.2c.

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Figure 3.2a: Chebyshev percent values graph

Figure 3.2b: Chebyshev percent values graph

Figure 3.2c: Chebyshev percent values graph

Example 3.13: According to Chebyshev's theorem at least what percentage of the data value lie between

(a)  $X \pm 4\sigma$ ,  $X \pm 4\sigma$ , (b)  $X \pm 2.3\sigma$ ,  $X \pm 2.3\sigma$

40

(a) Taking  $k = 4$ ,  $1 - \frac{1}{k^2} = 1 - \frac{1}{4^2} = 1 - \frac{1}{16} = \frac{15}{16}$

15

$\approx 0.9375 = 93.75\%$ . Thus, at least 93.75% of the data values lie between  $X \pm 4\sigma$ ,  $X \pm 4\sigma$ .

(b) Taking  $k = 2.3$ ,  $1 - \frac{1}{k^2} = 1 - \frac{1}{2.3^2} = 1 - \frac{1}{5.29} \approx 0.819 = 81.1\%$ . Therefore, at least 81.1% of the

data values fall within  $X \pm 2.3\sigma$  and  $X \pm 2.3\sigma$ .

Example 3.14: The average and standard deviation of a sample of size 150 are 15 and 2 respectively.

(a) At least what percentage of sample values lies between 9 and 21?

(b) How many sample values lie between 10 and 20?

(a) Given that

$n = 150$ ,  $X = 15$ ,  $\sigma = 2$ .

Let  $X \pm k\sigma = 21$ , which implies that  $k = \frac{21 - 15}{2} = \frac{6}{2} = 3$ ;  $X \pm k\sigma = 9$ , which implies that  $k = \frac{15 - 9}{2} = \frac{6}{2} = 3$ .

Thus, taking  $k = 3$  in Chebyshev's formula,

we have  $1 - \frac{1}{k^2} = 1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} \approx 0.89 = 89\%$ .

Thus, at least 89% of the sample values will fall within 9 and 21. That is, at least

$\approx 89\%$

$\approx 150 \times \frac{89}{100} = 133.5 \approx 134$  of the 150 values fall within 9 and 21.

$\approx 100\%$

(b) In this case, we want to find  $k$  so that  $X \pm k\sigma = 20$  which implies that  $k = \frac{20 - 15}{2} = \frac{5}{2} = 2.5$  and

$X \pm k\sigma = 10$ . Thus,  $k = \frac{15 - 10}{2} = \frac{5}{2} = 2.5$  and  $1 - \frac{1}{k^2} = 1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = \frac{5.25}{6.25} = \frac{21}{25}$

21

$\approx 0.84 = 84\%$ .

84

Hence, at least 84% of the  $150 = 100 \times \frac{84}{100} = 126$  of the 150 values fall within 10 and 20.

Assignment 3.4: According to Chebyshev's theorem at least what percentage of the data value lie

between (a)  $X = 8$  ,  $X = 8$  , (b)  $X = 6.3$  ,  $X = 6.3$

SOLUTION

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3.10 Z-scores or Standard Scores.

The Z-scores give the number of standard deviations the original value is from the mean. A z-score or

standard score is obtained on shifting the origin in the original measurement (X) to its mean and the scale

to its standard deviation. Standard scores is usually denoted by Z; given as

$Z = \frac{X - \text{Mean}}{\text{SD}}$

$Z = \frac{X - \text{Mean}}{\text{SD}}$

SD

where SD means standard deviation.

The Z-values are independent of the units of measurements or in other words, they are in standard unit.

Accordingly, the transformation of the Z-score formula from X-scores to Z-scores enables us to compare

two or more distributions with different means and standard deviations. Also, the Z-scores play a very

important role in normal distribution for computing areas under normal probability curves.

Example 3.15: Joseph and Stella are taking Statistics course from different colleges. Joseph's score in the

first year is 76, whereas the class average is 68 with a standard deviation of 5. Stella scored 82 in the first

year while class average score was 71 and standard deviation 8. Compare the performances of the two

students relative to their classes.

$X = X$

Score (X) Mean  $\pm$  X  $\pm$  SD  $\pm$  Z-score =

76  $\pm$  68

Joseph 76 68 5 Z 1  $\pm$  1.60

5

82  $\pm$  71

Stella 82 71 8 Z 2  $\pm$  1.38

8

76  $\pm$  68

Joseph's Z-score is Z 1  $\pm$  1.60 standard deviations above (because it is positive) his class

5

82  $\pm$  71

average and Stella's Z-scores is Z 2  $\pm$  1.38 standard deviations above his class average. Since

8

Z 1  $\pm$  Z 2 , Joseph performance (relative to their classes) is better than that of Stella in the first year

Statistics course.

Remark: on the face of it, Stella's score of 82 is higher than Joseph's score of 76, by  $82 - 76 = 6$  marks.

However, relative to their classes, Joseph's performance is better.

### 3.11 Variance and Standard Deviation using Assumed Mean and scaling factor.

The standard deviation of grouped data also can be calculated by "step deviation method". In this method

also, some arbitrary data value is chosen as the assumed mean,  $A$ . Then we calculate the deviations of all

data values by using  $d_i = x_i - A$ .

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$\sum n^2$

$\sum d^2$  and  $\sum d$

$n$

$\sum i$

$\sum i^2$  and the variance given as

$\frac{\sum d^2}{n}$

$\frac{\sum d^2}{n}$

$\frac{\sum d^2}{n}$

2

$n$

$\sum n$

$\sum d_i$  and  $\sum d_i$

2

$\sum i^2$  and  $\sum i$

$\sum n$

$\sum$

$\sum$

$d_i$

Or better still, for grouped data, we can scale down  $d$  to  $h_i$ , where  $c$  is the regular increment in the  $x$

$c$

values (Class size). Thus, the formula reduces to

$n$

$\sum f h_i^2$

$\sum f h_i$  and  $\sum f$

$\sum f$

$\sum h_i$

$i$

$\sum f h_i$  and  $\sum f$

2

$$\sum_{i=1}^n n_i$$

$$\sum_{i=1}^n n_i^2$$

$$f h_i$$

$$i$$

$$i$$

$$n$$

$$f_i$$

$$\sum_{i=1}^n f_i^2$$

$$\sum_{i=1}^n c^2$$
 for population, and

$$\sum_{i=1}^n$$

$$n$$

$$\sum_{i=1}^n$$

$$i$$

$$f_i$$

$$\sum_{i=1}^n$$

$$\sum_{i=1}^n f_i h_i$$

$$2$$

$$\sum_{i=1}^n n_i$$

$$\sum_{i=1}^n$$

$$f h_i$$

$$i$$

$$i$$

$$n$$

$$f_i$$

$$s^2 = \sum_{i=1}^n$$

$$\sum_{i=1}^n c^2$$
 for sample.

$$\sum_{i=1}^n$$

$$n$$

$$\sum_{i=1}^n f_i$$

$$\sum_{i=1}^n i$$

$$\sum_{i=1}^n$$

The population standard deviation and the sample standard deviation can be obtained as the square roots

of the above formula.

Example 3.16: Find the mean and standard deviation of the grouped data.

Class Interval 2-5 6-9 10 - 13 14 - 17 18 - 21

Frequency 7 15 22 14 2

The solution to the example 3.16 with the class size  $9.5 - 5.5 = 4$  can be obtained as

Class Interval 2-5 6-9 10 - 13 14 - 17 18 - 21 Total

Frequency 7 15 22 14 2 60

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Mid-point 3.5 7.5 11.5 15.5 19.5

$d \quad x \quad A \quad -8 \quad -4 \quad 0 \quad 4 \quad 8$

$h \quad -2 \quad -1 \quad 0 \quad 1 \quad 2$

$fh \quad -14 \quad -15 \quad 0 \quad 14 \quad 4 \quad -11$

$fh \quad 2 \quad 28 \quad 15 \quad 0 \quad 14 \quad 8 \quad 65$

$n$

$\sum fh \quad i \quad i$

$\sum 11 \quad 44$

Let  $A = 11.5$ . Then,  $x \quad A \quad i \quad i \quad 1$

$n$

$\sum c \quad 11.5 \quad \sum 4 \quad 11.5 \quad \sum 10.77 \quad .$

$\sum f$

60 60

$i$

$i \quad 1$

Thus, the variance is obtained as

$\sum n \quad \sum f \quad i \quad hi \quad \sum$

2

$\sum n \quad$

$\sum f \quad i \quad hi \quad \sum n \quad \sum$

$\sum i \quad 1$

$i \quad 1 \quad \sum fi \quad 65 \quad \sum \sum 60$

11 2

$\sum 65 \quad 2.02 \quad \sum 16 \quad 1.05 \quad \sum 16 \quad 16.8$

$s \quad \sum$

2  $i \quad 1$

$\sum \sum c \quad$

2

$\sum 42 \quad$

$\sum \sum$

$n$

60 60

$fi$

$\sum \sum i \quad 1$

$\sum \sum$

$s \quad 16.8 \quad 4.099 \quad .$

3.12 Coefficient of Variation.

Statistically, to compare the relative amounts of variation in populations with different means, the

coefficient of variation was developed. The Coefficient of variation (CV) is defined as

Standard Deviation  $S$

$CV = \frac{S}{\bar{X}} \times 100\%$

Mean  $\bar{X}$

The statistical value with the smallest CV among the variables is preferred to the one with highest CV.

### 3.13 Measure of partition: Quartiles, Deciles, and Percentiles

The Partition Values are the measures used to divide the total number of observations from a distribution

into a certain number of equal parts. Quartiles, Deciles, and Percentiles are some of the most often used

partition values. It is important here to note that the data should be sorted in either ascending or descending order before calculating the partition values. Quartiles divide the data into four equal parts;

decile divide the data into ten equal parts and percentiles divide the data into hundred equal parts.

#### 3.13.1 Quartiles and Interquartile Range

Quartiles are the summary measures that divide a ranked data set into four equal parts. Three measures

will divide any data set into four equal parts. These three measures are the first quartile (denoted by  $Q_1$ ),

the second quartile (denoted by  $Q_2$ ), and the third quartile (denoted by  $Q_3$ ). The data should be ranked in

increasing order before the quartiles are determined. The quartiles are defined as follows. Note that  $Q_1$

and  $Q_3$  are also called the lower and the upper quartiles, respectively.

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Quartiles are three values that divide a ranked data set into four equal parts. The second quartile is the

same as the median of a data set. The first quartile is the median of the observations that are less than the

median, and the third quartile is the median of the observations that are greater than the median.

Figure 3.3 shows the diagrammatic partition of the quartiles.

Figure 3.3: The quartiles

The difference between the third quartile and the first quartile for a data set is called the interquartile

range (IQR), which is a measure of dispersion. Calculating Interquartile Range The difference between

the third and the first quartiles gives the interquartile range; that is,

$IQR = \text{Interquartile} = Q_3 - Q_1$

$Q_3 - Q_1$

Semi- IQR = Interquartile =

2

Example 3.17: A sample of 12 commuter students was selected from a college. The following data give

the typical one-way commuting times (in minutes) from home to college for these 12 students.

29 14 39 17 7 47 63 37 42 18 24 55

- Find the values of the three quartiles.
- Where does the commuting time of 47 falls in relation to the three quartiles?
- Find the interquartile range.

Solution

- We perform the following steps to find the three quartiles.

Step 1. First we rank the given data in increasing order as follows:

7 14 17 18 24 29 37 39 42 47 55 63

Step 2. We find the second quartile, which is also the median. In a total of 12 data values, the median is between sixth and seventh terms. Thus, the median and, hence, the second quartile is given by the average of the sixth and seventh values in the ranked data set, that is the average of 29 and 37.

Thus, the second quartile is:

$$29 \square 37$$

$$Q2 \square \square 33$$

2

Note that  $Q2 = 33$  is also the value of the median.

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Step 3. We find the median of the data values that are smaller than  $Q2$ , and this gives the value of the first quartile. The values that is smaller than  $Q2$  are:

7 14 17 18 24 29

The value that divides these six data values in two equal parts is given by the average of the two middle values, 17 and 18. Thus, the first quartile is:

$$17 \square 18$$

$$Q1 \square \square 17.5$$

2

Step 4. We find the median of the data values that are larger than  $Q2$ , and this gives the value of the third quartile. The values that is larger than  $Q2$  are:

37 39 42 47 55 63

The value that divides these six data values in two equal parts is given by the average of the two middle values, 42 and 47. Thus, the third quartile is:

$$42 \square 47$$

$$Q3 \square \square 44.5$$

2

Now we can summarize the calculation of the three quartiles in the following figure.

The value of  $Q_1 = 17.5$  minutes indicates that 25% of these 12 students in this sample commute for less

than 17.5 minutes and 75% of them commute for more than 17.5 minutes.

Similarly,  $Q_2 = 33$  indicates that half of these 12 students commute for less than 33 minutes and the other

half of them commute for more than 33 minutes.

The value of  $Q_3 = 44.5$  minutes indicates that 75% of these 12 students in this sample commute for less

than 44.5 minutes and 25% of them commute for more than 44.5 minutes.

b. By looking at the position of 47 minutes, we can state that this value lies in the top 25% of the commuting times.

c. The interquartile range is given by the difference between the values of the third and the first quartiles. Thus,

$$IQR = \text{Interquartile range} = Q_3 - Q_1 = 44.5 - 17.5 = 27 \text{ minutes}$$

For a grouped data.

The Quartile for first quartile is given by

$$Q_1 = \frac{N}{4}$$

$$= \frac{46}{4} = 11.5$$

$$Q_1 = L_{Q_1} + \frac{\frac{N}{4} - C}{f_{Q_1}} \times C$$

$$11.5 = L_{Q_1} + \frac{11.5 - C}{f_{Q_1}} \times C$$

$$11.5 = L_{Q_1} + \frac{11.5 - C}{f_{Q_1}} \times C$$

where,

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$L_{Q_1}$  is the lower class boundary of the first quartile class,

$n$

$N = \sum f_i$  total frequency of the distribution,

$i = 1$

$Cf_{Q_1}$  is the cumulative frequency of the class just before or preceding the first quartile class

$f_{Q_1}$  is the frequency of the first quartile class,

$C$  is the class size.

The Quartile for third quartile is given by

$$Q_3 = \frac{3N}{4}$$

$$= \frac{3 \times 46}{4} = 34.5$$

$$Q_3 = L_{Q_3} + \frac{\frac{3N}{4} - C}{f_{Q_3}} \times C$$

$$34.5 = L_{Q_3} + \frac{34.5 - C}{f_{Q_3}} \times C$$

$$34.5 = L_{Q_3} + \frac{34.5 - C}{f_{Q_3}} \times C$$

where,

$L_{Q_3}$  is the lower class boundary of the third quartile class,

$n$

$N = \sum f_i$  total frequency of the distribution,

i = 1

Cf Q3 is the cumulative frequency of the class just before or preceding the third quartile

f Q3 is the frequency of the third quartile class,

C is the class size.

Example 3.18: Compute the quartiles of the following data set

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 5 12 2 15 13 20 7 5 1

The first quartile class is obtained as  $80/4 = 20$  and that of the third quartile is  $3N/4 = 240/4 = 60$ .

Thus, the first quartile class is 40 - 49 and the third quartile class is 60 - 69.

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 5 12 2 15 13 20 7 5 1

Cumulative 5 17 19 34 47 67 74 79 80

For the first quartile,

$L_{Q1} = 39.5$

47

n

$N = \sum f_i = 80$

i = 1

Cf Q1 = 19

f Q1 = 15

C = 10

$\frac{N}{4}$

$\frac{4}{4} = \text{Cf Q1} = 19$

$Q1 = L_{Q1} + \frac{\frac{N}{4} - \text{Cf Q1}}{f_{Q1}} \times C = 39.5 + \frac{10 - 19}{15} \times 10 = 40.2$

$\frac{f_{Q1}}{4} = 15$

$\frac{10}{4}$

For the third quartile

$L_{Q3} = 59.5$

n

$N = \sum f_i = 80$

i = 1

Cf Q3 = 47

f Q3 = 20

C = 10

$\frac{N}{4}$

$\frac{4}{4} = \text{Cf Q3} = 47$

$Q3 = L_{Q3} + \frac{\frac{N}{4} - \text{Cf Q3}}{f_{Q3}} \times C = 59.5 + \frac{10 - 47}{20} \times 10 = 66.0$

$\frac{f}{Q} = \frac{3}{20}$

$\frac{f}{Q}$

### 3.13.2 Deciles

A decile is a quantile that is used to divide a data set into 10 equal subsections. The 5<sup>th</sup> decile will be the

median for the dataset.

The (approximate) value of the  $k$ th decile, denoted by  $D_k$ , is

$\frac{k}{10}n$

$D_k$  = Value of the  $\frac{k}{10}n$  term in a ranked data set

$\frac{k}{10}n$

where  $k$  denotes the number of the decile and  $n$  represents the sample size.

For grouped data, we can apply the formula

$\frac{k}{10}n$

$\frac{k}{10}n$  =  $Cf_{D_k}$

$D_k = L_{D_k} + \frac{\frac{k}{10}n - Cf_{D_k}}{f_{D_k}} \times C$

$f_{D_k}$

$C$

$C$

where,

$L_{D_k}$  is the lower class boundary of the  $k$  decile class,

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$n$

$N = \sum f_i$  total frequency of the distribution,

$i = 1$

$Cf_{D_k}$  is the cumulative frequency of the class just before or preceding the decile class.

$f_{D_k}$  is the frequency of the decile class,

$C$  is the class size.

### 3.13.3 Percentiles

Percentiles are the summary measures that divide a ranked data set into 100 equal parts. Each (ranked)

data set has 99 percentiles that divide it into 100 equal parts. The data should be ranked in increasing

order to compute percentiles. The  $k$ th percentile is denoted by  $P_k$ , where  $k$  is an integer in the range 1 to 99.

For instance, the 25th percentile is denoted by  $P_{25}$ . Figure 3.4 shows the positions of the 99 percentiles.

Figure 3.4: Percentiles

Thus, the  $k$ th percentile,  $P_k$ , can be defined as a value in a data set such that about  $k\%$  of the measurements are smaller than the value of  $P_k$  and about  $(100 - k)\%$  of the measurements are greater

than the value of  $P_k$ .

The approximate value of the kth percentile is determined as explained next.

### Calculating Percentiles

The (approximate) value of the kth percentile, denoted by  $P_k$ , is

$$\left[ \frac{k}{100} n \right]$$

$P_k$  = Value of the  $\left[ \frac{k}{100} n \right]$  term in a ranked data set

$$\left[ \frac{k}{100} n \right]$$

where k denotes the number of the percentile and n represents the sample size.

$$\left[ \frac{k}{100} n \right]$$

If the value of  $\left[ \frac{k}{100} n \right]$  is fractional, always round it up to the next higher whole number.

$$\left[ \frac{k}{100} n \right]$$

For grouped data, we can apply the formula

$$P_k = L_{P_k} + \frac{\frac{k}{100} n - C_{f_{P_k}}}{f_{P_k}} \cdot C$$

$$\left[ \frac{k}{100} n \right] = C_{f_{P_k}}$$

$$P_k = L_{P_k} + \frac{\frac{k}{100} n - C_{f_{P_k}}}{f_{P_k}} \cdot C$$

$$\left[ \frac{k}{100} n \right]$$

$$\left[ \frac{k}{100} n \right]$$

$$\left[ \frac{k}{100} n \right]$$

where,

$L_{P_k}$  is the lower class boundary of the k percentile class,

$$\frac{k}{100} n$$

$$n$$

$N = \sum f_i$  total frequency of the distribution,

$$i = 1$$

$C_{f_{P_k}}$  is the cumulative frequency of the class just before or preceding the percentile class.

$f_{P_k}$  is the frequency of the percentile class,

C is the class size.

Example 3.19: Compute the 70th percentile of the following data set

A sample of 12 commuter students was selected from a college. The following data give the typical one-

way commuting times (in minutes) from home to college for these 12 students.

29 14 39 17 7 47 63 37 42 18 24 55

Arrange the data in increasing order, we have

7 14 17 18 24 29 37 39 42 47 55 63

Now, for  $k = 70$  and  $n = 12$ , we have

$$\left[ \frac{70}{100} \cdot 12 \right]$$

$$P_{70} = \text{Value of the } 8.4^{\text{th}} \text{ term.}$$

th

$$\left[ \frac{70}{100} \cdot 12 \right]$$

$P_{70} = \text{Value of the 9th term} = 42 \text{ minutes.}$

Example 3.20: Compute the 10th percentile of the following data set

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 5 12 2 15 13 20 7 5 1

$n = 10$   $n = 80$

Thus, the 10th percentile class is obtained as  $k = 10$ ,  $n = 80$ . This falls in the class

$n = 100$

interval 20 - 29, thus the P10

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 5 12 2 15 13 20 7 5 1

Cumulative 5 17 19 34 47 67 74 79 80

where,

$L_{Pk} = 19.5$

$n$

$N = \sum f_i = 80$

$i = 1$

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$Cf_{Pk} = 5$

$f_{Pk} = 12$

$C = 10$

$k = n = 10$

$n = 100$   $Cf_{Pk} = 8$   $5$

$P_k = L_{Pk} + \frac{(k - Cf_{Pk})}{f_{Pk}} \times C = 19.5 + \frac{10 - 5}{12} \times 10 = 22.0$

$f_{Pk} = 12$

$C = 10$

3.14 Skewness and Kurtosis.

The skewness (SK) of a statistical data is the degree of the asymmetry or departure from symmetry of a

distribution. Normally distributed data is symmetric about the mean, median and mode. This implies that

the mean, median and mode lie in the middle of the distribution and the value of skewness is zero. If the

smoothed frequency polygon of a distribution has a longer tail to the right of the central value than to the

left, the distribution is said to be skewed to the right, or to have positive skewness. If the reverse is true,

it is said to be skewed to the left or to have negative skewness. Measures of skewness are as follows:

Mean - Mode

SK =

Standard Deviation

$X - \text{Mode} \leq \text{Mean} \leq \text{Median}$

SK  $\leq 0$

SD Standard Deviation

SK  $\leq 0$

$3 \times \text{Median} \leq$

SD

$Q_3 - 2Q_2 \leq Q_1$

SK  $\leq 0$

$Q_3 - Q_1$

The kurtosis (K) is a measure of the tailedness of a distribution. Tailedness is how often outliers occur.

Excess kurtosis is the tailedness of a distribution relative to a normal distribution. Distributions with medium kurtosis (medium tails) are mesokurtic. It can be a measure of whether the data are heavy-tailed or

light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails,

or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers. A uniform distribution

would be the extreme case.

Mathematically,

$Q_3 - Q_1$

$K - 2$

$P_{90} - P_{10}$

where Q and P are the quartiles and percentile.

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Example 3.21: Compute the Kurtosis of the following data set

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 5 12 2 15 13 20 7 5 1

$Q_1 = 40.2$ ,  $Q_3 = 66.0$ ,  $P_{90} = 76.64$ ,  $P_{10} = 22$ . Thus,

$Q_3 - Q_1 = 66.0 - 40.2$

$K - 2 = 0.236$

$P_{90} - P_{10} = 76.64 - 22$

Assignment 3.5: Compute the quartiles, decile and percentile of the following data set

Class 10 - 19 20 - 29 30 - 39 40 - 49 50 - 59 60 - 69 70 - 79 80 - 89 90 - 99

interval

Frequency 15 20 12 13 14 25 17 25 11

SOLUTION

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## SOLUTION

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## Chapter Four

## 4.1 Introduction to Probability

The subject of probability theory is an aspect of mathematics that deals with study of possible outcomes

of a given events together with the outcomes relative likelihoods and distribution.

Probability is a numerical measure of the likelihood that a specific event will occur.

Suppose that an event E can happen in w ways out of a total of n possible equally likely ways. Then, the

probability of occurrence of the event called its success is denoted by

$$p = \Pr[E]$$

Number of ways E can occur

Total Number of possible outcomes

Simply put as

$$p = \frac{\Pr[E]}{w}$$

n

n

The probability of nonoccurrence of the event called its failure is denoted by

$$q = 1 - p$$

$$q = \Pr[\text{not } E] = 1 - \Pr[E]$$

h

n

Thus,  $\Pr[E] + \Pr[\text{not } E] = 1$ . That is  $p + q = 1$

## 4.2 Definitions of terms

In this section, we shall define some terms that will help us to understand the basic subject of probability.

a. Experiment: An experiment is a process that produces a result or an observation. Eg.

When a coin is tossed and throwing a die. In this case, the coin can appeared as head or tail.

Figure 4.1: Tree diagram for one toss of a coin.

The sample space is given as {H, T}. The number of sample is given as  $n(S) = 2$ . However, if the die is

tossed twice, the sample space will be {HH, HT, TH, TT}. The number of sample points is given as  $n(S)$

$$= 4.$$

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Figure 4.2: Tree diagram for one toss of a coin.

Suppose we throw a die once, we can obtain 1, 2, 3, 4, 5, and 6. The number of sample point is  $n(S) = 6$ .

However, if the die is rolled two times, we have

+ 1 2 3 4 5 6

1 2 3 4 5 6 7

2 3 4 5 6 7 8

3 4 5 6 7 8 9

4 5 6 7 8 9 10

5 6 7 8 9 10 11

6 7 8 9 10 11 12

The sample point  $S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ .

x 2 3 4 5 6 7 8 9 10 11 12 Total

$n(x)$  1 2 3 4 5 6 5 4 3 2 1 36

Thus,  $n(S) = 36$

b. Outcome: The outcome of any experiment is the particular result obtained when an experiment is performed.

c. Sample space: A sample space is a set of all possible outcomes of an experiment.

d. Sample points: These are the individual outcomes in a sample space.

e. Event: An event is any subset of the sample space. The number of sample points of A is denoted by  $n(A)$ .

f. Equally Likely Outcomes: Two or more outcomes that have the same probability of occurrence are said to be equally likely outcomes.

#### 4.3 Properties of Probability

There are two important properties of probability that we should always remember. These properties are mentioned below.

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1. The probability of an event always lies in the range 0 to 1.

Whether it is a simple or a compound event, the probability of an event is never less than 0 or greater than

1. We can write this property as follows.

$$0 \leq P(E_i) \leq 1$$

$$0 \leq P(A) \leq 1$$

An event that cannot occur has zero probability and is called an impossible (or null) event.

An event that is certain to occur has a probability equal to 1 and is called a sure (or certain) event.

In the following examples, the first event is an impossible event and the second one is a sure event.

$$P(\text{a tossed coin will stand on its edge}) = 0$$

$$P(\text{a child born today will eventually die}) = 1.0$$

There are very few events in real life that have probability equal to either zero or 1.0. Most of the events

in real life have probabilities that are between zero and 1.0. In other words, these probabilities are greater

than zero but less than 1.0. A higher probability such as .82 indicates that the event is more likely to occur. On the other hand, an event with a lower probability such as .12 is less likely to occur. Sometime events with very low (.05 or lower) probabilities are also called rare events.

2. The sum of the probabilities of all simple events (or final outcomes) for an experiment, denoted by  $\sum_{i=1}^n P(E_i)$  is always 1.

Example 4.1: Find the probability of obtaining an even number in one roll of a die.

This experiment of rolling a die once has a total of six outcomes: 1, 2, 3, 4, 5, and 6.

Given that the die is fair, these outcomes are equally likely. Let A be an event that an even number is

observed on the die. Event A includes three outcomes: 2, 4, and 6; that is,

$A = \text{an even number is obtained} = \{2, 4, 6\}$ .

If any one of these three numbers is obtained, event A is said to occur. Since three out of six outcomes are

included in the event that an even number is obtained, its probability is:

$P(A) =$

Number of outcomes included in A 3

$= \frac{3}{6} = 0.5$

Total number of outcome 6

#### 4.4 Some Probability Laws

There are several useful laws of probability properties. We shall consider some of them

a. Mutually Exclusive Events: In statistics, two or more events are said to be mutually exclusive if the occurrence of any one of them excludes the occurrence of the others. That is, the two events cannot happen at same time. Hence,  $P(E_1 \cap E_2) = 0$ .

However, if  $E_1 \cup E_2$  is the event that either  $E_1$  or  $E_2$  or both occur, then

$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

This can be extend to 3 events

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$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3)$

Nevertheless, for mutually exclusive event,

$P(E_1 \cup E_2) = P(E_1) + P(E_2)$

This is often called the addition law of probability.

Example 4.2: If  $E_1$  is the event of drawing an ace from a deck of cards and  $E_2$  is the event of drawing a

king. Obtain the probability of either drawing an ace or a king in a single draw.

There are 4 aces and 4 kings in the deck card with a total of 52. Thus, the probability of drawing an ace is

given as

$$\Pr[E_1] = \frac{13}{52}$$

Also, the probability of drawing a king is

$$\Pr[E_2] = \frac{4}{52}$$

Thus,

$$\Pr[E_1 \cap E_2] = \Pr[E_1] \cdot \Pr[E_2] = \frac{13}{52} \cdot \frac{4}{52} = \frac{13}{131} \cdot \frac{13}{132}$$

Example 4.3: If  $E_1$  is the event of drawing an ace from a deck of cards and  $E_2$  is the event of drawing a

spade. Obtain the probability of either drawing an ace or a spade in a single draw.

The probability of drawing an ace is given as

$$\Pr[E_1] = \frac{4}{52} \text{ . Also, the probability of drawing a spade is } \Pr[E_2] = \frac{13}{52} \text{ ,}$$

$$\text{Probability of } \Pr[E_1 \cap E_2] = \frac{1}{52}$$

Thus,

$$\Pr[E_1 \cup E_2] = \Pr[E_1] + \Pr[E_2] - \Pr[E_1 \cap E_2] = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52}$$

b. Independence Event: Suppose, we have events  $E_1$  and  $E_2$  . Thus, two events  $E_1$  and  $E_2$  are independent if the probability of the second event  $E_2$  is not affected by the occurrence or nonoccurrence of the first event  $E_1$  . Thus, if  $E_1$  and  $E_2$  are independent events, then

$$\Pr[E_1 \text{ and } E_2] = \Pr[E_1 \cap E_2] = \Pr[E_1] \cdot \Pr[E_2]$$

This is somewhat called the multiplication law of probability.

Example 4.4: Suppose two events  $E_1$  and  $E_2$  defined a sample space such that  $\Pr[E_2] = 0.2$  and  $\Pr[E_1 \cap E_2] = 0.75$  , Find  $\Pr[E_2]$  such that

1.  $E_1$  and  $E_2$  are independent
2.  $E_1$  and  $E_2$  are mutually exclusive.

Solution

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1. If  $E_1$  and  $E_2$  are independent, thus,  $\Pr[E_1 \cap E_2] = \Pr[E_1] \cdot \Pr[E_2]$  . Hence,

$$\Pr[E_1 \cap E_2] = \Pr[E_1] \cdot \Pr[E_2] = \Pr[E_1] \cdot 0.2$$

$$0.75 = \Pr[E_1] \cdot 0.2 \Rightarrow \Pr[E_1] = \frac{0.75}{0.2}$$

$$0.8 \Pr[E_1] = 0.75 \Rightarrow 0.2 = 0.5$$

$$\Pr[E_1] =$$

$$0.55$$

$$\Rightarrow 0.6875$$

$$0.8$$

2. If  $E_1$  and  $E_2$  are mutually exclusive, then  $\Pr[E_1 \cap E_2] = 0$  . Thus,

$$\Pr[E_1 \cap E_2] = \Pr[E_1] \cdot \Pr[E_2]$$

$$0.75 = \Pr[E_1] \cdot 0.2$$

$$\Pr[E_1] = 0.75 - 0.2 = 0.55$$

#### 4.5 Conditional Probability

If  $E_1$  and  $E_2$  are two events such that the probability that  $E_2$  occurs given that  $E_1$  has occurred is denoted by  $\Pr[E_1 \setminus E_2]$ , better still  $\Pr[E_1 \text{ give } E_2]$  is called conditional probability of  $E_2$  given that  $E_1$

has occurred. If the occurrence or nonoccurrence of  $E_1$  does not affect the probability of occurrence of

$E_2$ , then  $\Pr[E_1 \setminus E_2] = \Pr[E_2]$  and we say that  $E_1$  and  $E_2$  are independent events; otherwise dependent events.

If  $E_1 E_2$  is the event that both  $E_1$  and  $E_2$  occur, then

$$\Pr[E_1 E_2] = \Pr[E_1] \Pr[E_2 \setminus E_1]$$

However, the probability  $\Pr[E_1 E_2] = \Pr[E_1] \Pr[E_2]$  if independent events.

Example 4.5: Let  $E_1$  and  $E_2$  be the events heads on the fifth toss and head on the sixth toss of a coin

respectively. Thus,  $E_1$  and  $E_2$  are independent events, and thus the probability of heads on both the fifth

and sixth tosses is

$$\Pr[E_1 E_2] = \Pr[E_1] \Pr[E_2] =$$

$$\frac{1}{2} \cdot \frac{1}{2}$$

$$= \frac{1}{4}$$

$$\frac{1}{4}$$

Example 4.6: Suppose that a bag contains 3 white balls and 2 black balls. Let  $E_1$  be the event, first ball

drawn is black and  $E_2$  the event, second ball drawn is black. Where the balls are not replaced after being

drawn from the bag. Here,  $E_1$  and  $E_2$  are dependent events.

The probability that the first ball drawn is black is  $\Pr[E_1] = \frac{2}{5}$ . The probability that the second

ball drawn is black given that the first ball drawn was black is  $\Pr[E_2 \setminus E_1] = \frac{1}{4}$ . Thus, the probability that both balls drawn are black is

$$\Pr[E_1 E_2] = \Pr[E_1] \Pr[E_2 \setminus E_1] =$$

$$\frac{2}{5} \cdot \frac{1}{4}$$

$$= \frac{1}{10}$$

Example 4.7: A die is tossed two times. Compute the (1) probability of obtaining a sum of 9 (2) probability of getting a sum of 9 giving that the number on the 2nd toss is larger than the number on the first toss.

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$$+ 1 + 2 + 3 + 4 + 5 + 6$$

$$1 + 1, 1 + 2, 1 + 3, 1 + 4, 1 + 5, 1 + 6$$

2 2,1 2,2 2,3 2,4 2,5 2,6  
 3 3,1 3,2 3,3 3,4 3,5 3,6  
 4 4,1 4,2 4,3 4,4 4,5 4,6  
 5 5,1 5,2 5,3 5,4 5,5 5,6  
 6 6,1 6,2 6,3 6,4 6,5 6,6

1.  $\Pr(\text{sum of 9}) = 4/36 = 1/9$

2. Let  $E_1$  = the sum is 9 and  $E_2$  = 2nd toss number is larger than the first toss

$\Pr(E_1 \cap E_2)$

$\Pr(E_1 \setminus E_2)$

$\Pr(E_2)$

Thus,  $\Pr(E_2) = P(\text{2nd toss number is larger than the first toss}) = 15/36$ .

$\Pr(E_1 \cap E_2) = P(\text{sum of 9 and 2nd toss number is larger than the first toss}) = 2/36$

$\Pr(E_1 \cap E_2) = 2/36$

$\Pr(E_1 \setminus E_2)$

2

$\Pr(E_1 \cap E_2)$

$\Pr(E_2) = 15/36$

#### 4.6 Bayes's Theorem

The Bayes's theorem was developed by a renowned English Presbyterian minister called Reverend Thomas Bayes (1702 – 1761). He was able to expand the conditional probability as

$\Pr(E_i) = \Pr(E \setminus E_i)$

$\Pr(E_i \setminus E) = n$

$\Pr(E) = \Pr(E \setminus E_i)$

$i = 1$

$i = i$

Where  $E_i$  are all inclusive set of possible outcomes.

Example 4.8: A car repair firm employs three-sprayers, Tom, Dick and Harry. Owing to the different speeds of their respective production lines, Tom is responsible for painting 25 percent of all the cars

produced, Dick for 35 percent and Harry for the remaining 40 percent. On the basis of frequent quality

inspections it is discovered that, on average, 5 percent of the cars sprayed by Tom fall below the minimum acceptable standard (as regards the quality of painting), while the corresponding figure for Dick

is 9 percent and for Harry per cent. If a car is selected at a random from the firm's throughput of cars and

its paint finish is judged to be sub-standard, what is the probability that it was sprayed by Harry?

Let

$S$  = event that the car is sub-standard

$T$  = event that a car is sprayed by Tom

$D$  = event that a car is sprayed by Dick

H = event that a car is sprayed by Harry

Thus, the problem is to find  $p(H|S)$ .

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However,  $P(T) = 0.25$ , and  $P(S|T) = 0.05$ . That is, Tom sprays 25% of the cars and of these 5% are sub-

standard, on the average. Similarly, for Dick and Harry:

$P(D) = 0.35$ , and  $P(S|D) = 0.08$ ,

$P(H) = 0.40$ , and  $P(S|H) = 0.10$ .

Thus, applying the Bayes' rule, the probability that a randomly selected car, found to be sub-standard was

sprayed by Harry is

$$P(H \cap P(S) | H) = 0.40 \times 0.10$$

$$P(H | S) = \frac{P(H \cap P(S) | H)}{P(T \cap P(S) | T) + P(D \cap P(S) | D) + P(H \cap P(S) | H)}$$

$$P(T \cap P(S) | T) + P(D \cap P(S) | D) + P(H \cap P(S) | H) = 0.25 \times 0.05 + 0.35 \times 0.08 + 0.40 \times 0.10$$

$$0.04$$

$$= \frac{0.04}{0.04 + 0.028 + 0.04}$$

$$0.0805$$

Example 4.9: A product is being produced by Fupre Enterprises by three machines namely, M1, M2, M3.

These machines produce 40%, 35% and 25% of the product respectively. According, the defective products produced by these machines respectively are 7%, 10% and 12%. Find

a. The probability that a part selected at a random from the finished product is defective.

b. The probability of the defective product was produced by machine M1, M2, M3.

In the above example, let

$$P(M1) = 0.4$$

$$P(M2) = 0.35$$

$$P(M3) = 0.25$$

ND = Non defective product

D = Defective product

$$P(D|M1) = 0.07$$

$$P(D|M2) = 0.1$$

$$P(D|M3) = 0.12$$

$$a. P(D) = P(M1) P(D|M1) + P(M2) P(D|M2) + P(M3) P(D|M3)$$

$$0.4 \times 0.07 + 0.35 \times 0.1 + 0.25 \times 0.12 = 0.028 + 0.035 + 0.03 = 0.093$$

$$P(M1 | D) = \frac{P(M1 \cap P(D) | M1)}{P(D)}$$

$$b. P(M1 | D) = \frac{P(M1 \cap P(D) | M1)}{P(D)}$$

$$= \frac{0.028}{0.093}$$

$$= \frac{0.028}{0.093} = 0.3011$$

$$i = 1$$

$$i$$

$$i = 1$$

i i

Thus,

The probability that the defective is from machine 1 is given as

$$0.4 \times 0.07 = 0.028$$

$$P(M_1 | D) = \frac{0.028}{0.093} = 0.3011$$

$$0.093 = 0.093$$

The probability that the defective is from machine 2 is given as

$$0.35 \times 0.1 = 0.035$$

$$P(M_2 | D) = \frac{0.035}{0.093} = 0.3763$$

$$0.093 = 0.093$$

The probability that the defective is from machine 3 is given as

$$0.25 \times 0.12 = 0.03$$

$$P(M_3 | D) = \frac{0.03}{0.093} = 0.3226$$

$$0.093 = 0.093$$

## 4.7 Permutations and Combinations

### 4.7.1 Permutations

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Permutations refer to the number of ways in which a set of objects can be arranged in order (the order

being crucial). However, the key words for permutation are order or arrangement. Thus, the number of

possible arrangement =  $n!$  This can be expressed as  $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$ . Thus,  $nPr$

denotes the number of permutations of  $r$  objects out of a total of  $n$  objects, where  $n!$  is called  $n$  factorial.

Mathematically,

$$n!$$

$$n$$

$$Pr =$$

$$\frac{n!}{(n-r)!}$$

Example 4.10: Compute  $nPr$  if  $n = 5$  and  $r = 2$ .

$$n! = 5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$n$$

$$Pr = \frac{n!}{(n-r)!} = \frac{5!}{(5-2)!} = \frac{5!}{3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 20$$

$$\frac{n!}{(n-r)!} = \frac{5!}{3!} = 20$$

Example 4.11: In how many ways can 3 persons sit on 6 seats in a row?

In this example,  $n = 6$ ,  $r = 3$

$$n! = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$n$$

$$Pr = \frac{n!}{(n-r)!} = \frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 120$$

$$\frac{n!}{(n-r)!} = \frac{6!}{3!} = 120$$

#### 4.7.2 Combination

In the cast of permutations, the order in which the objects are arranged is important. However, if one is

interested only in what particular objects are selected when  $r$  objects are chosen from  $n$  objects, without

regard to their arrangement, then the unordered selection is called a combination. In this case, the number

of combinations is given by the formula:

$n$

$Pr\ n!$

$n$

$Cr\ \frac{n!}{r!(n-r)!}$

$r!(n-r)!$

Where  $nCr$  denotes the number of combinations possible in selecting  $r$  objects from  $n$  different objects.

Example 4.12: Find the number of ways in which 3 persons can be selected from a committee of 5 people.

In this example, 3 persons can be chosen from a committee of 5 in  $5C3$  ways

$n$

$Pr\ n! 5! 5! 5! 4$

Thus,  $nCr = \frac{n!}{r!(n-r)!} = \frac{5!}{3!2!} = 10$  ways

$r!(n-r)!$

Assignment 4.1: A product is being produced by Fupre Enterprises by three machines namely, M1, M2,

M3. These machines produce 20%, 65% and 15% of the product respectively. According, the defective

products produced by these machines respectively are 13%, 15% and 35%. Find

- The probability that a part selected at a random from the finished product is defective.
- The probability of the defective product was produced by machine M1, M2, M3.

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SOLUTION

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## Chapter Five

### 5.1 Discrete Probability Distribution.

In the previous chapter, we fairly dealt with probability of fairly simple events. However, many other

problems may be very complex in that they cannot be readily solved in same manner as those involving

repetitive experiments such as the inspection of components coming off an assembly line, tossing a coin

etc. Thus, a discrete probability distribution counts occurrences that have countable or finite outcomes.

Discrete distributions contrast with continuous distributions, where outcomes can fall anywhere on a

continuum. Common examples of discrete distribution include the binomial, Poisson, and Bernoulli distributions. The probability mass function (pmf) of a random variable  $x$  is denoted by  $p(x)$ . Various

discrete probability distributions have different pmfs.

#### 5.1.1 Discrete Random variable

A discrete random variable is a variable that can assume any set of possible values that can be quantify,

counted or listed. Example of discrete random variable is the outcomes when you roll a die.

#### 5.2 Mean or Expectation, and Variance of discrete probability distributions.

Let the probability mass function (pmf) of a discrete random variable  $x$  be given by  $p(x)$ . Then, the

mean or expectation of  $x$  denoted by  $E(x)$  is given by

$n$

$$E(x) = \sum_{i=1}^n x_i p(x_i)$$

$i=1$

The variance is given as

$n$

$$\text{Var}(x) = \sum_{i=1}^n x_i^2 p(x_i) - \left( \sum_{i=1}^n x_i p(x_i) \right)^2$$

$2$

$i=1$

The standard deviation is given as

$n$

$$\sigma = \sqrt{\sum_{i=1}^n x_i^2 p(x_i) - \left( \sum_{i=1}^n x_i p(x_i) \right)^2}$$

$2$

$i=1$

Example 5.1: Obtain the expectation of the data

$s/n \times P(x) \times p(x)$

$$1 \times 1 \times 0.17 = 0.17$$

$$2 \times 6 \times 0.13 = 0.78$$

$$3 \times 11 \times 0.26 = 2.86$$

$$4 \times 16 \times 0.06 = 0.96$$

$$5 \times 21 \times 0.38 = 7.98$$

$$\text{Total } n = 0.17 + 0.78 + 2.86 + 0.96 + 7.98 =$$

$$\sum_{i=1}^n p(x_i) = 1$$

$i=1$

$$= 12.75$$

$$63$$

$n$

$$\sum_{i=1}^n x_i p_i = 12.75$$

$$i=1$$

### 5.3 Bernoulli distribution.

A Bernoulli experiment is a statistical random experiment in which the outcome of the experiment is

classified into one of two mutually exclusive and exhaustive events, say either success or failure, false or

true, male or female, good or bad. Thus, a Bernoulli distribution is a probability distribution of a discrete

random variable named after Swiss mathematician Jacob Bernoulli that can have only two values: success and failure. If the probability of success is  $p$ , then the probability of failure is  $1 - p$ .

The probability mass (density) function (pdf) or (pmf) of the Bernoulli distribution is given as

$$f(x) = p^x (1-p)^{1-x}, \quad x = 0, 1$$

$$1 - x$$

$$\text{for } x = 0, 1$$

where  $x$  is a random variable that associate with the Bernoulli trial. Also, we say  $x$  has a Bernoulli distribution with parameter  $p$  if  $X \sim \text{Bernoulli}(p)$ . Thus, when an experiment of a boy or a girl is performed once, it is a Bernoulli experiment.

#### 5.3.1. The Expectation and Variance of the Bernoulli distribution

The mean or expectation of the Bernoulli distribution for a random variable  $x$  is obtained as

$$1$$

$$E(X) = \sum_{x=0}^1 x p^x (1-p)^{1-x} = 0 \cdot p^0 (1-p)^1 + 1 \cdot p^1 (1-p)^0 = p$$

The variance is given as

$$1$$

$$E(X^2) = \sum_{x=0}^1 x^2 p^x (1-p)^{1-x} = 0^2 \cdot p^0 (1-p)^1 + 1^2 \cdot p^1 (1-p)^0 = p$$

$$x=0$$

$$2$$

$$p^1 (1-p)^0 = p$$

The standard deviation is obtained by taking the square root of the variance as

$$\sqrt{pq}$$

Example 5.2: If  $X \sim \text{Bernoulli}(0.3)$ , find the mean and standard deviation.

In the above,  $p = 0.3$ . Thus,  $q = 1 - 0.3 = 0.7$

Thus,  $E(X) = p = 0.3$  and  $\sqrt{pq} = \sqrt{0.3 \cdot 0.7} = 0.46$ .

### 5.4 Binomial distribution.

Binomial distribution is a common probability distribution that models the probability of obtaining one of

two outcomes under a given number of parameters. It summarizes the number of trials when each trial has

the same chance of attaining one specific outcome. Suppose the random variable  $X$  has equal number of

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observation in  $n$  success in  $n$  Bernoulli trials with possible values  $x = 0, 1, 2, 3, \dots, n$ . Then the pdf of  $x$ ,

say  $f(x)$  is given as

$$f(x) = {}^n C_x p^x (1-p)^{n-x}$$

$n \geq x$

,  $x = 0, 1, 2, 3, \dots, n$ .

$n$

$n!$

where  ${}^n C_x = \frac{n!}{x!(n-x)!}$

$r! = n! / (n-r)!$

In summary, the number of ways of selecting  $x$  successful positions in  $n$  trials is a binomial distribution.

Thus, if  $X$  is a binomial distribution we write  $X \sim B(n, p)$  with  $n$  trials and parameter  $p$ .

Example 5.3: If  $X$  has a binomial distribution with the following.

a.  $n = 3, x = 2, p = 0.3$

b.  $n = 4, x = 0, p = 0.4$

Calculate the probability of  $x$

a. Using

$$f(x) = {}^n C_x p^x (1-p)^{n-x} = {}^3 C_2 (0.3)^2 (0.7)^{3-2} = 3 \times 0.09 \times 0.7 = 0.189$$

$$n! / x! (n-x)!$$

$$1!2!$$

$$f(x) = {}^n C_x p^x (1-p)^{n-x} = {}^4 C_0 (0.4)^0 (0.6)^{4-0} = 1 \times 1 \times 0.1296 = 0.1296$$

$$n! / x! (n-x)!$$

b.

$$0!4!$$

Example 5.4: A coin is loaded in such a way that it lands with a head showing 70% of the time. If it is

tossed five times, find the probability of getting.

a. 4 heads

b. No head

c. 3 heads

d. At least 3 heads

e. At most 2 heads

In the above example,  $p = 0.7, q = 0.3$ .

$$P(X = 4) = {}^5 C_4 (0.7)^4 (0.3)^{5-4} = 5 \times 0.7^4 \times 0.3 = 0.36015$$

$$4!5!4!$$

a.

$$P(X = 0) = {}^5 C_0 (0.7)^0 (0.3)^5$$

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0.00243

0

b.

$P(X = 3) = {}^5C_3 (0.7)^3 (0.3)^2 = 0.3087$

${}^5C_3 = 10$

c.

d.  $P(X = 3) = P(X = 3) = P(X = 4) = P(X = 5)$

$P(X = 5) = {}^5C_5 (0.7)^5 (0.3)^0$

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0.16807. Thus,

5

$P(X = 3) = P(X = 3) = P(X = 4) = P(X = 5) = 0.3087 + 0.36015 + 0.16807 + 0.83692$

e.  $P(X = 2) = 1 - P(X = 5) = 1 - 0.83692 = 0.16308$

Example 5.5: Ten unbiased coins are tossed simultaneously. Find the probability of obtaining i) Exactly

6 heads.

In the above example, let p be the probability of a head, then  $p = q = 1/2$  and  $n = 10$ . Thus,

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For  $x = 6$ , we have  $f(x) = {}^{10}C_6 p^6 q^4 = {}^{10}C_6 (1/2)^6 (1/2)^4 = 105$

$10 \times 6 \times 10!$

i. 0

4!6! 512

#### 5.4.1. Some properties of the Binomial Distribution

- The outcome of a any trial has two possible outcomes. It is either success or failure, boy or girl, male or female, good or bad.
- The experiment is repeated n times. That is, it has n trials
- The probability of success is p and that of failure is q. That is  $p + q = 1$ .

#### 5.4.2. The mean and Variance of the Binomial distribution

The mean or expectation of the Binomial distribution for a random variable x is obtained as

$n p$

$E(X) = \sum_{x=0}^n x \cdot {}^nC_x p^x q^{n-x}$

$x n! n!$

$p \times q n! x \cdot p \times q n! x$  for  $x \geq 0$ , the term is

$x \geq 0 \times 0! n! x \cdot 1 \times 1! n! x!$

zero.

$n p$

n

$n! 1! p \times 1! q n! x \cdot n p$

$x \geq 1 \times 1! n! x!$

The variance of the random variable  $x$  is obtained as

$n$

$$\text{Var } x = E X^2 - (E X)^2 = \sum_{i=0}^n x^i p_i - \left( \sum_{i=0}^n x^i p_i \right)^2$$

$i = 0$

$n$

Now, let  $E X = \sum_{i=0}^n x^i p_i$  and  $E X^2 = \sum_{i=0}^n x^2 p_i$ . Then,  $E X = np$  and  $E X^2 = np + np^2$ .

$x = 0$

However, since the sum of the first two term is zero, we have

$n$

$$p x q n = n p q = n p (1-p) = n p - n p^2$$

$E X^2 =$

$$x^2 p = x^2 p = n p (1-p) = n p - n p^2$$

$$n p (1-p) = n p - n p^2$$

$n$

$$n p (1-p) = n p - n p^2$$

$$x^2 p = x^2 p = n p (1-p) = n p - n p^2$$

Suppose  $k = x^2$ . Then, we have

$n$

$$E X^2 = \sum_{i=0}^n x^2 p_i = \sum_{i=0}^n x^2 p_i = n p (1-p) = n p - n p^2$$

$k = 0$

$$n p (1-p) = n p - n p^2 = n p (1-p)$$

Hence,

$$np(1-p) = np - np^2 = np(1-p)$$

Remark: the variance of the binomial distribution is less than the mean.

66

### 5.5 Poisson distribution.

The Poisson distribution is a discrete probability distribution. It gives the probability of an event happening a certain number of times ( $k$ ) within a given interval of time or space. The Poisson distribution

has only one parameter,  $\lambda$  (lambda), which is the mean number of events. The Poisson distribution was

derived in 1837 by a French mathematician called Simeon D. Poisson (1781 – 1840). It is the limiting

case of the Binomial probability distribution with the following conditions

- a. The number of trial is indefinite. That is  $n \rightarrow \infty$ .
- b. The constant probability  $p$  of success for each trial is indefinitely small. That is,  $p \rightarrow 0$ .
- c.  $np \rightarrow m$ , say is finite

The probability density function (pdf) of the Poisson distribution for a random variable  $x$  is given as

$$e^{-\lambda} \frac{\lambda^x}{x!}$$

$$f(x) = \Pr\{X = x\}, x = 0, 1, 2, 3, \dots$$

$$x!$$

Note that  $e \approx 2.71828$ .

Example 5.6: A building society branch manager notices that over a long period of time the number of

people using an automated cash point on a Saturday morning is on average, 30 people per hour.

What is

the probability that in say a 10 minute period:

- a. Nobody uses the machine?
- b. Three people use the machine?

In the example, we need to obtain the mean of the Poisson distribution before we proceed. Now,  $\lambda = 30$

$\lambda = 5$ . Thus,

$$5$$

$$e^{-\lambda} \frac{\lambda^x}{x!} = e^{-5} \frac{5^5}{5!}$$

$$P\{X = 0\} = e^{-5} \frac{5^0}{0!} = 2.7183^{-5} \approx 0.0067$$

$$5^5$$

$$a. \frac{5^0}{0!}$$

$$x! = 0!$$

$$\frac{5^0}{0!}$$

$$e^{-5} \frac{5^5}{5!}$$

$$b. P\{X = 3\} = e^{-5} \frac{5^3}{3!} \approx 0.1404$$

$$x! = 3!$$

Example 5.7: The number of accidents occurring on a bridge in Agbarho in a year has a Poisson distribution with mean 3. Compute the probability that in a year:

- a. Only one accident occurred
- b. At most one accident occurred
- c. At least one accident occurred.

The solution of the example can be obtained as follows:

$$e^{-\lambda} \frac{\lambda^x}{x!} = e^{-3} \frac{3^1}{1!}$$

$$a. P\{X = 1\} = e^{-3} \frac{3^1}{1!} \approx 0.149$$

$$x! = 1!$$

$$e^{-3} \frac{3^0}{0!} + e^{-3} \frac{3^1}{1!}$$

$$b. P\{X = 1\} = P\{X = 0\} + P\{X = 1\} = e^{-3} \frac{3^0}{0!} + e^{-3} \frac{3^1}{1!} \approx 0.199$$

$$0! = 1!$$

$$67$$

$e^{-3} \approx 0.050$

c.  $P(X = 1) = 1 - P(X = 0) = 1 - e^{-3} \approx 0.94979$

0!

### 5.6 Poisson Approximation to Binomial.

The Poisson distribution may be used to approximate the binomial if the probability of success is —small—

(such as 0.01) and the number of trials is —large— (such as 1,000).  $n$  is the number of trials, and  $p$  is the

probability of a —success.

### 5.7 Geometric distribution.

Geometric distribution can be defined as a discrete probability distribution that represents the probability

of getting the first success after having a consecutive number of failures. A geometric distribution can

have an indefinite number of trials until the first success is obtained.

The geometric distribution is another important distribution with the pdf given as

$$f(x) = pq^x \quad x = 1, 2, 3, \dots$$

The mean and the variance of the geometric distribution are obtained as

$$E(X) = \frac{1}{p} \quad \text{and} \quad \text{Var}(X) = \frac{1}{p^2}$$

1 - q

p

Example 5.8: An applicant for a driver's license has a probability of 75% of passing the road test on any

given independent trial. What is the probability that an applicant will finally pass the test on the fourth

try?

Let  $X$  denote the number of trials. Then  $p = 75\% = 0.75$  and  $q = 1 - 0.75 = 0.25$ . Thus,

$$f(x) = pq^x \quad x = 1 \quad 0.75(0.25)^3 = 0.75(0.015625) = 0.01171875$$

4 - 1 = 3

### 5.8 Hypergeometric distribution.

The hypergeometric distribution is used to determine the probability of a certain number of "successes" in

a series of draws made without replacement from a fixed population. The distribution depends on the size

of the population, the number of draws, and the number of "successes" in the population. In probability

theory and statistics, the hypergeometric distribution is a discrete probability distribution that describes

the probability of  $k$  successes (random draws for which the object drawn has a specified feature) in  $n$

draws, without replacement, from a finite population of size  $N$  that contains exactly  $K$  objects with that

feature, wherein each draw is either a success or a failure. In contrast, the binomial distribution describes

the probability of  $k$  successes in draws with replacement.

A random variable  $X$  follows the hypergeometric distribution if its probability mass function (pmf) is given as

$$p(y) = \frac{\binom{r}{y} \binom{N-r}{n-y}}{\binom{N}{n}}$$

$$0 \leq y \leq n$$

$$r \leq N$$

$$y \leq r$$

$$N \geq r$$

$$n \leq N$$

$$N \geq n$$

$$n \leq N$$

where

$$68$$

$N$  is the population size,

$r$  is the number of success states in the population,

$n$  is the number of draws (i.e. quantity drawn in each trial),

$y$  is the number of observed successes,

$\binom{a}{b}$  is a binomial coefficient.

a

b

If  $Y$  is a random variable with hypergeometric distribution. The mean and variance of  $Y$  are given by

$$E(Y) = n \frac{r}{N}$$

$$\text{Var}(Y) = n \frac{r}{N} \frac{N-r}{N} \frac{N-n}{N}$$

$$nr$$

. However, we express

$$N \geq n \text{ and } N \geq r$$

$$r \leq N$$

and  $1 - p = \frac{N-r}{N}$ , we have

$$N \geq n$$

$$N \geq n \text{ and } N \geq r$$

$E(Y) = np$  and  $\text{Var}(Y) = npq$  with  $q = 1 - p$  as the adjustment.

$$N \geq n \text{ and } N \geq r$$

**Example 5.9:** An important problem encountered by personnel directors and others faced with the selection of the best in a finite set of elements is exemplified by the following scenario. From a group of

20 Ph.D. engineers, 10 are randomly selected for employment. What is the probability that the 10 selected

include all the 5 best engineers in the group of 20?

For this example  $N = 20$ ,  $n = 10$ , and  $r = 5$ . That is, there are only 5 in the set of 5 best engineers, and we

seek the probability that  $Y = 5$ , where  $Y$  denotes the number of best engineers among the ten selected.

Then

$$\frac{n!}{r! (n-r)!}$$

$$\frac{5!}{20! \cdot 5!}$$

$$p^r (1-p)^{n-r}$$

$$\frac{10!}{5! \cdot 15!} \cdot$$

$$\frac{5!}{21}$$

$$=$$

$$0.162$$

$$\frac{n!}{r! (n-r)!} \cdot p^r (1-p)^{n-r}$$

### 5.9 Negative-binomial distribution.

A negative binomial distribution is the distribution of the sum of independent geometric random variables. The number of failures before the  $n$ th success in a sequence of draws of Bernoulli random variables, where the success probability is  $p$  in each draw, is a negative binomial random variable. Thus, in a more simpler form, In probability theory and statistics, the negative binomial distribution is a discrete probability distribution that models the number of failures in a sequence of independent and identically distributed Bernoulli trials before a specified (non-random) number of successes (denoted  $r$ ) occurs. For example, we can define rolling a 6 on a dice as a success, and rolling

any other number as a failure, and ask how many failure rolls will occur before we see the third success (

$r = 3$ ). In such a case, the probability distribution of the number of failures that appear will be a negative

binomial distribution.

The probability mass function of the negative binomial distribution is given as

$$f(x) = \binom{x+r-1}{r-1} p^r (1-p)^{x-r}$$

The mean of the negative binomial distribution is  $E[X] = \frac{r}{p}$

$$r \cdot q$$

$$\frac{r}{p}$$

$$p$$

$$69$$

The variance of the negative binomial distribution is  $\text{Var}[X] = \frac{r \cdot q}{p^2}$

$$r \cdot q$$

$$\frac{r}{p^2}$$

$$p^2$$

Here the mean is always greater than the variance. Mean > Variance.

Example 5.10: Jim is writing an exam with multiple-choice questions, and his probability of attempting

the question with the right answer is 60%. What is the probability that Jim gives the third correct answer

for the fifth attempted question?

Solution:

Probability of success  $P(s) = 60\% = 0.6$ , Probability of failure  $P(f) = 40\% = 0.4$ . It is given that Jim gives

the third correct answer for the fifth attempted question. Here we can use the concept of the negative

binomial distribution to find the third correct answer for the fifth attempted question. Here we have  $k = 5$ ,

$r = 3$ ,  $p = 0.6$ ,  $q = 0.4$

The formula for negative binomial distribution is

$$P(X = k) = \binom{k-1}{r-1} p^r q^{k-r}$$

$$P(X = 5, r = 3, p = 0.6, q = 0.4) = \binom{5-1}{3-1} p^3 q^{5-3} = \binom{4}{2} (0.6)^3 (0.4)^2 = 6 \times 0.216 \times 0.16 = 0.1296 \times 0.16 = 0.020736$$

$$= 0.020736$$

$$= 0.020736$$

Therefore the probability of Jim giving the third correct answer for his fifth attempted question is 0.02.

Example 5.11: A geological study indicates that an exploratory oil well drilled in a particular region should strike oil with probability 0.2. Find the probability that the third oil strike comes on the fifth well drilled.

Solution

Assuming independent drillings and probability 0.2 of striking oil with any one well, Let  $Y$  denotes the

number of the trial on which the third oil strike occurs. Then it is reasonable to assume that  $Y$  has a negative binomial distribution with  $p = 0.2$ . Because we are interested in  $r = 3$  and  $y = 5$ ,

$$P(Y = 5)$$

$$= \binom{5-1}{3-1} p^3 q^{5-3} = \binom{4}{2} (0.2)^3 (0.8)^2 = 6 \times 0.008 \times 0.64 = 0.03072$$

$$= 0.03072$$

$$= 0.03072$$

If  $r = 2, 3, 4, \dots$  and  $Y$  has a negative binomial distribution with success probability  $p$ ,  $P(Y = y_0) = p(y_0)$

can be found by using the R (or S-Plus) command `dnbinom(y0-r,r,p)`. If we wanted to use R to obtain  $p(5)$

in Example 5.10, we use the command `dnbinom(2,3,.2)`. Alternatively,  $P(Y \leq y_0)$  is found by using the R

(or S-Plus) command `pnbinom(y0-r,r,p)`. Note that the first argument in these commands is the value  $y_0 -$

$r$ , not the value  $y_0$ . This is because some authors prefer to define the negative binomial distribution to be

that of the random variable

$Y^*$  = the number of failures before the  $r$ th success. In our formulation, the negative binomial random

variable,  $Y$ , is interpreted as the number of the trial on which the  $r$ th success occurs. In Exercise 3.100,

you will see that  $Y^* = Y - r$ . Due to this relationship between the two versions of negative binomial random variables,

$P(Y = y_0) = P(Y - r = y_0 - r) = P(Y^* = y_0 - r)$ . R computes probabilities associated with  $Y^*$ , explaining

why the arguments for `dnbinom` and `pnbinom` are  $y_0 - r$  instead of  $y_0$ .

### 5.10 Moment Generating Functions (mgf).

In probability theory and statistics, the moment-generating function of a real-valued random variable is an

alternative specification of its probability distribution. Thus, it provides the basis of an alternative route to

analytical results compared with working directly with probability density functions or cumulative distribution functions. There are particularly simple results for the moment-generating functions of

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distributions defined by the weighted sums of random variables. However, not all random variables have

moment-generating functions.

As its name implies, the moment-generating function can be used to compute a distribution's moments:

the  $n$ th moment about 0 is the  $n$ th derivative of the moment-generating function, evaluated at 0.

In addition to real-valued distributions (univariate distributions), moment-generating functions can be

defined for vector- or matrix-valued random variables, and can even be extended to more general cases.

The moment-generating function of a real-valued distribution does not always exist, unlike the characteristic function. There are relations between the behaviour of the moment-generating function of a

distribution and properties of the distribution, such as the existence of moments.

Thus, the moment generating function  $M_X(t)$  for a continuous random variable is defined by

$n$

$$M_X(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

for  $t \in \mathbb{R}$

where  $f(x)$  is the pdf

Thus,

$$M_X'(t) = E[te^{tx}]$$

$$M_X''(t) = E[t^2 e^{tx}]$$

$$M_X'''(t) = E[t^3 e^{tx}]$$

$$M_X^{(4)}(t) = E[t^4 e^{tx}]$$

$$\int_0^{\infty} \dots f(x) dx$$

$$\int_0^{\infty} \dots$$

$$2! 3!$$

$$\int_0^{\infty} \dots$$

$$t^2 x^2 t^3 x^3$$

$$\int_0^{\infty} f(x) dx \int_0^{\infty} t x f(x) dx \int_0^{\infty} f(x) dx \int_0^{\infty} f(x) dx \dots$$

$$\int_0^{\infty} \dots$$

$$2! \dots$$

$$3!$$

$$t^2 t^3$$

$$M_X(t) = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \dots$$

$$2! 3!$$

Example 5.12: Find the moment generating function for the continuous random variable.

$$f(x) = 4e^{-4x}$$

$$0 \leq x < \infty$$

$$f(x) = 0$$

$$0 \text{ elsewhere}$$

The solution to the question is obtained as

$$\int_0^{\infty} \dots$$

$$\int_0^{\infty} e^{-4x} t x dx$$

$$M_X(t) = E[etx] = \int_0^{\infty} etx f(x) dx = \int_0^{\infty} etx 4e^{-4x} dx = 4 \int_0^{\infty} t x e^{-4x} dx = 4 \int_0^{\infty} t x e^{-(4-t)x} dx$$

$$\int_0^{\infty} \dots$$

$$\int_0^{\infty} e^{-4x} t x dx = \int_0^{\infty} e^{-4x} t x dx = \int_0^{\infty} e^{-4x} t x dx = \int_0^{\infty} e^{-4x} t x dx$$

$$\int_0^{\infty} \dots$$

$$\int_0^{\infty} \dots$$

The mean, second moment, and third moment of the question can be obtained as

$$71$$

$$4$$

$$M_X(t) = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \dots$$

$$\int_0^{\infty} \dots$$

The mean is obtained by taking the first derivative of the mfg.

$$0 \leq x < \infty$$

$$M'_X(t) = \int_0^{\infty} \dots$$

$$4$$

$$\int_0^{\infty} \dots$$

$$2$$

$$M'_X(0) = \int_0^{\infty} \dots$$

$$4$$

$$\int_0^{\infty} \dots$$

$$2$$

The second moment is obtained by taking the second derivative of the first derivative of the mfg.

$$0 \leq t < 4 \quad \frac{d}{dt} M_X(t) = 8t^3$$

$$M_X''(t) = 24t^2$$

$$8$$

$$0 \leq t < 4$$

$$0 \leq t < 4$$

$$0 \leq t < 4 \quad \frac{d}{dt} M_X(t) = 8t^3$$

$$4$$

$$M_X''(0) = 0$$

$$8$$

$$0 \leq t < 4$$

$$3$$

The second moment is obtained by taking the third derivative of the second derivative of the mfg.

$$0 \leq t < 4 \quad \frac{d}{dt} M_X''(t) = 48t$$

$$2$$

$$M_X'''(t) = 48t$$

$$0 \leq t < 4 \quad \frac{d}{dt} M_X''(t) = 48t$$

$$M_X'''(0) = 0$$

$$24$$

$$256$$

### Assignment 5.1

$$f(x) = m e^{-x/2} \quad x \geq 0$$

1. Given the pdf of a continuous random variable  $X$  as  $f(x) = m e^{-x/2}$ .

$f(x) = 0$  elsewhere

Find (a) the constant value  $m$  (b) The expectation of  $X$  (c)  $P(1 \leq X \leq 4)$ .

2. Let  $X$  be a continuous random variable  $X$  with the probability density function

$$f(x) = 2e^{-4x}$$

. Compute (a)  $E[X]$  (b)  $3E[X]$  (c)  $P(1 \leq X \leq 3)$ .

$$0 \leq x < \infty$$

$$f(x) = 0 \text{ elsewhere}$$

$f(x) = 0$  elsewhere

3. Let  $X$  be a random variable with pdf of  $f(x) = 2x$

$$2x$$

,  $0 \leq x \leq 5$  and zero elsewhere. Obtain (a)

$$25$$

$$E[X] \text{ (b) } E[X^2]$$

$$(c) E[X^3] \text{ (d) } \text{Var}[X] = E[X^2] - (E[X])^2$$

4. Find the expected value of the gamma random variable using the mgf approach.

$$x \geq 0$$

5. Find the moment generating function of the function  $f(x) = x e^{-x}$

1

 $\exp\left(-\frac{x}{k}\right), 0 \leq x < \infty.$ 
 $k = k_0$ 

6. Using question 5 above, compute the mean, second moment, third moment, and the variance.

7. The daily expenses of a company has a probability density function given as

 $f(x) = \frac{3}{2}x^2, 0 \leq x \leq 2$ 

$f(x) = 0$  elsewhere. What is the expected amount to be budgeted for 5 days in

thousands of naira?

72

8. If the moment generating function of a random variable X is given as

 $3t$ 
 $e^{2t}$ 
 $M_X(t) =$ 
 $e$ 

, where  $t \geq 0$ . Compute the mean, second moment, third moment, and the

t

variance.

 $x^2$ 

9. Let X denotes a continuous random variable with pdf given as  $f(x) =$

1

 $\exp\left(-\frac{x}{3}\right), 0 \leq x < \infty.$ 
 $3e^{-x/3}$ 

Obtain (a)  $P(X \leq 9)$  (b)  $P(2 \leq X \leq 4)$ .

 $1 - e^{-x/3}$ 

10. Find the mgf of the continuous random variable X with pdf given as  $f(x) =$

 $0$  elsewhere

. Compute the mean, second moment, third moment, and the variance.

11. Obtain the moment generating function of the Normal distribution

12. Obtain the moment generating function of the Gompertz distribution

13. Obtain the moment generating function of the Alpha Power Gompertz distribution

14. Obtain the moment generating function of the Alpha Power Muth-G distribution

15. Obtain the moment generating function of the exponential distribution

16. Obtain the moment generating function of the Weibull distribution

17. Obtain the moment generating function of the Alpha Power Gompertz distribution

18. Obtain the moment generating function of the Chi-Square distribution

19. Obtain the moment generating function of the Cauchy distribution

20. Obtain the moment generating function of the shifted Gompertz-G distribution

SOLUTION

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SOLUTION

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SOLUTION

75

SOLUTION

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SOLUTION

77

SOLUTION

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### 5.11 Probability Generating Functions

In probability theory, the probability generating function of a discrete random variable is a power series

representation (the generating function) of the probability mass function of the random variable.

Probability generating functions are often employed for their succinct description of the sequence of

probabilities  $\Pr(X = i)$  in the probability mass function for a random variable  $X$ , and to make available the

well-developed theory of power series with non-negative coefficients.

The probability generating function gives an alternative method of finding the mathematical expectation

of a discrete random variable. The probability generating function helps in finding the probability distributions and properties of the discrete random variable.

The probability generating function of a random variable  $X$  is given as

$$P(t) = E[t^X] = \sum_{x=0}^{\infty} t^x f(x).$$

$x \geq 0$

If  $t = f(x)$  repeatedly, we have

Suppose we differentiate  $P(t) = E[t^X]$  w.r.t  $t$

$x \geq 0$

$\frac{d}{dt}$

$$P'(t) = E[X t^{X-1}] = \sum_{x=0}^{\infty} x t^{x-1} f(x)$$

$x \geq 0$

$\frac{d}{dt}$

$$P''(t) = E[X(X-1)t^{X-2}] = \sum_{x=0}^{\infty} x(x-1)t^{x-2} f(x)$$

$x \geq 0$

$\frac{d}{dt}$

$$P'''(t) = E[X(X-1)(X-2)t^{X-3}] = \sum_{x=0}^{\infty} x(x-1)(x-2)t^{x-3} f(x)$$

$x \geq 0$

$\frac{d}{dt}$

$$P\{k \leq t \leq E[X] \leq 1 \leq X \leq 3 \dots\} = X \leq k \leq 1 \leq t \leq X \leq k \leq x \leq x \leq 1 \leq x \leq 3 \dots\} = X \leq k \leq 1 \leq t \leq x \leq k \leq f(x) \leq x \leq 0$$

Setting  $t = 1$ , we obtain

$$P' \leq 1 \leq E[X] \leq 1$$

$$P' \leq 1 \leq E[X] \leq X \leq 1 \leq E[X^2] \leq E[X] \leq 2$$

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$$P' \leq t \leq E[X] \leq X \leq 1 \leq X \leq 2 \leq 3$$

$$P\{k \leq t \leq E[X] \leq X \leq 1 \leq X \leq 3 \dots\} = X \leq k \leq 1 \leq t \leq X \leq k \leq k$$

Thus, we can now deduce  $P_i \leq t \leq$  to find the mean and variance.

Hence,

$$P' \leq 1 \leq E[X] \leq 1 = \text{Mean}$$

$$\leq \leq \leq$$

$$P' \leq 1 \leq E[X] \leq X \leq 1 \leq E[X^2] \leq E[X] \leq E[X^2] \leq P' \leq 1 \leq$$

But,  $E[X^2] \leq P' \leq 1 \leq E[X] \leq P' \leq 1 \leq P' \leq 1 \leq$ , and the variance is given as

$$\text{Var}[X] \leq E[X^2] \leq E[X]^2$$

2

Hence,

$$\text{Var}[X] \leq E[X^2] \leq E[X]^2 \leq P' \leq 1 \leq P' \leq 1 \leq P' \leq 1 \leq$$

2 2

Example 5.13: Find the pgf and the mean for the geometric random variable.

The pmf of the geometric distribution is given as  $f(x) \leq pq \leq x \leq 1$ . Thus, the solution of the example is

given as

$$\leq \leq \leq$$

$$\leq \leq$$

$$P\{t \leq E[t] \leq t \leq pq \leq x \leq 1 \leq \leq \leq qt \leq x \leq p \leq qt \leq q \leq 2t \leq 2 \leq q \leq 3t \leq 3 \leq \dots$$

p

$$x \leq 1 \leq x \leq 1 \leq q \leq q$$

However, since the term in the series is an infinite geometric progression, we have for  $t \leq 1$ ,  $qt \leq 1$ .

Thus,

$$qt$$

$$qt \leq q \leq 2t \leq 2 \leq q \leq 3t \leq 3 \leq \dots \leq$$

$$\leq 1 \leq qt \leq$$

$$p \leq qt \leq$$

$$P\{t \leq \leq$$

$$pt$$

$$\leq \leq \leq$$

$$q \leq 1 \leq qt \leq 1 \leq qt$$

The mean is obtained by taking the first derivative as

$$d \leq p t \leq 1 \leq q t \leq p \leq q p t$$

P' t            

$$dt \quad \square \quad 1 \quad \square \quad qt \quad \square \square$$

□1 □ qt □2

Now, setting  $t = 1$ , we have

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P' 1

$$1 \oplus q \oplus p \oplus qp \oplus p^2 \oplus pq \oplus p \oplus p \oplus q \oplus 1$$

$$p_1 \leq q \leq p_2 \leq p_2 \leq p$$

Example 5.14: Find the probability generating function for the binomial random variable with parameter

$n$ ,  $p$  and obtain the mean and variance.

The pdf of the binomial random variable is given as  $n C x p^x q^{n-x}$ . Thus, the

n n

$P_t \leq E t x \leq t x C x p x q n \leq x \leq n C x \leq t p \leq q n \leq x \leq q \leq p t$ ,  $\dots$

$$n \times n$$

$$x \neq 0 \quad x \neq 0$$

The mean is obtained as

$$P' \leq t \leq E \leq X \leq g \leq p t \leq n \leq n p \leq g \leq p t \leq n \leq 1.$$

d

$dt$

Setting  $t \leq 1$ , we have

$$P' \rightarrow 1 \rightarrow E \rightarrow X \rightarrow np \rightarrow q \rightarrow p$$

$$n \geq 1$$

$$\|np, \text{ since } q \leq p \leq 1.$$

The variance is obtained as

$$\text{Var}[X] = E[X^2] - E[X]^2 = P' - (P')^2 = P(1-P).$$

22

But

P' ' t

d

$dt$

□ □

$$np^2q \leq pt \leq n^2 \leq 1 + p^2 \leq q \leq pt.$$

$$n \geq 1 \quad n \geq 2$$

Setting  $t \leq 1$ , we have

$$P' = \frac{t}{n} + \frac{n-1}{n} p^2 + \frac{n-2}{n} p^2 + \dots + np^2.$$

Hence,

$\text{Var } X = P' - (P')^2 = 1 - (1 - p)^2 = 1 - (1 - 2p + p^2) = 2p - p^2 = p(2 - p)$

Example 5.15: Find the probability generating function when the pdf of  $X$  is defined by

$f(x) =$

$x = 1, 2, 3, 4$ .

10

The solution is obtained as

$f(t) =$

$n$

$P(t) = E t^x = \sum_{x=1}^4 t^x f(x) = t + 2t^2 + 3t^3 + 4t^4$

$x = 1, 2, 3, 4$

$5 \times 10^{-4} + 3 \times 10^{-3} + 2 \times 10^{-2} + 1 \times 10^{-1}$

For  $x = 1, 2, 3, 4$ ,  $x = 2, 3, 4$ ,  $x = 3, 4$ ,

$10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}$

81

$5 \times 10^{-4}$

$x = 4, 0.1$

$10^{-1}$

$5 \times 10^{-4}$

4

$E t^x = t + 2t^2 + 3t^3 + 4t^4$

$x = 1, 10$

## Assignment 5.2

1. Find the pgf when the pdf of a random variable  $X$  is defined by

$f(x) =$ ,  $x = 1, 2, 3$ .

$x$

a.

6

$f(x) = 0.3 \times 0.7^x$ ,  $x = 0, 1$

$x = 1, x$

b.

## SOLUTION

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## 5.12 Sample Distribution

In this section, we presented methods for finding the distributions of functions of random variables.

Throughout this chapter, we will be working with functions of the variables  $Y_1, Y_2, Y_3, \dots, Y_n$  observed in a

random sample selected from a population of interest.

The random variables  $Y_1, Y_2, Y_3, \dots, Y_n$  are independent and have the same distribution. Certain functions of

the random variables observed in a sample are used to estimate or make decisions about unknown population parameters.

For example, suppose that we want to estimate a population mean  $\mu$ . If we obtain a random sample of  $n$

observations,  $y_1, y_2, y_3, \dots, y_n$ , it seems reasonable to estimate  $\mu$  with the sample mean

$\bar{y}$

$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$

$y_i$

$n$

The goodness of this estimate depends on the behaviour of the random variables  $Y_1, Y_2, Y_3, \dots, Y_n$  and the effect that

$n$

$\bar{y}$

this behaviour has on  $y_i$ . Notice that the random variable  $\bar{Y}$  is a function of (only) the random variables

$n$

$Y_1, Y_2, Y_3, \dots, Y_n$  and the (constant) sample size  $n$ . The random variable  $\bar{Y}$  is therefore an example of a statistic.

A statistic is a function of the observable random variables in a sample and known constants.

Thus,

$\bar{Y} = \bar{Y}(y_1, y_2, \dots, y_n)$

$E(\bar{Y}) = E\left(\frac{1}{n} \sum_{i=1}^n y_i\right) = \frac{1}{n} \sum_{i=1}^n E(y_i)$

$E(\bar{Y}) = \frac{1}{n} \sum_{i=1}^n E(y_i)$

$E(y_i) = \mu$

$E(\bar{Y}) = \frac{1}{n} \sum_{i=1}^n \mu = \mu$

$\bar{Y}$

$n$

$\text{Var}(\bar{Y}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n y_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(y_i)$

$\text{Var}(y_i) = \sigma^2$

$\text{Var}(\bar{Y}) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}$

$\bar{Y}$

$n$

$\bar{Y}$

$\bar{Y}$

### 5.13 The Central Limit Theorem

If  $\bar{Y}$  is normally distributed with mean  $\mu$  and standard deviation  $\sigma/\sqrt{n}$ , the Z-scores that correspond to estimating

with mean  $\mu$  is

$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}$

$Z$

$n$

Example 5.16: A single fair die is tossed once. Let  $Y$  be the number facing up. Find the expected value

and variance of  $Y$ . Also, suppose a balanced die is tossed three times. Let  $Y_1$ ,  $Y_2$ , and  $Y_3$  denote the number of

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spots observed on the upper face for tosses 1, 2, and 3, respectively. Suppose we are interested in  $Y = Y_1 + Y_2 + Y_3$

$Y = 1 + 2 + 3$  the average number of spots observed in a sample of size 3.

3

$n =$

$y_i = 1, 1, 2, 3, 4, 5, 6, 1$

$E[Y] = \sum_{i=1}^n y_i = 1 + 1 + 2 + 3 + 4 + 5 + 6 = 22$

$1 + 1 + 1$

$= 3.5$

$n = 6, 6, 6, 6, 6, 6, 6$

$= 6$

The variance can be obtained as

$n = 2$

$y_i = 1$

$\text{Var}[Y] = \sum_{i=1}^n y_i^2$

$1 + 1 + 1 + 1$

$= 2 + 12 + 22 + 32 + 42 + 52 + 62 = 187$

$n = 6, 6, 6, 6, 6, 6$

$= 6$

$n =$

$y_i = n = y$

$E[Y] = E[y] = 1 + 1 + 1 + 1 = 4$

$n = 1, n =$

$= 6$

$n =$

$y_i = n = y_i = n \text{ Var}[y_i] = n^2 = 2^2 = 4$

$\text{Var}[Y] = \text{Var}[y] = 1$

$n = 1$

$\text{Var}[Y] = \text{Var}[y] = 2^2 = 4$

$n = 1, n =$

2

$i = 1, n = 3$

$= 6$

Example 5.17: Let the random 2, 7, 8, 10, and 15 been drawn from a bag. Compute  $E[Y]$  and  $\text{Var}[Y]$ .

$= 6$

The solution is obtained as

$n$

yi n y 2 7 8 10 15

EY E i 1 E i 20.4

$$n \leq i \leq n + 5$$

□ □ □ □

□ n □

□ □ yi □ n

□□ □y □

$$\text{Var } Y = \text{Var} \left[ \sum_{i=1}^n \text{Var} \left[ \frac{1}{n} \right] \right]$$

□ 2 632.24

□ □ 126.448 .

$$n \leq i \leq n-5$$

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## 5.14 The Law of Large Numbers

Considering any shape of a population, the law of large numbers makes us understand that as the sample

size say  $n'$  increases; the sample mean gets closer, and closer to the population mean. In other words, the

larger the sample, the more likely the sample mean gets closer to the population mean. Thus, if an experiment is repeated again and again, the probability of an event obtained from the relative frequency

approaches the actual or theoretical probability.

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#### 5.14.1.1 The Law of Large Numbers of Binomial Trials

If  $X$  is the number of successes in  $n$  numbers of Binomial trials with probability of success  $p$  on each of

the trial, the mean,  $np$  and variance,  $npq$ . We could deduce that the fraction of success  $\frac{X}{n}$  can provide us

with an estimate of  $p$ . we expect as  $n$  becomes large the estimate moves closer to  $p$ .

☐ X ☐

Using the chebychev's inequality, we note that  $P\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \geq \epsilon\right] \leq \frac{\sigma^2}{n\epsilon^2}$  where  $\epsilon > 0$ .

□ N □

Thus, using the chebychev's inequality  $P\{|X - \mu| \geq k\sigma\} \leq \frac{1}{k^2}$  and substituting  $X = x$  and  $s^2 = \frac{1}{n}$ ,

k

We have  $P(X \leq k) \leq n^{-2} \leq 1$

k

Let  $0 \leq k \leq n$  and  $k \leq \lfloor \frac{n}{2} \rfloor$ .

$\square \square n$

Then,

$\square \square \square \square \square n$

2

$$P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 2$$

$$\frac{1}{\sqrt{n}} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

2

$$P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 2$$

Next, taking the limit as  $n$  tends to infinity, we have

$$\frac{1}{\sqrt{n}} \rightarrow 0$$

$$\frac{1}{\sqrt{n}} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\frac{1}{\sqrt{n}} \rightarrow 0$$

Therefore,

$$\lim_{n \rightarrow \infty} P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 2$$

lim

$$P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 0.$$

Also, since we know that

$$P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 1 - P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 0 \text{ as } n \rightarrow \infty.$$

$$P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 1 - P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 0 \text{ as } n \rightarrow \infty.$$

Therefore,

$$P(X \leq \bar{X} \leq \bar{X} + \frac{1}{\sqrt{n}}) = 1.$$

Thus, the proof shows that for large number  $n$ , the sample mean  $\bar{X}$  will be very close or converges to

population mean  $\mu$ .

**Example 5.18:** Let  $X$  be the mean of a random sample of size 20 from a distribution whose pdf is  $f(x) = 3e^{-3x}$ ,  $0 \leq x < \infty$ . Find

2

$$1. P(X \leq 1.3)$$

85

$$2. P(X \leq 1.42)$$

$$3. P(1.2 \leq X \leq 1.45)$$

The solution is obtained as follows:

From the pdf  $f(x) = 3e^{-3x}$ ,  $0 \leq x < \infty$ , the mean is obtained as

2

$$\int_0^{\infty} x \cdot 3e^{-3x} dx$$

2

$$\int_0^{\infty} x \cdot 3e^{-3x} dx = \frac{1}{3} \int_0^{\infty} x \cdot e^{-x} dx = \frac{1}{3} \cdot 1 = \frac{1}{3} \approx 0.33$$

$$\int_0^{\infty} x \cdot e^{-x} dx = 1, \text{ and also,}$$

$$\int_0^{\infty} x^2 \cdot e^{-x} dx = 2$$

$$\int_0^{\infty} x^3 \cdot e^{-x} dx = 6$$

2

$$E(X) = \int_0^{\infty} x \cdot f(x) dx = \int_0^{\infty} x \cdot 3e^{-3x} dx = \frac{1}{3}$$

3 2 2 2 3 2 4 12

8 0 8 0 8 5 0 5

Thus,

2

12 12

$\text{Var}[X] = 0.15$ .

5 8

Therefore,  $\mu = 1.51$ ,  $\sigma^2 = 0.15$ .

$1.3 - 1.51 = -0.2$

1.  $P[X = 1.3] = P[Z = \frac{P - \mu}{\sigma} = \frac{1.3 - 1.51}{\sqrt{0.15}}] = P[Z = -2.31] = 0.0104$ .

$0.15 = 0.0866$

20

$1.42 - 1.51 = -0.08$

2.  $P[X = 1.42] = 1 - P[Z = \frac{1.42 - 1.51}{\sqrt{0.15}}] = 1 - P[Z = -0.74] = 0.7704$

$0.15 = 0.0866$

20

$1 - P[Z = -0.92] = 1 - 0.1788 = 0.8212$ .

$1.2 - 1.51 = -0.31$

3.  $P[1.2 < X < 1.45] = P[\frac{1.2 - 1.51}{\sqrt{0.15}} < Z < \frac{1.45 - 1.51}{\sqrt{0.15}}] = P[-2.46 < Z < -0.58] = 0.0104 - 0.0054 = 0.0050$

$0.15 = 0.0866$

20 20

$P[Z < -0.58] = P[Z < -3.46] = 0.0003 + 0.2807 = 0.2810$ .

Example 5.19: Let  $X_1, X_2, \dots, X_{25}$  denote the random sample of size 25 from the uniform distribution

having  $E[X_i] = 15$  and  $\text{Var}[X_i] = 101$ , if  $Y$  is the sum of all the random sample. That is,  $Y = X_1 + X_2 + \dots + X_{25}$ . Compute

a.  $P[Y < 6.35]$

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b.  $P[4.25 < Y < 5.90]$

25

$x_i = x \cdot n$

i

$Z = \frac{\bar{y} - \mu}{\sigma/\sqrt{n}}$

$n = 25$

$\mu = 5$

$\sigma = 1$

and  $Y = X_1 + X_2 + \dots + X_{25}$ , Thus,  $n = 25$ ,  $\mu = 5$ ,  $\sigma = 1$ .

$\sigma = 1$

Thus, the probability that the sum of the random sample will be less than or equal to 6.35 is given as

$P(Y \leq 6.35)$

$= P\left(\frac{Y - \mu}{\sigma/\sqrt{n}} \leq \frac{6.35 - 5}{1/\sqrt{25}}\right)$

$= P(Z \leq 6.35 - 5.0)$

a.  $P(Y \leq 6.35) = P\left(\frac{Y - \mu}{\sigma/\sqrt{n}} \leq \frac{6.35 - 5}{1/\sqrt{25}}\right) = P(Z \leq 0.85) = 0.8023$ .

$= 0.8023$

$P(Y \leq 6.35)$

$= 0.8023$

$= 0.8023$

b.  $P(4.25 \leq Y \leq 5.90) = P\left(\frac{4.25 - 5}{1/\sqrt{25}} \leq Z \leq \frac{5.90 - 5}{1/\sqrt{25}}\right)$

$= P(-1.581 \leq Z \leq 1.581)$

$= P(Z \leq 1.581) - P(Z \leq -1.581) = 0.9429 - 0.0571 = 0.8858$ .

Assignment 5.2

1. Suppose the height of workers are normally distributed with  $\mu = 1.69m$ , and the  $\sigma = 0.32$ .

Compute

i. The probability that the sum of the height of a group of 16 workers will be great than 1.55m

ii. The probability that the sum of the heights of 20 workers will be between 29.6m and 34.5m

iii. The probability that the sum of the height of a group of 16 workers will be less than 1.55m.

$P(X > 1.55)$

2. Find  $P(\bar{x} \leq 48)$  for a sample size of 81 drawn from a population with a mean 45 and standard deviation 9.

SOLUTION

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SOLUTION

88

Chapter Six

6.1 Introduction to Continuous Probability Distributions

The previous dealt with the discrete random variable. However, in this chapter, we shall consider the

continuous random variable. A random variable  $X$  is said to be a continuous random variable if it takes

on any value in an interval whose set of possible values are uncountable. That is  $\infty < x < \infty$ . A good example is the height of statistics students in the department of statistics in federal University of

Petroleum Resources, Effurun Delta State can take any countable infinity of values in an interval of real numbers.

### 6.1.1 Probability Distribution

Suppose  $X$  is a random variable. Then, the probability distribution function for the continuous random

variable  $X$  is defined as

$$F(x) = \int_{-\infty}^x f(t) dt.$$

However, the probability that  $X$  will fall within the interval  $[a, b]$  is given as

$b$

$$F(b) - F(a) = \int_a^b f(x) dx.$$

$a$

The probability density function is a theoretical model for the frequency distribution (histogram) of a

population of measurements. For example, observations of the lengths of life of washers of a particular

brand will generate measurements that can be characterized by a relative frequency histogram.

Conceptually, the experiment could be repeated ad infinitum, thereby generating a relative frequency

distribution (a smooth curve) that would characterize the population of interest to the manufacturer. This

theoretical relative frequency distribution corresponds to the probability density function for the length of

life of a single machine,  $Y$ . This is shown in Figure 6.1.

Figure 6.1: The distribution function

Features of the probability distribution function

- $0 \leq f(x) \leq 1$
- $F(x)$  is a non-negative function
- $F(-\infty) = 0$  and  $F(\infty) = 1$
- $F(xb) \leq F(xa)$  if  $xb \leq xa$
- $F(x) = \int_{-\infty}^x f(t) dt$  is the cumulative distribution function

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$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$dF(x) =$$

$$f(x) dx = F'(x) dx.$$

$dx$

Example 6.1: A function  $f(x)$  is defined as  $f(x) = kx^2$ ,  $0 \leq x \leq 3$ . Find the constant  $k$ ,  $F(x)$ ,  $P(1 \leq X \leq 3)$ .

The  $k$  is obtained as

$$kx^3$$

3

$$\frac{1}{3} \int_0^3 3 \, dy = 1$$

$$\frac{1}{3} \int_0^3 3 \, dy = 1$$

3

$$\frac{1}{3} \int_0^3 3 \, dy = 1$$

2

$$\frac{1}{3} \int_0^3 3 \, dy = 1$$

1

$$\frac{1}{3} \int_0^3 3 \, dy = 1$$

9

The cumulative distribution function is obtained as

$$F(y) = \int_0^y f(t) \, dt$$

$$F(y) = \int_0^y 3t^2 \, dt$$

$$F(y) = \int_0^y 3t^2 \, dt$$

$$F(y) = \int_0^y 3t^2 \, dt = t^3 \Big|_0^y = y^3$$

$$F(y) = \int_0^y 3t^2 \, dt = t^3 \Big|_0^y = y^3$$

$$F(y) = \int_0^y 3t^2 \, dt = t^3 \Big|_0^y = y^3$$

$$F(y) = \int_0^y 3t^2 \, dt = t^3 \Big|_0^y = y^3$$

$$F(y) = \int_0^y 3t^2 \, dt = t^3 \Big|_0^y = y^3$$

$$F(y) = \int_0^y 3t^2 \, dt = t^3 \Big|_0^y = y^3$$

3

$$P(1 \leq X \leq 3) = F(3) - F(1) = 3^3 - 1^3 = 26$$

$$P(1 \leq X \leq 3) = F(3) - F(1) = 3^3 - 1^3 = 26$$

$$P(1 \leq X \leq 3) = F(3) - F(1) = 3^3 - 1^3 = 26$$

$$P(1 \leq X \leq 3) = F(3) - F(1) = 3^3 - 1^3 = 26$$

$$P(1 \leq X \leq 3) = F(3) - F(1) = 3^3 - 1^3 = 26$$

$$P(1 \leq X \leq 3) = F(3) - F(1) = 3^3 - 1^3 = 26$$

$$P(1 \leq X \leq 3) = F(3) - F(1) = 3^3 - 1^3 = 26$$

Example 6.2: Let

$$f(y) = 0 \text{ for } y < 0,$$

$$f(y) = 3y^2 \text{ for } 0 \leq y \leq 1,$$

$$f(y) = 0 \text{ for } y > 1.$$

Find the probability density function for Y and graph it.

Note that the function  $f(y)$  is the derivative of  $F(y)$ . Thus,

$$f(y) = \frac{d}{dy} F(y)$$

$$f(y) = \frac{d}{dy} F(y) = 3y^2 \text{ for } 0 \leq y \leq 1,$$

$$f(y) = \frac{d}{dy} F(y) = 3y^2 \text{ for } 0 \leq y \leq 1,$$

$$f(y) = \frac{d}{dy} F(y) = 3y^2 \text{ for } 0 \leq y \leq 1,$$

$$f(y) = \frac{d}{dy} F(y) = 3y^2 \text{ for } 0 \leq y \leq 1,$$

$$f(y) = \frac{d}{dy} F(y) = 3y^2 \text{ for } 0 \leq y \leq 1,$$

$\leq 0$  for  $y \leq 1$ .

$\leq dy$

and  $f(y)$  is undefined at  $y = 0$  and  $y = 1$ . A graph of  $F(y)$  is shown in Figure 6.2.

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Figure 6.3: Distribution function of  $F(y)$

Figure 6.4: Distribution function of  $f(y)$

The graph of  $f(y)$  for Example 6.2 is shown in Figure 6.4. Notice that the distribution and density functions given in Example 6.2 have all the properties required of distribution and density functions,

respectively. Moreover,  $F(y)$  is a continuous function of  $y$ , but  $f(y)$  is discontinuous at the points  $y = 0$ ,

1. In general, the distribution function for a continuous random variable must be continuous, but the

density function need not be everywhere continuous.

Example 6.3: Let  $Y$  be a continuous random variable with probability density function given by

$f(y) = 3y^2$  for  $0 \leq y \leq 1$ ,

$f(y) = 0$  elsewhere

$f(y) = 0$  elsewhere

Find  $F(y)$ . Graph both  $f(y)$  and  $F(y)$ .

x

The solution is obtained by finding  $F(x) = \int_{-\infty}^x f(t) dt$

$\leq$ .

Thus,

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$\leq y$

$\leq 0$  for  $y \leq 0$ ,

$\leq$

$\leq y$

$F(y) = \int_{-\infty}^y 3t^2 dt = 0$  for  $y \leq 0$ , for  $0 \leq y \leq 1$ ,

$\leq 0$

$\leq 1$

$\leq 0$  for  $y \geq 1$ , for  $1 \leq y$

$\leq 1$

$\leq 0$

Notice that some of the integrals that we evaluated yield a value of 0. These are included for completeness in this initial example. In future calculations, we will not explicitly display any integral that

has value 0. The graph of  $F(y)$  is given in Figure 6.5.

Figure 6.5: Density function of  $f(y)$

If the random variable  $Y$  has density function  $f(y)$  and  $a \leq b$  then the probability that  $Y$  falls in the

interval  $[a, b]$  is

$$P[a \leq Y \leq b] = F(b) - F(a) = \int_a^b f(y) dy.$$

$b$

$a$

This probability is the shaded area in Figure 6.6.

Figure 6.5: Density function of  $P[a \leq Y \leq b]$ .

Example 6.4: Let  $Y$  be a random variable with  $p(y)$  given in the table below.

$y$  1 2 3 4

$p(y)$  0.4 0.3 0.2 0.1

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a. Give the distribution function,  $F(y)$ . Be sure to specify the value of  $p(y)$  for all  $y$ ,  $-\infty < y < \infty$ .

b. Sketch the distribution function given in part (a).

The solution is obtained as

$F(y) = 0$   $y < 1$

$F(y) = 0.4$   $1 \leq y < 2$

a.  $F(x) = 0.7$   $2 \leq y < 3$

$F(y) = 0.9$   $3 \leq y < 4$

$F(y) = 1$   $y \geq 4$

b. The graph of the distribution function is given as

Assignment 6.1

1. A box contains five keys, only one of which will open a lock. Keys are randomly selected and tried, one at a time; until the lock is opened (keys that do not work are discarded before another is tried). Let  $Y$  be the number of the trial on which the lock is opened.

a. Find the probability function for  $Y$ .

b. Give the corresponding distribution function.

c. What is  $P(Y \leq 3)$ ?  $P(Y = 3)$ ?  $P(Y > 3)$ ?

d. If  $Y$  is a continuous random variable, we argued that, for all  $-\infty < a < \infty$ ,  $P(Y = a) = 0$ . Do any of your answers in part (c) contradict this claim? Why?

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SOLUTION

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SOLUTION

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## 6.2 Mean or Expectation, and Variance of continuous probability distributions

The next step in the study of continuous random variables is to find their means, variances, and standard

deviations, thereby acquiring numerical descriptive measures associated with their distributions.

Many

times it is difficult to find the probability distribution for a random variable  $Y$  or a function of a random

variable,  $g(Y)$ . Even if the density function for a random variable is known, it can be difficult to evaluate

appropriate integrals (we will see this to be the case when a random variable has a gamma distribution,

Section 4.6). When we encounter these situations, the approximate behavior of variables of interest can be

established by using their moments and the empirical rule or Tchebysheff's theorem.

The expectation of a random variable  $X$  is given as

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx,$$

provided that the integral exists, and where the function  $f(x)$  is the probability density function.

In more general form, let  $g(X)$  be a function of  $X$ ; then the expected value of  $g(X)$  is given as

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

The variance of a random variable  $X$  is given as

$$\sigma^2$$

$$2$$

$$\text{Var}[X] = \sigma^2 = E[X^2] - (E[X])^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - (E[X])^2 = E[X^2] - (E[X])^2$$

$$2$$

Example 6.5: Suppose that the length of the wood assumes a continuous random variable with pdf given

as  $f(x) = \frac{3}{2}x^2$ ,  $0 \leq x \leq 1$ ,  $f(x) = 0$ , elsewhere. Find the expected value and the variance.

The solution is obtained as

a. The expectation is given

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

$$1$$

$$E[X] = \int_0^1 x \cdot \frac{3}{2}x^2 dx = \frac{3}{2} \int_0^1 x^3 dx = \frac{3}{2} \left[ \frac{x^4}{4} \right]_0^1 = \frac{3}{2} \cdot \frac{1}{4} = \frac{3}{8}$$

$$= 0.375$$

$$= 0.375$$

$$0$$

$$= \frac{3}{8} = 0.375$$

$$= 0.375$$

$$= 0.375$$

b. The variance is given as  $\text{Var}[X] = E[X^2] - (E[X])^2$ . Thus, we obtain the

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)dx$$

$$= \int_0^1 x^2 \cdot \frac{3}{2}x^2 dx = \frac{3}{2} \int_0^1 x^4 dx = \frac{3}{2} \left[ \frac{x^5}{5} \right]_0^1 = \frac{3}{2} \cdot \frac{1}{5} = \frac{3}{10}$$

$$E[X^2] = \frac{3}{10} = 0.3$$

$$1$$

$\frac{1}{2} \times 0$

$\frac{1}{2} \times \frac{1}{2}$

$\frac{1}{2} \times 2$

$\frac{1}{2} \times 2 \times 5 \times 4 \times 0 \times 10 \times 4$

11

$\frac{1}{2} \times 0.55$  .

20

Thus,

$\text{Var}[X] = \frac{1}{2} \times 0^2 + \frac{1}{2} \times 2^2 + \frac{1}{2} \times 5^2 + \frac{1}{2} \times 4^2 + \frac{1}{2} \times 0^2 + \frac{1}{2} \times 10^2 + \frac{1}{2} \times 4^2 - 0.55^2 = 0.708 + 2 \times 0.55 \times 0.5013$  .

96

6.2.1 Some Properties of Mean and Variance of a random variables  $X$  , and  $Y$  , and  $a$  and  $b$  are constants.

a.  $E[a] = a$

b.  $E[X + b] = E[X] + b$

c.  $E[aX] = aE[X]$

d.  $E[aX + bY] = aE[X] + bE[Y]$

e.  $\text{Var}[a] = 0$

f.  $\text{Var}[aX] = a^2 \text{Var}[X]$

g.  $\text{Var}[X + b] = \text{Var}[X]$

Assignment 6.2

Let  $X$  be a random variable with pdf given as  $p(x) = 4 - 2x$ ,  $1 \leq x \leq 2$  and zero elsewhere . Find  $E[X]$ ,  $E[X^2]$  , and  $\text{Var}[X]$  .

SOLUTION

97

6.3 Normal distribution

The most widely used continuous probability distribution is the normal distribution, a distribution with

the familiar bell shape that was discussed in connection with the empirical rule. The examples and exercises in this section illustrate some of the many random variables that have distributions that are

closely approximated by a normal probability distribution. In Chapter 7 we will present an argument that

at least partially explains the common occurrence of normal distributions of data in nature. The normal

density function is as follows:

A random variable  $Y$  is said to have a normal probability distribution if and only if, for  $\mu \in \mathbb{R}$  and

$\sigma^2 > 0$  , the density function of  $Y$  is

$f(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$

$f(y) =$

1

$\exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

$\mu$  and  $\sigma$

The normal density function contains two parameters,  $\mu$  and  $\sigma$ . Figure 6.6 shows the normal density function

Figure 6.6: The normal Probability density function

Areas under the normal density function corresponding to  $P(a \leq Y \leq b)$  require evaluation of the integral

$$\int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy$$

$$\frac{1}{\sigma\sqrt{2\pi}} \int_a^b \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right] dy$$

$\mu$  and  $\sigma$

Unfortunately, a closed-form expression for this integral does not exist; hence, its evaluation requires the use of numerical integration techniques.

The normal density function is symmetric around the value  $\mu$ , so areas need be tabulated on only one side

of the mean. The tabulated areas are to the right of points  $z$ , where  $z$  is the distance from the mean,

measured in standard deviations. This area is shaded in Figure 6.7.

Figure 6.7: Tabulated area for the normal Probability density function

98

### 6.3.1 Properties of the Normal Distribution

1. The normal distribution is symmetric about the mean  $\mu$
2. It is centered at the mean  $\mu$
3.  $\int_{-\infty}^{\infty} f(x) dx = 1$  that is, total area under the normal distribution curve is 1
4. Approximately 68% of a normal population lies within 1 standard deviation of the mean
5. Approximately 95% of a normal population lies within 2 standard deviation of the mean
6. The probability that a randomly selected member  $X$  of a normal population lies between two values  $x_L$  and  $x_R$   $P(x_L \leq X \leq x_R)$  is precisely equal to the area under the normal curve between  $x_L$  and  $x_R$ .

If  $X$  is a normally distributed random variable with parameters,  $\mu$  and  $\sigma$ . Then the mean and variance of  $X$  are given as

$$E[X] = \mu \text{ and } \text{Var}[X] = \sigma^2.$$

The moment generating function of the normal distribution is given as

$$M_X(t) = \exp\left[\mu t + \frac{\sigma^2 t^2}{2}\right]$$

$$M_X(t) = \exp\left[\mu t + \frac{\sigma^2 t^2}{2}\right]$$

$$Z = \frac{X - \mu}{\sigma}$$

We can always transform a normal random variable  $X$  to a standard normal random variable  $Z$  by using

the relationship

$X \sim$

$Z \sim$ .

Thus,

$\sigma^2 = 1$

$f(y) =$

1

$\exp\left[-\frac{1}{2}z^2\right]$ ,  $z = \frac{y - \mu}{\sigma}$ .

$2\sigma^2 = 2$

This is often called the standard normal distribution with mean zero and standard deviation 1.

### 6.3.2 Some important features of the standard normal distribution

- $P(Z \leq a) = 1 - P(Z \geq a)$ .
- $P(Z \leq -a) = 1 - P(Z \leq a)$ .
- $P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a)$ .
- $P(Z \leq -a) = 1 - P(Z \leq a)$ .

Example 6.6: Let  $Z$  denote a normal random variable with mean 0 and standard deviation 1.

- $P(Z \leq 2.6) = 0.9953$
- $P(Z \leq -0.26) = 1 - P(Z \leq 0.26) = 1 - 0.9953 = 0.0047$
- $P(Z \leq -0.26) = P(Z \leq 0.26) = 0.9953$
- $P(0.5 \leq Z \leq 2.5) = P(Z \leq 2.5) - P(Z \leq 0.5) = 0.9938 - 0.6915 = 0.3023$
- $P(-2 \leq Z \leq 2) = P(Z \leq 2) - P(Z \leq -2)$

99

$1 - 2(1 - P(Z \leq 2)) = 1 - 2(0.0228) = 0.9544$ . The area is shown in Figure 6.8.

Figure 6.8: Desired area

Example 6.7: The achievement scores for a college entrance examination are normally distributed with

mean 75 and standard deviation 10. What fraction of the scores lies between 80 and 90?

Recall that  $z$  is the distance from the mean of a normal distribution expressed in units of standard deviation. Thus, we convert the random variable to the desired  $z$  score using

$X \sim$

$Z \sim$

Thus, the desired fraction of the population is given by the area between

$80 - 75 = 5$  and  $90 - 75 = 15$

$Z_1 = \frac{5}{10} = 0.5$  and  $Z_2 = \frac{15}{10} = 1.5$

$10$

Figure 6.9 gives the shape

Figure 6.9: Required area

$P(Z \leq 1.5) - P(Z \leq 0.5) = 0.9332 - 0.6915 = 0.2417$

### Assignment 6.3

- Assuming that the height of maize  $X$  is normally distributed with mean of 300cm and standard

deviation 60cm. what is the probability that a randomly selected maize

- Height is greater than 360cm?
  - Height is between 150cm and 400cm?
  - Height is between 240cm and 300cm?
  - Height is less than 400cm?
2. If  $Z$  is a standard normal random variable, find the value  $z_0$  such that
- $P(Z > z_0) = .5$ .
  - $P(Z < z_0) = .8643$ .
  - $P(-z_0 < Z < z_0) = .90$ .
  - $P(-z_0 < Z < z_0) = .99$ .

100

SOLUTION

101

SOLUTION

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#### 6.4 Uniform distribution.

In probability theory and statistics, the continuous uniform distributions or rectangular distributions are a

family of symmetric probability distributions. Such a distribution describes an experiment where there is

an arbitrary outcome that lies between certain bounds. The bounds are defined by the parameters,  $a$  and

$b$  which are the minimum and maximum values. The interval can either be closed (i.e.  $[a, b]$ ) or open

(i.e.  $(a, b)$ ). Therefore, the distribution is often abbreviated  $U[a, b]$  where  $U$  stands for uniform distribution. The difference between the bounds defines the interval length; all intervals of the same

length on the distribution's support are equally probable. It is the maximum entropy probability distribution for a random variable  $X$  under no constraint other than that it is contained in the distribution's support.

The pdf of the random variable  $X$  is said to be a uniform distribution on the interval  $[a, b]$  if

$$f(x) = \frac{1}{b-a}$$

1

,  $a \leq x \leq b$ .

$b \geq a$

The figure below indicated the uniform distribution with parameters  $a$  and  $b$  as real such that  $b \geq a$ .

Figure 6.10: probability density function diagram of the uniform distribution

The mean of the uniform distribution is obtained as

$$\frac{a+b}{2}$$

$\frac{a+b}{2}$

$b \geq a$

$$E[X] = \int_a^b x f(x) dx$$

$$x = 1$$

$$a$$

$$b = a$$

$$dx = 1$$

$$b = a$$

$$x dx = x^2/2$$

$$2b = 2a \Rightarrow 2b = 2a$$

$$b^2 = a^2$$

$$2$$

Example 6.7: Arrivals of customers at a checkout counter follow a Poisson distribution. It is known that,

during a given 30-minute period, one customer arrived at the counter. Find the probability that the customer arrived during the last 5 minutes of the 30-minute period.

The solution is obtained as just mentioned, the actual time of arrival follows a uniform distribution over

the interval of (0, 30). If Y denotes the arrival time, then

$$30 = 25 + 5$$

$$P(25 \leq Y \leq 30) = \frac{30 - 25}{30 - 0}$$

$$= \frac{5}{30}$$

$$= \frac{1}{6}$$

$$\frac{25 - 30}{30 - 0}$$

The probability of the arrival occurring in any other 5-minute interval is also 1/6.

Assignment 6.4

$$103$$

1. Show that the variance of the uniform distribution is given as  $\text{Var}[X] = \frac{(b-a)^2}{12}$ .

$$\frac{(b-a)^2}{12}$$

$$12$$

$$\exp(-bt) = \exp(-at)$$

2. Show that the mgf of the uniform distribution is  $M_X(t) = \frac{e^{bt} - e^{at}}{(b-a)t}$ .

$$t = \frac{b-a}{t}$$

SOLUTION

$$104$$

6.5 Exponential distribution.

The exponential distribution for the continuous random variable is closely related to the Poisson distribution of the discrete case.

A random variable X has an exponential distribution if and only if its probability density function is defined as

$$f(x) = \lambda e^{-\lambda x}, x \geq 0$$

$$f(x) = 0, x < 0$$

0 elsewhere

However, in some cases, it can be represented as

$$1 - x$$

$$\exp(-x), x \geq 0$$

$$f(x) = \begin{cases} 1 - x & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$0$$

elsewhere

If  $X$  is a random variable that has an exponential distribution, then the moment generating function, mean and variance of  $X$  is given as

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad E[X] = \frac{1}{\lambda} \text{ and } \text{Var}[X] = \frac{1}{\lambda^2}.$$

$$1$$

$$1 - e^{-\lambda t}$$

Example 6.8: Suppose  $X$  has an exponential distribution with mean 20. Find

a.  $P(X \leq 18)$

b.  $P(X \geq 18)$

The solution is as follows

$$f(x) = \begin{cases} \frac{1}{20} e^{-x/20} & x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$18$$

$$P(X \leq 18) = \int_0^{18} \frac{1}{20} e^{-x/20} dx$$

$$= 1 - e^{-18/20}$$

$$= 1 - \exp\left(-\frac{18}{20}\right) = 1 - \exp(-0.9) = 1 - 0.4066 = 0.5934$$

a.

$$0.5934$$

$$P(X \geq 18) = 1 - P(X \leq 18) = 1 - 0.5934 = 0.4066$$

$$P(X \leq 18)$$

$$= 1 - e^{-18/20} = 1 - 0.4066 = 0.5934$$

$$20$$

b.  $P(X \geq 18) = 1 - P(X \leq 18) = 1 - 0.5934 = 0.4066$ .

## 6.6 Gamma distribution.

Some random variables are always nonnegative and for various reasons yield distributions of data that are

skewed (non-symmetric) to the right. That is, most of the area under the density function is located near

the origin, and the density function drops gradually as  $y$  increases. A skewed probability density function

is shown in Figure 6.11.

The lengths of time between malfunctions for aircraft engines possess a skewed frequency distribution, as

do the lengths of time between arrivals at a supermarket checkout queue (that is, the line at the checkout

counter). Similarly, the lengths of time to complete a maintenance check-up for an automobile or aircraft

engine possess a skewed frequency distribution. The populations associated with these random variables

frequently possess density functions that are adequately modelled by a gamma density function.

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A random variable  $Y$  is said to have a gamma distribution with parameters  $\alpha > 0$  and  $\beta > 0$  if and only

if the density function of  $Y$  is

$$f(y) = \frac{1}{\beta^\alpha \Gamma(\alpha)} e^{-y/\beta} y^{\alpha-1}$$

$y$

$$0 \leq y < \infty$$

$$f(y) = 0 \text{ elsewhere}$$

where

where

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy.$$

0

The quantity  $\Gamma(\alpha)$  is known as the gamma function. A direct integration validate that  $\Gamma(\alpha) > 0$ . Also, we

noted that  $\Gamma(\alpha) = \frac{1}{\alpha} \Gamma(\alpha+1)$  for any  $\alpha > 0$  and  $\Gamma(n) = (n-1)!$ , provided that  $n$  is an integer. The

graphs of the gamma function for  $\alpha = 1, 2, 4$  and  $\beta = 1$  are given in Figure 6.12.

For this reason,  $\alpha$  is sometimes called the shape parameter associated with a gamma distribution. The

parameter  $\beta$  is generally called the scale parameter because multiplying a gamma-distributed random

variable by a positive constant (and thereby changing the scale on which the measurement is made)

produces a random variable that also has a gamma distribution with the same value of  $\alpha$  (shape parameter) but with an altered value of  $\beta$ .

Figure 6.11: Gamma probability density function for  $\alpha = 1$

If  $Y$  is a gamma random variable with parameters  $\alpha$  and  $\beta$ . Then the mean and variance of the gamma distribution is given as

$$E(Y) = \alpha\beta \text{ and } \text{Var}(Y) = \alpha\beta^2.$$

This can be proven as shown below.

By definition, the gamma density function is such that

$y$

$$f(y) = \frac{1}{\beta^\alpha \Gamma(\alpha)} e^{-y/\beta} y^{\alpha-1}$$

$$0 \leq y < \infty \text{ and } \int_0^\infty f(y) dy = 1.$$

Hence,

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$\frac{dy}{dx} = \frac{1}{x^2}$  ,

$\frac{dy}{dx} = \frac{1}{x^2}$

0

Thus,

$y$

$y = \frac{1}{x}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$E(Y) = \int_0^1 \frac{1}{x} dx$

$1 = \int_0^1 \frac{1}{x} dx$

$y$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

0 =  $\int_0^1 \frac{1}{x} dx$

$\frac{dy}{dx} = \frac{1}{x^2}$

1

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

The variance is given as  $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$ . Thus,  $\frac{dy}{dx} = \frac{1}{x^2}$

$y$

$y = \frac{1}{x}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$y$

$E(Y^2) = \int_0^1 \frac{1}{x^2} dx$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

0 =  $\int_0^1 \frac{1}{x^2} dx$

$\frac{dy}{dx} = \frac{1}{x^2}$

1

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

$\frac{dy}{dx} = \frac{1}{x^2}$

Thus,

$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \int_0^1 \frac{1}{x^2} dx - 1 = 1 - 1 = 0$

2

$$\frac{1}{\Gamma(2)} 2^{-2} x^{2-1} e^{-2x} = \frac{1}{1} 2^{-2} x^{1} e^{-2x} = \frac{1}{4} x e^{-2x}$$

Example 6.9: Suppose  $X$  has a gamma distribution with mean  $\mu = 4$  and  $\sigma^2 = 3$ . Find the (a) mean, (b)

variance, (c) standard deviation, (d)  $P(X \leq 5)$ , and (e) write out the pdf of  $X$ .

$$f(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$$

$$\Gamma(k) = (k-1)!$$

$$E[X] = k\theta$$

$$\sigma^2 = k\theta^2$$

a.  $\sigma = \sqrt{3} \approx 1.732$ .

$$\theta = \frac{\mu}{k} = \frac{4}{2} = 2$$

$$f(x) = \frac{1}{\Gamma(2)2^2} x^{2-1} e^{-x/2} = \frac{1}{4} x e^{-x/2}$$

$$P(X \leq 5) = \int_0^5 \frac{1}{4} x e^{-x/2} dx$$

b.  $\sigma^2 = 3$

$$E[X^2] = k(k+1)\theta^2 = 2(2+1)2^2 = 24$$

$$\sigma^2 = 3$$

$$\sigma = \sqrt{3}$$

$$\sigma = \sqrt{3}$$

$$\sigma = \sqrt{3}$$

$$\sigma = \sqrt{3}$$

$$\sigma = \sqrt{3}$$

$$\sigma = \sqrt{3}$$

$$\sigma = \sqrt{3}$$

$$\sigma^2 = 3$$

$$\sigma^2 = 3$$

$$\sigma^2 = 3$$

$$12 \pm 3 \pm 12 \pm 9 \pm 12 \pm 9$$

2

$$\text{Var}(X) = k\theta^2 = 2(2)^2 = 8$$

$$\sigma^2 = 8$$

$$\sigma = \sqrt{8} = 2\sqrt{2}$$

$$16 \pm 4 \pm 16 \pm 16$$

c.  $\text{Std}(X) = \sqrt{\text{Var}(X)} = \sqrt{E[X^2] - (E[X])^2} = \sqrt{24 - 16} = \sqrt{8} = 2\sqrt{2}$

$$16$$

$$\sigma = 2\sqrt{2}$$

$$f(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$$

$$3 \pm 4 \pm x$$

$$4 \pm x \pm e$$

$$2 \pm 4 \pm x$$

$$x \pm e$$

$$d. P(X \leq 5) = \int_0^5 \frac{64}{5^2} dx = \frac{64}{5^2} \cdot 5 = \frac{64}{5}$$

$$= \frac{64}{5} \approx 12.8$$

$$e^{4x}$$

Let  $u = x^2$ ;  $du = 2x dx$ , and  $v = e^{4x}$ ;  $dv = 4e^{4x} dx$ . Thus,

$$\int \frac{1}{x^2} e^{4x} dx = \int u^{-1/2} v dv$$

$$= \frac{1}{2} \int u^{-1/2} dv$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{u}} dv = \frac{1}{2} \int \frac{1}{\sqrt{x^2}} 4e^{4x} dx$$

$$= \frac{1}{2} \int \frac{4e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx$$

$$= 2 \int \frac{e^{4x}}{x} dx = 2 \left( \frac{1}{4} e^{4x} \right) = \frac{1}{2} e^{4x}$$

$$= \frac{1}{2} e^{4x} + C$$

$$= \frac{1}{2} e^{4x} + C$$

$$= \frac{1}{2} e^{4x} + C$$

$$= \frac{1}{2} e^{4x} + C$$

$$= \frac{1}{2} e^{4x} + C$$

$$= \frac{1}{2} e^{4x} + C$$

e. The pdf will be

$$f(x) = \frac{1}{\Gamma(4)} \exp(-x) x^{4-1} = \frac{1}{6} \exp(-x) x^3$$

$$f(x) = \frac{1}{6} \exp(-x) x^3, 0 \leq x < \infty$$

$$f(x) = \frac{1}{6} \exp(-x) x^3$$

6.7 Weibull distribution.

The Weibull distribution is a two parameter continuous distribution of positive random variables that is

commonly used to describe the failure time of physical entities. It models a broad range of random variables, largely in the nature of a time to failure or time between events. Examples are maximum one-

day rainfalls and the time a user spends on a web page. The distribution is named after Swedish mathematician Waloddi Weibull, who described it in detail in 1939, although it was first identified by

Maurice René Fréchet and first applied by Rosin and Rammler (1933) to describe a particle size distribution.

A random variable  $X$  has a Weibull distribution if and only if its probability density function is given as

$f(x)$

$$f(x) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{x}{\lambda}\right)^k\right] \quad x \geq 0$$

The graph of the pdf is shown in Figure 6.12.

Figure 6.12: The graph of the Weibull pdf

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The cumulative distribution function (cdf) is given as

$$F(x) = 1 - \exp\left[-\left(\frac{x}{\lambda}\right)^k\right]$$

with  $k > 0$  and  $\lambda > 0$  as the shape and scale parameters respectively.

The cdf of the Weibull generalized family of distribution can be obtained from

$$G(x) = \int_0^x g(t) dt$$

$$G(x) = \int_0^x g(t) dt$$

$$F(x) = \int_0^x f(t) dt$$

$$1 - G(x) = \int_x^\infty g(t) dt$$

$$\frac{d}{dx} [1 - G(x)] = -g(x)$$

$$\exp\left[-\left(\frac{x}{\lambda}\right)^k\right] = 1 - G(x)$$

$$\frac{d}{dx} \exp\left[-\left(\frac{x}{\lambda}\right)^k\right] = -g(x)$$

$$\frac{d}{dx} \exp\left[-\left(\frac{x}{\lambda}\right)^k\right] = -g(x)$$

$$0 = -g(x)$$

$$1 - G(x) = \exp\left[-\left(\frac{x}{\lambda}\right)^k\right]$$

$$G(x) = 1 - \exp\left[-\left(\frac{x}{\lambda}\right)^k\right]$$

This can be obtained by substituting for  $x$  in the cdf of the Weibull distribution.

$$1 - G(x) = \exp\left[-\left(\frac{x}{\lambda}\right)^k\right]$$

The pdf of the Weibull generalized family is obtained by differentiating the cdf with respect to  $x$  as

$$\frac{d}{dx} [1 - G(x)] = -g(x)$$

$$f(x) = \frac{d}{dx} F(x) = g(x)$$

$$1 - G(x) = \exp\left[-\left(\frac{x}{\lambda}\right)^k\right] \Rightarrow \frac{d}{dx} [1 - G(x)] = -g(x)$$

Note that  $G(x)$  is the cdf of the parent or baseline distribution, and  $g(x)$  is the corresponding pdf of the

baseline distribution.

With the cdf and pdf of the generalized family many new classes of the family of the Weibull can be developed.

### Assignment 6.5

- Develop the pdf and cdf of Weibull exponential distribution
- Develop the pdf and cdf of Weibull Frechet distribution
- Develop the pdf and cdf of Weibull Lomax distribution
- Develop the pdf and cdf of Weibull Gompertz distribution
- Develop the pdf and cdf of Weibull Teissier distribution

- f. Develop the pdf and cdf of Weibull Kumaraswamy distribution
- g. Develop the pdf and cdf of Weibull Inverse exponential distribution
- h. Develop the pdf and cdf of Weibull alpha power distribution
- i. Develop the pdf and cdf of Weibull beta distribution

SOLUTION

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SOLUTION

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SOLUTION

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SOLUTION

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SOLUTION

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### 6.8 Gompertz distribution

In probability and statistics, the Gompertz distribution is a continuous probability distribution, named

after Benjamin Gompertz. The Gompertz distribution is often applied to describe the distribution of adult

lifespans by demographers, and actuaries. Related fields of science such as biology and gerontology also

considered the Gompertz distribution for the analysis of survival. More recently, computer scientists have

also started to model the failure rates of computer code by the Gompertz distribution. In Marketing Science, it has been used as an individual-level simulation for customer lifetime value modeling. In network theory, particularly the Erdos-Renyi model, the walk length of a random self-avoiding walk (SAW) is distributed according to the Gompertz distribution.

A random variable  $X$  has a Gompertz distribution if and only if its probability density function is given

as

$f(x)$

$= \lambda e^{-\lambda x} \exp\left(-e^{-\lambda x}\right), \quad x \geq 0, \lambda > 0.$

$f(x)$

The cdf is given as

$F(x)$

$= 1 - \exp\left(-e^{-\lambda x}\right), \quad x \geq 0, \lambda > 0.$

$F(x)$

The graph of the pdf is shown in Figure 6.13.

Figure 6.12: The graph of the Gompertz pdf

The cdf of Gompertz-G family of distributions is obtained as

$$W = G(x) = \int_0^x r(t) dt$$

$$F(x) = \int_0^x r(t) dt$$

0

Where  $r(t)$  is the pdf of the Gompertz distribution, and  $W = G(x) = \int_0^x r(t) dt = 1 - \exp\{-1 - G(x)\}$ . Thus,

$$e^{-1 - G(x)} = 1 - F(x) = \int_0^x r(t) dt$$

$$\frac{d}{dx} e^{-1 - G(x)} = -r(x)$$

$$F(x) = \int_0^x r(t) dt$$

$$\frac{d}{dx} \log[1 - G(x)] = -r(x)$$

$$\frac{d}{dx} e^{-1 - G(x)} = -r(x)$$

$$\frac{d}{dx} F(x) = r(x)$$

0

$$\frac{d}{dx} F(x) = r(x)$$

The corresponding pdf can be derived as follows:

$$f(x) = \frac{d}{dx} F(x)$$

$$f(x) = \frac{d}{dx} F(x)$$

$$f(x) = \frac{d}{dx} F(x) = r(x)$$

$$\frac{d}{dx} F(x) = r(x)$$

On simplification, we have

$$f(x) = \frac{d}{dx} F(x) = r(x)$$

$$\frac{d}{dx} F(x) = r(x)$$

$$\frac{d}{dx} F(x) = r(x)$$

$$\frac{d}{dx} F(x) = r(x)$$

$$\exp\{-1 - G(x)\} = 1 - F(x)$$

$$\frac{d}{dx} \exp\{-1 - G(x)\} = -r(x)$$

### Assignment 6.6

- Obtain the mean and variance of the Gompertz distribution
- Develop the pdf and cdf of Gompertz exponential distribution
- Develop the pdf and cdf of Gompertz Frechet distribution
- Develop the pdf and cdf of Gompertz Lomax distribution
- Develop the pdf and cdf of Gompertz Gompertz distribution
- Develop the pdf and cdf of Gompertz Teissier distribution
- Develop the pdf and cdf of Gompertz Kumaraswamy distribution
- Develop the pdf and cdf of Gompertz Inverse exponential distribution
- Develop the pdf and cdf of Gompertz alpha power distribution
- Develop the pdf and cdf of Gompertz beta distribution

SOLUTION

$$f(x) = \frac{d}{dx} F(x)$$

SOLUTION

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SOLUTION

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SOLUTION

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SOLUTION

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6.9 Cauchy distribution.

The Cauchy distribution, named after Augustin Cauchy, is a continuous probability distribution. It is also

known, especially among physicists, as the Lorentz distribution (after Hendrik Lorentz), Cauchy-Lorentz

distribution, Lorentz(ian) function, or Breit-Wigner distribution. The Cauchy distribution is the distribution of the x-intercept of a ray issuing from with a uniformly distributed angle. It is also the distribution of the ratio of two independent normally distributed random variables with mean zero. The Cauchy distribution is often used in statistics as the canonical example of a "pathological" distribution since both its expected value and its variance are undefined. The Cauchy distribution does not

have finite moments of order greater than or equal to one; only fractional absolute moments exist. The

Cauchy distribution has no moment generating function.

In mathematics, it is closely related to the Poisson kernel, which is the fundamental solution for the Laplace equation in the upper half-plane. x-analytically, the others being the normal distribution and the

Levy distribution.

A random variable  $X$  is said to have a Cauchy distribution if its probability density function is defined as

$$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$$

1

,  $\frac{1}{\pi} \frac{1}{1+x^2}$ .

$\frac{1}{\pi} \frac{1}{1+x^2}$

$$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$$

The distribution has one parameter.

Assignment 6.7:

1. Suppose  $X$  has a Cauchy distribution with  $\mu = 4$ . Compute  $P(X \leq 7)$ .
2. Suppose  $X$  has a Cauchy distribution with  $\mu = 9$ . Compute  $P(1 \leq X \leq 7)$ .

SOLUTION

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6.10 Chi-Square distribution.

n 1

A Chi-square distribution is a special case of the gamma distribution with parameters  $\alpha = \frac{n}{2}$  and  $\beta = 2$ .

$\frac{n}{2}$

The useful part of this distribution is the fact that the sum of squares of  $n$  independent normal random

variables has a Chi-square distribution with  $n$ -degrees of freedom. The pdf of the Chi-square with  $n$ -degrees of freedom is given as

$$f(x) =$$

$$\frac{1}{2^{n/2} \Gamma(n/2)}$$

$$x^{n/2-1} \exp\left(-\frac{x}{2}\right)$$

$$\exp\left(-\frac{x}{2}\right), x \geq 0.$$

$$x \geq 0.$$

$$x \geq 0.$$

$$x \geq 0.$$

$$x \geq 0.$$

$$x \geq 0.$$

$$x \geq 0.$$

$$x \geq 0.$$

$$x \geq 0.$$

It has its expectation and variance as  $E[X] = n$  and  $2n$ .

### 6.11 Beta distribution.

A random variable  $X$  has a Beta distribution if and only if its probability density function is given by

$$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, \text{ for } 0 < x < 1, a > 0, b > 0$$

$$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, \text{ for } 0 < x < 1, a > 0, b > 0$$

$$x \in (0, 1)$$

$$a, b > 0$$

The mean and variance of the Beta distribution are given as

$$E[X] = \frac{a}{a+b}$$

$$\text{Var}[X] = \frac{ab}{(a+b)^2(a+b+1)}$$

$$E[X] = \frac{a}{a+b}$$

$$a, b > 0$$

### 6.12 Quantile Functions and Order statistic of Continuous Probability distributions.

The quantile function is another way of describing a distribution other than the pdf or cdf. It is obtained as

the inverse cdf and calculated as:

$$Q(u) = F^{-1}(u)$$

For example, obtain the quantile function for the exponential distribution.

The cdf of the exponential distribution is given as

$$F(x) = 1 - \exp(-\lambda x)$$

$$u = 1 - \exp(-\lambda x)$$

$$1 - u = \exp(-\lambda x)$$

$$\log_{1-u} x$$

$$x = -\log_{1-u} u$$

$$1$$

$$121$$

Hence, the quantile function of the exponential distribution can be written as

$$Q(u) = -\log_{1-u} u,$$

$$1$$

where  $u \in [0,1]$  a uniform distribution.

The quantile function is very much useful for simulation purposes. This implies that random numbers can

be generated for the exponential distribution using the expression  $Q(u) = -\log_{1-u} u$ . The median is

$$1$$

$$\text{in } Q(u) = -\log_{1-u} u.$$

$$1/1$$

obtained if  $u =$

$$2^{-1}$$

#### 6.12.1 Survival function

It can also be called reliability function. It is a function that gives the probability that a device or object of

interest will survive beyond any specified time. It is calculated as:

$$S(x) = \Pr\{X > x\}$$

$$= 1 - \Pr\{X \leq x\}$$

Thus,

$$S(x) = 1 - F(x)$$

#### 6.12.2 Hazard Function

It can also be called hazard rate or failure rate. It is a conditional density given that the event under consideration has not yet occurred prior to time  $t$ . It is calculated as the ratio of the pdf to the survival

function:

$$f(x)$$

$$h(x) =$$

$$S(x)$$

#### 6.12.3 Cumulative Hazard Function

It is obtained as the integral of the hazard function:

$$t$$

$$H(t) = \int_0^t h(u) du$$

$$0$$

$$H(t) = -\ln S(t)$$

Also,

$S(t) = e^{-H(t)}$

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$f(t) = h(t) e^{-H(t)}$

#### 6.12.4 Reversed Hazard Function

It is calculated as the ratio of the pdf to the cdf as:

$f(x)$

$r(x)$

$F(x)$

#### 6.12.5 Odds Function

It is calculated as the ratio of the cdf to the survival function as:

$F(x)$

$O(x)$

$S(x)$

#### 6.12.6 Order Statistics

Suppose that a random sample of size  $n$  was drawn from an infinite population with a continuous pdf,

and that the values are arranged as in order of magnitude such that the smallest of the  $x$ 's corresponds to the value of the random variable (r.v), the next has a r.v and the largest has a r.v.

The r.v.s just defined is called order statistics. So, is the first order statistics (minimum order statistics),

is the second order statistics and is the maximum order statistics (last order statistics).

Theorem: For random samples from an infinite population which has the value  $f(x)$  at  $x$ , the p.d.f of the

$k$ th order statistics is:

$n! / (k-1)! (n-k)!$

$g(k) = y_k \cdot F(y_k)^{k-1} \cdot f(y_k) \cdot (1 - F(y_k))^{n-k}$  (23)

$k = 1, 2, \dots, n$

For  $y_k$

i. The distribution of minimum order statistics is obtained when  $k = 1$

ii. The distribution of maximum order statistics is obtained when  $k = n$

iii. When  $n$  is odd,  $n = 2m + 1$ , set  $k = m + 1$ , then the distribution of median is;

$g(m+1) = y_{m+1} \cdot F(y_{m+1})^m \cdot f(y_{m+1}) \cdot (1 - F(y_{m+1}))^m$

$2m! / (m!)^2$

$m! / (m!)^2$

$m! / (m!)^2$

iv. When  $n$  is even,  $n = 2m$

Exercise 6:7

1. Let  $X_1, X_2, \dots, X_n$  be the order statistics of a random sample of size  $n=8$  from the

distribution with pdf;  $f(x) = 2x^2 - x^3, 0 \leq x \leq 1$ . Compute (i) the pdf of  $X \sim U(0,1)$  (ii) the pdf of  $X \sim N(0,1)$  (iii)  $\Pr(1 \leq X \leq 5) = 4/5$

2. (i) Find the sampling distribution of  $Y_1$  and  $Y_n$ , the minimum and maximum order statistics of random samples of size  $n$  drawn from a continuous uniform distribution with parameters  $a = 0$  and  $b = 1$ . (ii) If  $n$  is odd, find the sampling distribution of the median. (iii) If  $n = 5$ , obtain the distribution of the median.

3. Obtain the distribution of minimum and maximum order statistics for the Weibull Exponential distribution.

#### Assignment 6.8

- Obtain the quantile function of the Gompertz distribution
- Develop the quantile function of the exponential distribution
- Develop the quantile function of the Frechet distribution
- Develop the quantile function of the Lomax distribution
- Develop the quantile function of the Weibull distribution
- Develop the quantile function of the e pdf and cdf of Gompertz Teissier distribution
- Develop the quantile function of the Kumaraswamy distribution
- Develop the quantile function of the exponential distribution
- Develop the quantile function of the alpha power distribution
- Develop the quantile function of the beta distribution

#### SOLUTION

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#### SOLUTION

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#### SOLUTION

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#### SOLUTION

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#### 6.13 Maximum Likelihood estimation of parameters of Continuous Probability distributions.

Let  $w_1, w_2, \dots, w_n$  be random variable obtained from a population with Weibull alpha power inverted

exponential distribution with a pdf given as:

$$f(w) = \frac{1}{w^2} \exp\left(-\frac{1}{w}\right)$$

$$a = \exp\left(-\frac{1}{w}\right) \log\left(\frac{1}{w}\right) \exp\left(-\frac{1}{w}\right)$$

$$f(w) = \frac{1}{w^2} \exp\left(-\frac{1}{w}\right)$$

$$\frac{1}{w^2}$$

$$\frac{1}{w^2} \exp\left(-\frac{1}{w}\right)$$

$a$

$a \neq 1$

$\exp\left(-\frac{w}{a}\right)$

$\frac{1}{a} \exp\left(-\frac{w}{a}\right)$ ,

$w \geq 0$

$a > 0$

$a \neq 1, a > 0, a \neq 1$ ,

Then, the log-likelihood of exponential alpha power inverted exponential distribution for vector

$\mathbf{w}$   $a, \alpha, \beta$  can be represented as

$T$

$n$

$w_i, \alpha$  as

$n$

$w_i, \alpha \log a - n \log \alpha$

$n$

$\log \alpha$

$w_i \log \alpha \log \alpha - n \exp\left(-\frac{w_i}{a}\right)$

$i=1, \dots, n$

$\log \alpha$

$n \exp\left(-\frac{w_i}{a}\right) \log \alpha - n \log \alpha - \log w_i^2 - a^{-1}$

$w_i \log \alpha$

$\exp\left(-\frac{w_i}{a}\right)$

$\log \alpha \log \alpha - \log \alpha - \log \alpha$ .

$w_i$

$\alpha$

$\alpha$

Taking partial derivative of Equation with respect to the parameters  $a, \alpha, \beta$  and equating to zero. The

solution to the nonlinear equation for the parameters can be obtained using R software, MATLAB, and

MAPLE. Thus, yield the maximum likelihood estimate  $\hat{a}, \hat{\alpha}, \hat{\beta}$ .

### 6.13.1 How to Plot Graphs for PDF and CDF Using R-software

Here is the R code for plotting the Gompertz Inverse Exponential distribution. Its pdf and cdf are:

$\alpha > 0$

$\alpha \neq 1$

$x \geq 0$

$\alpha \neq 1, \alpha > 0$

$\alpha$

$\alpha \neq 1, \alpha > 0$

$f(x) = \frac{2}{\alpha} \exp\left(-\frac{x}{\alpha}\right) \exp\left(-\frac{x}{\alpha}\right)$

```

[] []
e [] []
x [] []
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and
[] [] [] []
[] [] [] x [] []
[]1[] []1[] e [] []
[] [] [] []
F [] x[] [] 1[] e [] [] [] []

```

respectively.

Follow this R-code for pdf of:

```

rm(list=ls())

f<-function(x,al,be,th)
al*(th/x^2)* (exp(-(th/x))) *(1-exp(-(th/x)))^(-be-1) * exp((al/be)* (1-(1-exp(-(th/x)))^(-be )))

#PDF GoIE
curve(f(x,2.0,1.7,2.6),main="", ylab="f(x)",xlab="x",ylim=c(0,0.8),0,6,lwd=2)
curve(f(x,1.0,2.4,2.5),lty=1,col=2,add=T,lwd=2)
curve(f(x,0.5,0.3,0.6),lty=1,col=3,add=T,lwd=2)
curve(f(x,1,0.5,2.5),lty=1,col=4,add=T,lwd=2)
curve(f(x,0.6,4.3,3.4),lty=1,col=5,add=T,lwd=2)

legend("topright",title=expression(" "),
c(expression(alpha*"=2.0","*~beta*"=1.7","*~theta*"=2.6" ),
expression(alpha*"=1.0","*~beta*"=2.4","*~theta*"=2.5" ),
expression(alpha*"=0.5","*~beta*"=0.3","*~theta*"=0.6" ),
expression(alpha*"=1.0","*~beta*"=0.5","*~theta*"=2.5" ),
expression(alpha*"=0.6","*~beta*"=4.3","*~theta*"=3.4" )),
cex=1.1,lty=c(1) ,lwd=2,col=c(1,2,3,4,5))

```

```

##### CDF

```

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```

rm(list=ls())

f<-function(x,al,be,th)
1- exp((al/be)* (1-(1-exp(-(th/x)))^(-be )))

#CDF GoIE
curve(f(x,2.0,1.7,2.6),main="", ylab="F(x)",xlab="x",ylim=c(0,1.0),0,6,lwd=2)
curve(f(x,1.0,2.4,2.5),lty=1,col=2,add=T,lwd=2)
curve(f(x,0.5,0.3,0.6),lty=1,col=3,add=T,lwd=2)
curve(f(x,1,0.5,2.5),lty=1,col=4,add=T,lwd=2)
curve(f(x,0.6,4.3,3.4),lty=1,col=5,add=T,lwd=2)

```

```

legend("bottomright",title=expression(" "),
c(expression(alpha*"=2.0","*~beta*"=1.7,"*~theta*"=2.6" ),
expression(alpha*"=1.0","*~beta*"=2.4,"*~theta*"=2.5" ),
expression(alpha*"=0.5","*~beta*"=0.3,"*~theta*"=0.6" ),
expression(alpha*"=1.0","*~beta*"=0.5,"*~theta*"=2.5" ),
expression(alpha*"=0.6","*~beta*"=4.3,"*~theta*"=3.4" )),
cex=1.1,lty=c(1) ,lwd=2,col=c(1,2,3,4,5))

```

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## Chapter Seven

### 7.1 Introduction to Statistical Estimation.

The purpose of statistics is to use the information contained in a sample to make inferences about the

population from which the sample is taken. Because populations are characterized by numerical descriptive measures called parameters, the objective of many statistical investigations is to estimate the

value of one or more relevant parameters. As you will see, the sampling distributions derived in Chapter 6

play an important role in the development of the estimation procedures that are the focus of this chapter.

Estimation has many practical applications. For example, a manufacturer of washing machines might be

interested in estimating the proportion  $p$  of washers that can be expected to fail prior to the expiration of a

1-year guarantee time. Other important population parameters are the population mean, variance, and

standard deviation. For example, we might wish to estimate the mean waiting time  $\mu$  at a supermarket

checkout station or the standard deviation of the error of measurement  $\sigma$  of an electronic instrument. To

simplify our terminology, we will call the parameter of interest in the experiment the target parameter.

Suppose that we wish to estimate the average amount of mercury  $\mu$  that a newly developed process can

remove from 1 ounce of ore obtained at a geographic location. We could give our estimate in two distinct

forms. First, we could use a single number for instance 0.13 ounce that we think is close to the unknown

population mean  $\mu$ . This type of estimate is called a point estimate because a single value, or point, is

given as the estimate of  $\mu$ . Second, we might say that  $\mu$  will fall between two numbers; for example,

between 0.07 and 0.19 ounce. In this second type of estimation procedure, the two values that we give

may be used to construct an interval (0.07, 0.19) that is intended to enclose the parameter of interest; thus,

the estimate is called an interval estimate.

The information in the sample can be used to calculate the value of a point estimate, an interval estimate,

or both. In any case, the actual estimation is accomplished by using an estimator for the target parameter.

Definition: Estimation is the assignment of value(s) to a population parameter based on a value of the

corresponding sample statistic is called estimation.

Definition: Estimate and Estimator: The value(s) assigned to a population parameter based on the value

of a sample statistic is called an estimate. An estimator is a rule, often expressed as a formula that tells

how to calculate the value of an estimate based on the measurements contained in a sample.

For example, the samples mean,

$\bar{x}$

$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

$\bar{x}$

Is one possible point estimator of the population mean  $\mu$ . Clearly, the expression for  $\bar{x}$  is both a rule and a formula. It tells us to sum the sample observations and divide by the sample size  $n$ .

An experimenter who wants an interval estimate of a parameter must use the sample data to calculate two

values, chosen so that the interval formed by the two values includes the target parameter with a specified

probability. Examples of interval estimators will be given in subsequent sections. Many different estimators (rules for estimating) may be obtained for the same population parameter. This should not be

surprising. Ten engineers each assigned to estimate the cost of a large construction job could use different

methods of estimation and thereby arrive at different estimates of the total cost. Such engineers, called

estimators in the construction industry, base their estimates on specified fixed guidelines and intuition.

Each estimator represents a unique human subjective rule for obtaining a single estimate. This brings us

to a most important point: Some estimators are considered good, and others, bad. The management of a

construction firm must define good and bad as they relate to the estimation of the cost of a job. How can

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we establish criteria of goodness to compare statistical estimators? The following sections contain some answers to this question.

### 7.2 Statistical Point Estimation.

An estimate may be a point estimate or an interval estimate. These two types of estimates are described.

If we select a sample and compute the value of a sample statistic for this sample, then this value gives the point estimate of the corresponding population parameter.

Point Estimate: The value of a sample statistic that is used to estimate a population parameter is called a point estimate.

Thus, the value computed for the sample mean,  $\bar{x}$ , from a sample is a point estimate of the corresponding

population mean,  $\mu$ . For the example mentioned earlier, suppose the Census Bureau takes a random

sample of 10,000 households and determines that the mean housing expenditure per month,  $\bar{x}$ , for this

sample is \$2970. Then, using  $\bar{x}$  as a point estimate of  $\mu$ , the Bureau can state that the mean housing

expenditure per month,  $\mu$ , for all households is about \$2970. Thus,

Point estimate of a population parameter = Value of the corresponding sample statistic

Each sample selected from a population is expected to yield a different value of the sample statistic. Thus,

the value assigned to a population mean,  $\mu$ , based on a point estimate depends on which of the samples is

drawn. Consequently, the point estimate assigns a value to  $\mu$  that almost always differs from the true

value of the population mean.

### 7.3 Sampling distribution of a statistic.

Suppose we take a sample of size  $n$  from a population of size  $N$ , then there are  $N$  combinations  $n = k$ , ie.

$N$

$C_n = k$ , say possible sample. We can compute the statistic, say,  $t$  for each of these samples. Let  $t_1, t_2, t_3, \dots, t_k$  be values of the statistic  $t$  for these  $k$  possible samples. Thus,

The mean and variance can be computed as

$\frac{1}{k}$

$E(t) = \frac{1}{k} \sum_{i=1}^k t_i$

$\frac{1}{k} \sum_{i=1}^k t_i^2$

$\frac{1}{k}$

$Var(t) = \frac{1}{k} \sum_{i=1}^k t_i^2 - E(t)^2$

$\frac{1}{k} \sum_{i=1}^k t_i^2$

With precision of

1

$t^*$

S.E.  $t^*$

With S.E. as the standard error given as

S.E.  $\pm$  Var  $t^*$

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#### 7.4 Statistical Interval Estimation.

In the case of interval estimation, instead of assigning a single value to a population parameter, an interval

is constructed around the point estimate, and then a probabilistic statement that this interval contains the

corresponding population parameter is made.

Interval Estimation: In interval estimation, an interval is constructed around the point estimate, and it is

stated that this interval contains the corresponding population parameter with a certain confidence level.

For the example about the mean housing expenditure, instead of saying that the mean housing expenditure

per month for all households is ₦2970, we may obtain an interval by subtracting a number from ₦2970

and adding the same number to ₦2970. Then we state that this interval contains the population mean,  $\mu$ .

For purposes of illustration, suppose we subtract ₦340 from ₦2970 and add ₦340 to ₦2970. Consequently,

we obtain the interval (₦2970 – ₦340) to (₦2970 + ₦340), or ₦2630 to ₦3310. Then we state that the

interval ₦2630 to ₦3310 is likely to contain the population mean,  $\mu$ , and that the mean housing expenditure per month for all households in Nigeria is between ₦2630 and ₦3310. This procedure is called

interval estimation. The value ₦2630 is called the lower limit of the interval, and ₦3310 is called the upper

limit of the interval. The number we add to and subtract from the point estimate is called the margin of

error or the maximum error of the estimate.

The question arises: What number should we subtract from and add to a point estimate to obtain an interval estimate? The answer to this question depends on two considerations:

1. The standard deviation  $\sigma_x$  of the sample mean,  $\bar{x}$
2. The level of confidence to be attached to the interval

First, the larger the standard deviation of  $\bar{x}$ , the greater is the number subtracted from and added to the

point estimate. Thus, it is obvious that if the range over which  $\bar{x}$  can assume values is larger, then the

interval constructed around  $\bar{x}$  must be wider to include  $\mu$ .

Second, the quantity subtracted and added must be larger if we want to have a higher confidence in our

interval. We always attach a probabilistic statement to the interval estimation. This probabilistic statement

is given by the confidence level. An interval constructed based on this confidence level is called a confidence interval.

Confidence Level and Confidence Interval: Each interval is constructed with regard to a given confidence

level and is called a confidence interval.

The confidence interval is given as

Point estimate  $\pm$  Margin of error

The confidence level associated with a confidence interval states how much confidence we have that this

interval contains the true population parameter. The confidence level is denoted by  $(1 - \alpha) 100\%$ .

As mentioned above, the confidence level is denoted by  $(1 - \alpha) 100\%$ , where  $\alpha$  is the Greek letter alpha.

When expressed as probability, it is called the confidence coefficient and is denoted by  $1 - \alpha$ . In passing,

note that  $\alpha$  is called the significance level, this will be explained later.

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Although any value of the confidence level can be chosen to construct a confidence interval, the more

common values are 90%, 95%, and 99%. The corresponding confidence coefficients are 0.90, 0.95, and

0.99, respectively. The next section describes how to construct a confidence interval for the population

mean when the population standard deviation,  $\sigma$ , is known.

7.4.1 Estimation of a Population mean when,  $\sigma$ , is known.

This subsection explains how to construct a confidence interval for the population mean  $\mu$  when the population standard deviation  $\sigma$  is known. Here, there are three possible cases, as follows.

Case I. If the following three conditions are fulfilled:

1. The population standard deviation  $\sigma$  is known
2. The sample size is small (i.e.,  $n < 30$ )
3. The population from which the sample is selected is approximately normally distributed, then we use

the normal distribution to make the confidence interval for  $\mu$  the sampling distribution of  $\bar{x}$  is normal

with its mean equal to  $\mu$  and the standard deviation equal to  $\sigma \times \frac{1}{\sqrt{n}}$ , assuming that  $n/N \leq 0.05$ .

Case II. If the following two conditions are fulfilled:

1. The population standard deviation  $\sigma$  is known

2. The sample size is large (i.e.,  $n \geq 30$ ), then, again, we use the normal distribution to make the confidence interval for  $\mu$  because from the central limit theorem, the sampling distribution of  $\bar{x}$  is (approximately) normal with its mean equal to  $\mu$  and the standard deviation equal to  $\sigma/\sqrt{n}$ , assuming that  $n/N \leq 0.05$ .

Case III. If the following three conditions are fulfilled:

1. The population standard deviation  $\sigma$  is known
2. The sample size is small (i.e.,  $n < 30$ )
3. The population from which the sample is selected is not normally distributed (or its distribution is unknown), then we use a nonparametric method to make the confidence interval for  $\mu$ .

This section will cover the first two cases. The procedure for making a confidence interval for  $\mu$  is the

same in both these cases. Note that in Case I, the population does not have to be exactly normally distributed. As long as it is close to the normal distribution without any outliers, we can use the normal

distribution procedure. In Case II, although 30 is considered a large sample, if the population distribution

is very different from the normal distribution, then 30 may not be a large enough sample size for the

sampling distribution of  $\bar{x}$  to be normal and, hence, to use the normal distribution.

The following chart summarizes the above three cases.

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The confidence Interval for  $\mu$ . The  $(1 - \alpha) 100\%$  confidence interval for  $\mu$  under Cases 1 and 11 is

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where

$$\frac{\sigma}{\sqrt{n}}$$

The value of  $z$  used here is obtained from the standard normal distribution table.

The quantity  $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$  in the confidence interval formula is called the margin of error and is denoted by  $E$ .

Margin of Error: The margin of error for the estimate of  $\mu$ , denoted by  $E$ , is the quantity that is subtracted from and added to the value of  $\bar{x}$  to obtain a confidence interval for  $\mu$ . Thus,

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

The value of  $z$  in the confidence interval formula is obtained from the standard normal distribution table

the given confidence level. To illustrate, suppose we want to construct a 95% confidence interval for  $\mu$ .

A 95% confidence level means that the total area under the standard normal curve between two points (at

the same distance) on different sides of  $\mu$  is 95%, or .95, as shown in Figure 7.2. Note that we have denoted these two points by  $-z$  and  $z$  in Figure 8.1. To find the value of  $z$  for a 95% confidence level, we

first find the areas to the left of these two points,  $z$  and  $-z$ . Then we find the  $z$  values for these two areas

from the normal distribution table. Note that these two values of  $z$  will be the same but with opposite

signs. To find these values of  $z$ , we perform the following two steps:

Figure 7.1: Finding  $z$  for a 95% confidence level.

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1. The first step is to find the areas to the left of  $-z$  and  $z$ , respectively. Note that the area between  $-z$

and  $z$  is denoted by  $1 - \alpha$ . Hence, the total area in the two tails is  $\alpha$  because the total area under the curve is 1.0. Therefore, the area in each tail, as shown in Figure 7.2, is  $\alpha/2$ . In our example,  $1 - \alpha = 0.95$ . Hence, the total area in both tails is  $\alpha = 1 - 0.95 = 0.05$ . Consequently, the area in each tail is  $\alpha/2 = 0.05/2 = 0.025$ . Then, the area to the left of  $-z$  is 0.0250, and the area to the left of  $z$  is  $0.0250 + .95 = .9750$ .

2. Now find the  $z$  values from the standard normal distribution table such that the areas to the left of

$-z$  and  $z$  are 0.0250 and 0.9750, respectively. These  $z$  values are  $-1.96$  and  $1.96$ , respectively.

Thus, for a confidence level of 95%, we will use  $z = 1.96$  in the confidence interval formula.

Figure 7.2: Area in the tails

Table 7.1 lists the  $z$  values for some of the most commonly used confidence levels. Note that we always

use the positive value of  $z$  in the formula.

Table 7.1:  $z$  Values for Commonly Used Confidence Levels

Confidence Level Areas to Look for in Table  $z$  Value

90% 0.0500 and 0.9500 1.64 or 1.65

95% 0.0250 and .9750 1.96

96% 0.0200 and .9800 2.05

97% 0.0150 and 0.9850 2.17

98% 0.0100 and 0.9900 2.33

99% .0050 and .9950 2.57 or 2.58

Example 7.1: A publishing company has just published a new college textbook. Before the company decides the price at which to sell this textbook, it wants to know the average price of all such textbooks in

the market. The research department at the company took a random sample of 25 comparable textbooks

and collected information on their prices. This information produced a mean price of \$145 for this sample. It is known that the standard deviation of the prices of all such textbooks is \$35 and the population distribution of such prices is approximately normal.

a. What is the point estimate of the mean price of all such college textbooks?

b. Construct a 90% confidence interval for the mean price of all such college textbooks.

Solution Here,  $\sigma$  is known and, although  $n < 30$ , the population is approximately normally distributed.

Hence, we can use the normal distribution. From the given information,

$n = 25$ ,  $\bar{x} = \$145$ ,  $\sigma = \$35$

The standard deviation of  $\bar{x}$  is

$\sigma_{\bar{x}} =$

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\$35}{\sqrt{25}} = \$7.00$

$n = 25$

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a. The point estimate of the mean price of all such college textbooks is \$145; that is, point estimate of  $\mu = \$145$ .

b. The confidence level is 90%, or .90. First we find the  $z$  value for a 90% confidence level.

Here, the area in each tail of the normal distribution curve is  $\alpha/2 = (1 - 0.90)/2 = 0.05$ .

Now in the standard normal table, look for the areas 0.0500 and 0.9500 and find the corresponding values of  $z$ . These values are (approximately)  $z = -1.65$  and  $z = 1.65$

Next, we substitute all the values in the confidence interval formula for  $\mu$ . The 90% confidence interval for  $\mu$  is

$\bar{x} \pm z_{\alpha/2} \sigma_{\bar{x}} = 145 \pm 1.65(7.00) = 145 \pm 11.55 = (145 - 11.55) \text{ to } (145 + 11.55) = \$133.45 \text{ to } \$156.55$

Thus, we are 90% confident that the mean price of all such college textbooks is between \$133.45 and

\$156.55. Note that we cannot say for sure whether the interval \$133.45 to \$156.55 contains the true

population mean or not. Since  $\mu$  is a constant, we cannot say that the probability is .90 that this interval

contains  $\mu$  because either it contains  $\mu$  or it does not. Consequently, the probability that this interval

contains  $\mu$  is either 1.0 or 0. All we can say is that we are 90% confident that the mean price of all such

college textbooks is between \$133.45 and \$156.55.

In the above estimate, \$11.55 is called the margin of error or give-and-take figure.

Figure 7.3: Confidence intervals.

How do we interpret a 90% confidence level? In terms of Example 7.1, if we take all possible samples of

25 such college textbooks each and construct a 90% confidence interval for  $\mu$  around each sample mean,

we can expect that 90% of these intervals will include  $\mu$  and 10% will not. In Figure 7.3 we show means

$\bar{x}_1$ ,  $\bar{x}_2$  and  $\bar{x}_3$  of three different samples of the same size drawn from the same population. Also shown in

this figure are the 90% confidence intervals constructed around these three sample means. As we observe,

the 90% confidence intervals constructed around  $\bar{x}_1$ ,  $\bar{x}_2$  include  $\mu$ , but the one constructed around  $\bar{x}_3$

does not. We can state for a 90% confidence level that if we take many samples of the same size from a

population and construct 90% confidence intervals around the means of these samples, then we expect

90% of these confidence intervals will be like the ones around  $x_1$ ,  $x_2$  in Figure 7.3, which include  $\mu$ , and

10% will be like the one around  $x_3$ , which does not include  $\mu$ .

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Assignment 7.1: The standard deviation for a population is  $\sigma = 14.8$ . A random sample of 25 observations selected from this population gave a mean equal to 143.72. The population is known to have

a normal distribution.

- Make a 99% confidence interval for  $\mu$ .
- Construct a 95% confidence interval for  $\mu$ .
- Determine a 90% confidence interval for  $\mu$ .
- Does the width of the confidence intervals constructed in parts a through c decrease

SOLUTION

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SOLUTION

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#### 7.4.2 Determining the Sample Size for the Estimation of Mean

Determining the Sample Size for the Estimation of  $\mu$  given the confidence level and the standard deviation of the population, the sample size that will produce a predetermined margin of error  $E$  of the

confidence interval estimate of  $\mu$  is

$$z_{\alpha/2}^2 \frac{\sigma^2}{n}$$

$$n = \frac{z_{\alpha/2}^2 \sigma^2}{E^2}$$

$$E^2$$

Assignment 7.2: An alumni association wants to estimate the mean debt of this year's college graduates.

It is known that the population standard deviation of the debts of this year's college graduates is \$11,800.

How large a sample should be selected so that a 99% confidence interval of the estimate is within \$800 of

the population mean?

SOLUTION

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#### 7.4.3 Confidence Interval for the Mean for Unknown Variance and Small Sample Size

In the previous section, we have seen that we can use the normal distribution to construct confidence

intervals for a population mean  $\mu$  when  $\sigma$  is known provided that the underlying population is normally

distributed with large sample size larger than 30.

Case I. If the following three conditions are fulfilled:

1. The population standard deviation  $\sigma$  is not known
2. The sample size is small (i.e.,  $n < 30$ )
3. The population from which the sample is selected is approximately normally distributed, and then we

use the t- distribution to make the confidence interval for  $\mu$ .

Case II. If the following two conditions are fulfilled:

1. The population standard deviation  $\sigma$  is not known
2. The sample size is large (i.e.,  $n \geq 30$ ), then again we use the t distribution to make the confidence interval for  $\mu$ .

Case III. If the following three conditions are fulfilled:

1. The population standard deviation  $\sigma$  is not known
2. The sample size is small (i.e.,  $n < 30$ )
3. The population from which the sample is selected is not normally distributed (or its distribution is unknown),

However, we can construct a confidence interval for the population mean when the standard deviation is

not known. Thus, in place of z, the standard normal, we use the test statistic t with n-1 degrees of freedom. The t-statistic is defined as

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

However, if the sample size  $n > 30$ , we can approximate the t-distribution to normal distribution.

The confidence Interval for  $\mu$  can be obtained using the  $(1 - \alpha)$  100% confidence interval for  $\mu$  under

Cases 1 and 11 is

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$141$$

where

$$s$$

$$s_x = \frac{s}{\sqrt{n}}$$

$$n$$

The value of t is obtained from the t distribution table for  $n - 1$  degrees of freedom and the given confidence level. Here  $t_{\alpha/2}$  is the margin of error or the maximum error of the estimate; that is,

$$E = t_{\alpha/2} \frac{s}{\sqrt{n}}$$

Example 7.2: A sample of 16 private colleges around the state shows a mean cost for a year's tuition of

\$25000 with a standard deviation of \$4500. Find a 90% confidence interval for the average tuition costs at

private colleges.

$n = 16$ ,  $\bar{x} = \$25000$ ,  $s = \$4500$ , we calculate.

4500

$s = 1125$

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Since the sample size  $n = 16 < 31$ , we use a t-distribution with  $n - 1 = 15$  degrees of freedom.

$\bar{x} \pm 1.753 \frac{s}{\sqrt{n}} = 25000 \pm 1.753 \frac{1125}{\sqrt{16}} = 23027.875, 26972.125$

s

$\bar{x} \pm t$

n

#### 7.4.4 Confidence Interval for Proportion

Often we want to estimate the population proportion or percentage. (Recall that a percentage is obtained

by multiplying the proportion by 100.) For example, the production manager of a company may want to

estimate the proportion of defective items produced on a machine. A bank manager may want to find the

percentage of customers who are satisfied with the service provided by the bank.

Again, if we can conduct a census each time we want to find the value of a population proportion, there is

no need to learn the procedures discussed in this section. However, we usually derive our results from

sample surveys. Hence, to take into account the variability in the results obtained from different sample

surveys, we need to know the procedures for estimating a population proportion.

Recall from Chapter 7 that the population proportion is denoted by  $p$ , and the sample proportion is denoted by  $\hat{p}$ . This section explains how to estimate the population proportion,  $p$ , using the sample proportion,  $\hat{p}$ . The sample proportion,  $\hat{p}$ , is a sample statistic, and it possesses a sampling distribution.

We know that:

1. The sampling distribution of the sample proportion  $\hat{p}$  is approximately normal for a large sample.
2. The mean of the sampling distribution of  $\hat{p}$ ,  $\mu_{\hat{p}}$ , is equal to the population proportion,  $p$ .

pq

3. The standard deviation of the sampling distribution of  $\hat{p}$  is  $\sigma_{\hat{p}} = \sqrt{pq}$ , where  $q = 1 - p$ , given

n

n

that  $\alpha = 0.05$

N

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In the case of a proportion, a sample is considered to be large if  $np$  and  $nq$  are both greater than 5. If  $p$  and  $q$  are not known, then  $np^{\wedge}$  and  $nq^{\wedge}$  should each be greater than 5 for the sample to be large.

When estimating the value of a population proportion, we do not know the values of  $p$  and  $q$ . Consequently, we cannot compute  $\sigma_{\hat{p}}$ . Therefore, in the estimation of a population proportion, we use the value of  $s_{\hat{p}}$  as an estimate of  $\sigma_{\hat{p}}$ . The value of  $s_{\hat{p}}$  is calculated using the following formula.

Estimator of the Standard Deviation of  $\hat{p}$  The value of  $s_{\hat{p}}$ , which gives a point estimate of  $\sigma_{\hat{p}}$ , is calculated as follows. Here,  $s_{\hat{p}}$  is an estimator of  $\sigma_{\hat{p}}$ .

$$s_{\hat{p}} = \sqrt{\frac{p^{\wedge} q^{\wedge}}{n}}$$

$$s_{\hat{p}} = \sqrt{\frac{p^{\wedge} q^{\wedge}}{n}}$$

$$s_{\hat{p}} = \sqrt{\frac{p^{\wedge} q^{\wedge}}{n}}$$

$$s_{\hat{p}} = \sqrt{\frac{p^{\wedge} q^{\wedge}}{n}}$$

Note that the condition  $n \geq 0.05$  must hold true to use this formula.

N

The sample proportion,  $\hat{p}$ , is the point estimator of the corresponding population proportion  $p$ .

Then to find the confidence interval for  $p$ , we add to and subtract from  $\hat{p}$  a number that is called the margin of error,  $E$ .

Confidence Interval for the Population Proportion,  $p$  For a large sample, the  $(1 - \alpha)100\%$  confidence

interval for the population proportion,  $p$ , is

$$p^{\wedge} \pm z s_{\hat{p}}$$

The value of  $z$  is obtained from the standard normal distribution table for the given confidence level, and

$$s_{\hat{p}} = \sqrt{\frac{p^{\wedge} q^{\wedge}}{n}}$$

$s_{\hat{p}} = \sqrt{\frac{p^{\wedge} q^{\wedge}}{n}}$ . The term  $z s_{\hat{p}}$  is called the margin of error, or the maximum error of the estimate, and is

denoted by  $E$ .

Assignment 7.3: According to a Gallup-Purdue University study of college graduates conducted during

February 4 to March 7, 2014, 63% of college graduates polled said that they had at least one college

professor who made them feel excited about learning ([www.gallup.com](http://www.gallup.com)). Suppose that this study was

based on a random sample of 2000 college graduates. Construct a 97% confidence interval for the corresponding population proportion.

SOLUTION

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#### 7.4.5 Confidence Interval for Variances and Standard Deviations

When random samples are drawn from a normal population of a known variance  $\sigma^2$ , the quantity  $\frac{(n-1)s^2}{\sigma^2}$  possesses a probability distribution that is known as chi-square distribution. The calculation

$$\frac{(n-1)s^2}{\sigma^2}$$

follows as:

$$\frac{(n-1)s^2}{\sigma^2}$$

$$\frac{(n-1)s^2}{\sigma^2}$$

$$\frac{(n-1)s^2}{\sigma^2}$$

Making  $\sigma^2$  the subject of the formula, we have

$$\sigma^2 = \frac{(n-1)s^2}{\chi^2}$$

$$\sigma^2 = \frac{(n-1)s^2}{\chi^2}$$

Where  $s^2$  is the sample variance,  $n$  is the sample size, and  $\sigma^2$  is the value of the population variance with

$(n-1)$  degrees of freedom.

The confidence interval is

$$\left[ \frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \right]$$

$$\left[ \frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \right]$$

$$\left[ \frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \right]$$

$$\left[ \frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \right]$$

$$\left[ \frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \right]$$

$$\left[ \frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \right]$$

Assignment 7.4: Find  $\sigma^2$  at 25, 0.025

SOLUTION

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## Chapter Eight

### Introduction to Statistical Test of Hypotheses

#### 8.0 Introduction

A hypothesis is an assertion or conjecture about the population parameter(s). For example  $H_0: \mu = 10$ ,  $H_1: \mu > 10$ , etc. These are statements about the population parameters. Sometimes, a hypothesis could be conjectures about parameters of two or more populations. For instance,  $H_0: \mu_1 = \mu_2$ ,  $H_1: \mu_1 > \mu_2$ , etc.

There are two types of hypothesis, namely, null hypothesis and alternative hypothesis. The null hypothesis is the hypothesis which is to be tested for acceptance or rejection. It is usually denoted as  $H_0$ .

The alternative hypothesis, denoted as  $H_1$  or  $H_A$ , is a conjecture about a population parameter which

gives an alternative to the null hypothesis. For instance, if  $H_0: \mu = 10$ , the alternatives are  $H_1: \mu > 10$ ,  $H_1: \mu < 10$ ,  $H_1: \mu \neq 10$ .

$H_1: \mu > 10$ ,  $H_1: \mu < 10$ ,  $H_1: \mu \neq 10$

#### 8.1 Introduction to Statistical Test of Hypotheses - Simple and Alternative.

Test of hypothesis is the statistical analysis carried out in order to decide whether to accept or reject the

null hypothesis. The method of hypothesis testing uses tests of significance to determine the likelihood

that a statement (often related to the mean or variance of a given distribution) is true, and at what likelihood we would, as statisticians, accept the statement as true. While understanding the mathematical

concepts that go into the formulation of these tests is important, knowledge of how to appropriately use

each test (and when to use which test) is equally important.

Once the data is collected, tests of hypotheses follow the following steps:

1. Using the sampling distribution of an appropriate test statistic, determine a critical region of size
2. Determine the value of the test statistic from the sample data.
3. Check whether the value of the test statistic falls within the critical region; if yes, we reject the null in favor of the alternative hypothesis, and if no, we fail to reject the null hypothesis.

These three steps are what we will focus on for every test; namely, what the appropriate sampling distribution for each test is and what test statistic we use (the third step is done by simply comparing values).

Next, we give definitions and explanations of important concepts in statistical test of hypotheses.

#### 8.1.1 Types of Errors in Hypothesis Testing:

There is always some possibility of committing an error in taking decision about the hypothesis. These

errors are grouped into two, type I error and type II error.

A type I error is committed when we reject a correct  $H_0$  when  $H_0$  is actually true and  $H_1$  is false. On the

other hand, a type II error is committed when we accept an incorrect  $H_0$ . when  $H_1$  is actually correct. A

type II error is considered more severe than the type I error. Hence, in drawing conclusion (decision)

about the  $H_0$ , the practice is to ensure that type II error can be minimized even at a certain risk of committing a type I error.

**8.1.2 Levels of Significance:** This is defined as the quantity of risk of the type I error which we are willing to tolerate in making a decision about  $H_0$ . In other words, it is the probability of committing a type

I error of a tolerable level or degree. It is denoted by  $\alpha$  and is conventionally chosen as 0.05 (5%) or 0.01

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(1%). Significance level of  $\alpha = 0.01$  is used for high precision while that of  $\alpha = 0.05$  is used for moderate precision.

#### 8.1.3 P-value (Probability Value):

This is defined as the smallest level of  $\alpha$  at which  $H_0$  is significant, that is the smallest level of  $\alpha$  at which

$H_0$  is rejected. It is a number that denotes the likelihood of your data having occurred under the null

hypothesis of your statistical test. The level of statistical significance is usually represented as a P-value

or probability value between 0 and 1. The smaller the p-value, the more likely it is that you would reject

the null hypothesis.

P-Value enables an individual to decide for himself how significant the data are. It avoids the imposition

of a fixed level of significance about the acceptance or rejection of  $H_0$ .

#### 8.1.4 Critical Region (C.R)

It is the region within which the test statistic lies or falls for the  $H_0$  to be rejected. A statistic that is used to

test the  $H_0$  follows some known distribution. In a test, the area under the probability density curve is

divided into two regions, namely, the region of acceptance and the region of rejection also known as the

critical region.

The area of the critical region is equal to the level of significance  $\alpha$ . Critical region is always on the tail(s)

of the distribution curve depending on whether the test is one-tailed or two-tailed.

If the alternative hypothesis,  $H_1$ , is of the type  $> 0$ , then the critical region lies on right tail of the probability density curve and the test is called a one-tailed test. On the other hand, if  $H_1$ , is of the type

$< 0$ , then the critical region lies on left tail of the probability density curve and the test is also called a one-

tailed test.

If  $H_1$  is of the type  $\neq 0$ , the critical region lies on both tails of the distribution curve and the test is called a two-tailed test.

In a two-tailed test, the critical region is still equal to  $\alpha$  but  $\alpha/2$  of it lies on each tails for a test of significant level  $\alpha$ .

Fig: 8.1: One-Sided Right Tailed Critical Region

Fig 2: One-Sided Left Tailed Critical Region

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Fig.: Two-Tailed Critical Regions.

8.1.5 Size of A Test: This is defined as the probability rejecting the null hypothesis when it is true. It is

usually denoted by  $\alpha$ . Therefore

$$P(\text{Reject } H_0 / H_0 \text{ is true}) = \alpha \quad (8.1)$$

8.1.6 Power of A Test: This is defined as the probability of rejecting  $H_0$  when  $H_0$  is actually false and  $H_1$

is true.

Power ( )

Power ( )

Power ( ),

Power (8.2)

8.1.7 Degrees of Freedom: Degree of freedom (d.f) is defined as the number of independent observations

in a set. The table values for the distribution of the test statistics are provided on statistical tables for

various levels of significance and degrees of freedom. These tables of values enable us to decide about the

rejection or otherwise of  $H_0$ .

8.1.8 Point Estimate of a Population Parameter

A point estimate is a single value obtained from a sample and used as an estimate of a parameter. For

instance, a sample mean is a point estimate of a population mean.

8.1.9 Confidence Interval (CL) for a Population Mean

The confidence interval (CL) for a population mean is the interval within which the mean of a population

is expected to lie with a given level of confidence. Generally, the ( ) confidence interval is obtained using the formula:

$$\bar{x} \pm t_{\alpha/2} (s/\sqrt{n}) \quad (8.3)$$

where  $\bar{x}$  is the mean of a sample drawn from the population,  $t_{\alpha/2}$  is the table value of an appropriate test statistic at a specified level of significance  $\alpha$ , (where applicable),  $n$  is the degree of freedom and  $s/\sqrt{n}$  is the standard error of the sample mean.

8.2 Small Sample Tests for Population Mean and Difference of Two Population Means

We consider tests of hypotheses for one sample and two samples when dealing with samples of small

size. In both of the cases, the student t -test is applied.

Assumptions about the test include the following:

- The random variable  $X$  follows a normal distribution
- All the observations in the sample are independent
- The sample size is not large, i.e. 30
- The assumed value  $\mu_0$  of the population mean is the correct value.

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- The sample values are correctly taken and recorded.

In case the above assumptions do not hold, the reliability of the test decreases.

8.2.1 Test of Hypothesis about Population Mean for One Sample of Small Size

The t-distribution is used in the test of hypothesis about the population mean when the sample size is

small. It is defined as the deviation of the estimated sample mean from its population mean expressed in

terms of standard deviation.

Suppose a small random sample  $(X_1, X_2, \dots, X_n)$  of size  $n$  has been drawn from a normal population having a mean  $\mu$  and an unknown variance  $\sigma^2$ . We want to test the hypothesis:

$H_0$ : against  $H_1$ : where  $\mu_0$  is some assumed value for  $\mu$ .

Let the observed values on the random sample  $(X_1, X_2, \dots, X_n)$  be  $(x_1, x_2, \dots, x_n)$ . The statistic for student t

test is given by:

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} \quad (8.4)$$

where the estimate of the population standard deviation is obtained using sample standard deviation

given by:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (8.5)$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

or

$$s^2 = \frac{1}{n-1} \left\{ \sum_{i=1}^n x_i^2 - n\bar{x}^2 \right\} \quad (8.6)$$

$$s = \sqrt{s^2}$$

If calculated value of  $t$  from (8.3) is greater or equal to the table value of  $t_{\alpha, n-1}$  at a given  $\alpha$  and for  $(n-1)$  d.f, we reject  $H_0$ , and therefore conclude that there is the difference between the sample mean and population mean is significant, otherwise accept  $H_0$ .

Note that for a right tailed test where the alternative hypothesis takes the form  $H_1: \mu > \mu_0$ ,  $H_0$  is rejected if calculated  $t$  is greater or equal to the table value of  $t_{\alpha, n-1}$  at the specified  $\alpha$  and a  $(n-1)$  d.f.

Similarly, for a left tailed test where the alternative hypothesis takes the form  $H_1: \mu < \mu_0$ ,  $H_0$  is rejected if calculated  $t$  is less or equal to the negative table value of  $t_{\alpha, n-1}$  at the specified  $\alpha$  and a  $(n-1)$  d.f.

The rules for rejection of  $H_0$  is summarized below:

- Given reject if  $t \geq t_{\alpha, n-1}$ , other do not reject
- Given reject if  $t \geq t_{\alpha, n-1}$ , other do not reject (Right tail test)
- Given reject if  $t \leq -t_{\alpha, n-1}$ , other do not reject (Left tail test)

for a  $t$ -test, the  $(1-\alpha)$  confidence interval (CI) is given as:

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \quad (8.7)$$

where the lower and upper limit are respectively given as  $\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$  and  $\bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$ , and

$$\frac{s}{\sqrt{n}}$$

is the table value of  $t$  at a given  $\alpha$  and d.f of  $(n-1)$ .

Example 8.1: In a modulation study on a certain breed of crop, a sample of eleven plants gave the following shoot lengths. 10.1, 21.5, 11.7, 12.9, 14.8, 11.0, 19.2, 11.4, 22.6, 10.8, 10.2.

- An earlier study reported that the mean shoot length is 15cm. Test whether the experimental data

confirm the old view at 5% level of significance?

- Determine the p-value of the test statistic and take a decision based on it
- Calculate the confidence interval for
- Determine the upper and lower limits of

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Solution

(a) Since the sample size is small, a  $t$ -test is performed where we have to test the hypothesis:

$H_0: \mu = 15$  against  $H_1: \mu \neq 15$

The test statistic is:

$$t = \frac{15 - 14.2}{4.6861 / \sqrt{11}}, \text{ standard deviation, } = 4.6861.$$

Therefore,

$$t = 0.5662$$

At 5% level of significance,  $\alpha = 0.05/2 = 0.025$ . The table value of  $t$  for 10 d.f is 2.228

Since  $H_0$  is not rejected at 5% level of significance. We conclude that the experimental data confirms the old view at 5% level of significance.

(b) here we want to show how to compute a  $p$ -value in the absence of specialized calculators and computer software.

In the distribution table, the calculated  $t$  for 10 d.f is seen to lie between 0.260 (under  $t = 0.260$ ) and 0.7000 (under  $t = 0.7000$ ).

We interpolate between these values to get value of  $p$ , (say  $p$ ) corresponding to 0.5662 as follows:

For a two-tailed test, since  $p > 0.025$ , we do not reject  $H_0$  and therefore conclude that the experimental data confirms the old view at 5% level of significance.

(c) The confidence interval (CL) for the population mean is:

$$\bar{x} \pm t_{\alpha/2} \left( \frac{s}{\sqrt{n}} \right)$$

$$CL = 14.2 \pm (2.228) \left( \frac{4.6861}{\sqrt{11}} \right)$$

$$CL = 14.2 \pm 3.178$$

$$CL = (11.0520, 17.3480),$$

$$\text{Lower limit} = 11.0520,$$

$$\text{Upper limit} = 17.3480.$$

Example 8.2: A researcher claimed that the life expectancy of people living in country A is expected to

be 50 years. A survey of life expectancy was conducted in eleven states of the country and the data obtained is as follows:

54.2 50.4 44.2 49.7 55.4 57.0 58.2 56.6 61.9 57.5 53.4

(a) Do the data support the researcher's claim at 95% level of significance?

(b) Find the 95% confidence interval for the population mean,  $\mu$ .

Solution

(a) We have to test

against

Test statistic  $t$  is given by:

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

$$t = \frac{54.2 - 50}{4.6861 / \sqrt{11}}$$

$$t = 3.178$$

$$t = 3.178$$

Without using the value of  $t$ , we can compute with equation (8.6) given as:

$$\left\{ \sum (\bar{x} - \mu_0)^2 \right\}$$

Therefore,

$$t = 3.178$$

$\sqrt{}$  ( )

The table value of t at and a d.f of 10 for a two-tailed test is 2.228.

Since , we reject and conclude that life expectancy of the people in the said country is not 50years.

(b). Confidence interval (CI) is given as:

$\bar{x} \pm t_{\alpha/2} \cdot s_{\bar{x}}$  ( ),

Therefore,

( )

Upper limit =  $54.41 + 3.26 = 57.67$ ,

Lower limit =  $54.41 - 3.26 = 51.15$

### 8.2.2 Tests of Hypothesis about Difference in Population Means for Samples of Small Sizes

The t-distribution is also used in the test of hypothesis involving the difference in two population means

when the sample size is small. It is defined as the deviation of the estimated sample means expressed in

terms of the sum of their standard deviations.

Here, we assume that the samples are from two normal populations with mean , and we desire to test the hypothesis:

vs

which is equivalently,

vs

CASE 1: When . That is the two populations are distributed with the same variance.

If two independent samples of sizes and are taken, respectively, from the two populations, the test statistic under (that is ) is given by:

—

, (8.7)

with a d.f of ( ), where  $\bar{x}_1$  and  $\bar{x}_2$  are the means of sample I and sample II given as

( ) and ( ), respectively, and the pooled standard deviation or standard

error ( ) is given by:

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$\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2$

( )

, (8.8)

If the sample variances, , are known or have been calculated earlier, then is computed using the formula:

( ) ( )

( )

, (8.9)

Decision about at level of significance is taken according to the following rules:

(I) Given reject if ( ) , or ( ) otherwise is not

rejected;

(II) Given reject if ( ) , otherwise  $H_0$  is not rejected;

(III) Given reject if ( ) , otherwise  $H_0$  is not rejected.

From (8.7) for difference of two means, the ( ) confidence interval (CI) is given as:

$(\bar{x} - \bar{y}) \pm t_{\alpha/2, n-1} \sqrt{s_p^2 \left( \frac{1}{n} + \frac{1}{m} \right)}$ , (8.10)

where the upper and lower limits are respectively given by:

$(\bar{x} - \bar{y}) + t_{\alpha/2, n-1} \sqrt{s_p^2 \left( \frac{1}{n} + \frac{1}{m} \right)}$ , and  $(\bar{x} - \bar{y}) - t_{\alpha/2, n-1} \sqrt{s_p^2 \left( \frac{1}{n} + \frac{1}{m} \right)}$ .

Example 8.3: The table below gives the average value of two random samples representing the monthly

sales of a certain product in two countries X and Y.

Table 8.1. Data for Example 8.3

Month X Y

Jan 363 536

Feb 404 474

Mar 518 556

Apr 521 549

May 613 479

Jun 587 422

Jul 365 315

Aug 412 414

Sep 469 505

Oct 468 552

Nov 371 492

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Dec 330 507

(a). Assume that the variances in sales of the product are the same in X and Y, test whether the average

daily sales in the two countries are equal

(b). determine the 95% confidence interval for the difference in the means of X and Y.

Solution:

(a) Let  $\mu_X$  and  $\mu_Y$  be the means of the population from which X and Y, respectively, were drawn. The hypothesis to test is given by:

against

Under the assumption of equal variances, the test statistic is given by:

—

$\Sigma$ ,

—,

$$\begin{aligned} & \sum, \\ & \sum, \\ & -, \\ & \sum, \\ & \{ \sum (\sum) \} \{ \sum (\sum) \} \\ & ( ) \\ & \{ ( ) \} \{ ( ) \} \end{aligned}$$

Therefore,

The table value of t at  $\alpha$  and a d.f of 22 is 2.074. that is ( )

Since ( )  $H_0$  is not rejected. We conclude that the mean daily sales of the product is the same in the two countries.

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(b) The  $(1 - \alpha)$  confidence interval for difference of means under the assumption of equal variances is given by:

$$(\bar{x} - \bar{y}) \pm t_{\alpha/2} \sqrt{s_p^2 \left( \frac{1}{n} + \frac{1}{m} \right)},$$

Therefore,

$$CI = (451.75 - 483.42) \pm 2.074 \sqrt{100 \left( \frac{1}{10} + \frac{1}{10} \right)},$$

$$CI = -31.67 \pm 41.48,$$

With upper limit and lower limit of 38.20 and -104.54.

CASE 2: When  $\sigma_1^2 \neq \sigma_2^2$ .

In this situation, the variance cannot be pooled. We use the Behrens-Fisher test statistic given by:

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{s_1^2/n + s_2^2/m}}, \quad (8.11)$$

where  $\bar{x}$  and  $\bar{y}$  are estimates of  $\mu_1$  and  $\mu_2$ , respectively.

The tabulated value of  $t$  denoted by  $t_{\alpha/2, \nu}$  is obtained from Cochran's approximation using the formula:

$$t_{\alpha/2, \nu} = t_{\alpha/2, \nu_1} + \frac{t_{\alpha/2, \nu_2} - t_{\alpha/2, \nu_1}}{\nu_2 - \nu_1}, \quad (8.12)$$

where  $t_{\alpha/2, \nu_1}$  and  $t_{\alpha/2, \nu_2}$  are the table values of  $t$  at a prefixed level of significance with  $\nu_1$  and  $\nu_2$  d.f, respectively.

If  $t > t_{\alpha/2, \nu}$ , we obtain:

$$t > t_{\alpha/2, \nu}, \quad (8.13)$$

with  $\nu$  d.f.

For decision about  $H_0$ , if  $|t| > t_{\alpha/2, \nu}$ , reject  $H_0$ , otherwise  $H_0$  is not rejected.

From (8.11), the  $(1 - \alpha)$  confidence interval (CI) for difference of two population means when  $\sigma_1^2 \neq \sigma_2^2$  is given by:

$$(\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{s_1^2/n + s_2^2/m}, \quad (8.14)$$

where the upper and lower limits are respectively given by:

$$(\bar{x} - \bar{y}) + t_{\alpha/2, \nu} \sqrt{s_1^2/n + s_2^2/m}, \text{ and } (\bar{x} - \bar{y}) - t_{\alpha/2, \nu} \sqrt{s_1^2/n + s_2^2/m}.$$

Example 8.4: The table below gives the gain in body weight (kilograms) per heifer of two different

breeds under a grazing treatment

Table 8.2: data for Example 8.4

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Breed 1 57.3 26.9 53.2 16.8 44.8 54.2 71.4

Breed 2 64.2 52.2 48.6 26.6 44.5 71.8

At 95% level of significance, test the equality of the mean gain in body weight of the two breeds under

the assumption that the population variances are different.

Solution

We want to test the hypothesis

against

Under the assumption of unequal population variances, we have:

$$\Sigma \bar{x}, \Sigma s^2$$

$$\Sigma \bar{y}, \Sigma s^2$$

$$* ( ) ,$$

$$* ( ) ,$$

Therefore,

, and ,

Therefore,

The calculated value of therefore,  $H_0$  is not rejected. We conclude that the average gain in body weight of the two breeds is equal.

### 8.3 Large Sample Tests for Population Mean and Difference of Two Population Means

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We apply the Z-test when we are dealing with large samples, that is sample sizes of equal to or greater

than 30.

#### 8.3.1 Large Sample Tests for Population Mean

To test the  $H_0$ : against the alternative hypothesis when sample size is large, we use the Z-test (also referred to as the normal test) given by:

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \quad (8.15)$$

The population variance may be estimated from the sample variance .

Example 8.5: The table below shows the income (in thousand naira) of 36 randomly selected persons

from a particular class of people in Nigeria.

Table 8.3: Data for Example 8.5

6.5 10.5 12.7 13.8 13.2 11.4

5.5 8.0 9.6 9.1 9.0 8.5

4.8 7.3 8.4 8.7 7.3 7.4

5.5 6.8 6.9 6.8 6.1 6.5

4.0 6.4 6.4 8.0 6.6 6.2

4.7 7.4 8.0 8.3 7.6 6.7

on the basis of the above data, can it be said that the mean income of persons in this class is N10,000 per

year?

Solution

We have to test the hypothesis

against

Since the sample size is large, we use the Z test (normal test)

— , ,

Therefore,

$\sqrt{(\ )}$

$Z = -5.81$ .

The table value of Z at for a two-tailed test is 1.96. Since Zcal is less than -1.96, we reject  $H_0$  and conclude that the mean income is not N10,000.

### 8.3.2 Large Sample Test for Difference in Two Population Means

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To test the hypothesis  $H_0 : \mu_1 = \mu_2$  vs  $\mu_1 \neq \mu_2$ , we use the Z-test where the test statistic is

given by:

$\frac{\bar{x}_1 - \bar{x}_2 - \mu_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

, (8.16)

where  $\mu_0$  is the hypothesized difference between the two population means.

In the event that the population variances are unknown, then their estimates are obtained from the samples variances  $s_1^2$  and  $s_2^2$ , respectively.

Decision about  $H_0$  is taken according to the following rules:

I. Given or reject  $H_0$  if or if

II. Given  $\alpha$ , reject  $H_0$  if

III. Given  $\alpha$ , reject  $H_0$  if  $|Z| > Z_{\alpha/2}$ .

Example 8.6: Two samples were drawn from two normal populations ( ) and ( ). The following information was available on these samples regarding the expenditure in naira per family

Sample 1:  $\bar{x}_1 = 1500$ ,  $s_1^2 = 400$ ,  $n_1 = 10$

Sample 2:  $\bar{x}_2 = 1200$ ,  $s_2^2 = 300$ ,  $n_2 = 10$

Is there significant difference between the mean expenditure of the two families?

Solution

We want to test the hypothesis

against

Since sample sizes are large,  $H_0$  is tested by the statistic,

—

The table value of Z at is 1.96. Since calculated Z is greater than tabulated Z, we reject  $H_0$  and conclude that the difference in mean expenditure is significant.

#### 8.4 Tests of Hypothesis for Proportion (One Sample Case)

If the observations on various items are categorized into two classes, and (binomial population), we often want to test the hypothesis whether the proportion of items in a particular class, say is or not. Thus the hypothesis:

against or or

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Can be tested using the Z-test where P is the actual proportion of items in the population belonging to

class .

Let  $\hat{p}$  be the sample estimate of P.  $H_0$  can be tested by the statistic

, (8.17)

where n is the sample size,  $\hat{p}$  is the proportion obtained from the sample, p is the hypothesized or assumed value of the population proportion, and  $\hat{p}$  .

Decision about  $H_0$  can be taken as follows:

I. Given  $\alpha$ , reject  $H_0$  if or

II. Given  $\alpha$ , reject  $H_0$  if ,

III. Given  $\alpha$ , reject  $H_0$  if

At and

At and

Example 8.7: To test the claim of the management that 60% employees support a new bonus scheme, a

sample of 150 employees was drawn. It was discovered that only 55 of the employees support the new

bonus scheme. Is the management right in their claim at

Solution:

The hypothesized value of  $P = 60\% = 60/100 = 0.6$ ,

$n = 150$ ,  $\hat{p}$  .

We want to test the hypothesis that:

Vs

$\sqrt{(\ )}$

$Z = -0.5825$ .

At  $\alpha$ , the table value of Z is 2.58. since calculated Z is less than  $\alpha$ , we reject  $H_0$  and conclude that 60% of the employees do not support the new bonus scheme.

#### 8.5 Test of Equality of Two Proportions (Two Sample Case)

If we have two populations, and each items of a population belongs to either of the two classes and .

A researcher may be interested to know whether the proportions of items in class in both populations

are the same or not. Here, we wish to test the hypothesis that:

against or or .

where are the proportions of items in the two populations belonging to class .

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When it is assumed that the population proportions are equal, that is ,  $H_0$  can be tested against using the statistic

$| |$

, (8.18)

$\sqrt{\hat{p}(\hat{p} - \frac{1}{2})}$

where  $\hat{p}$  , and

$\hat{p}$  , (8.19)

The standard error of ( ) is by  $\sqrt{\hat{p}(\hat{p} - \frac{1}{2})}$  .

Under the assumption that , the test statistic becomes:

$| |$

, (8.20)

$\sqrt{\hat{p}(\hat{p} - \frac{1}{2})}$

Decisions about  $H_0$  in equations (8.18 and 8.20) are the same as in equation (8.17) above for one sample case.

Example 8.7: A sample of 400 families in City A is randomly selected, and another sample of 500 families is selected in City B. The number of TV owners in City A and City B are 48 and 120, respectively. Test the hypothesis that the number of TV owners in both cities are the same at .

Solution:

, ,

Under the assumption that ,

$( ) ( )$

$\hat{p}$  ,

$\hat{p}(\hat{p} - \frac{1}{2})$  ,

$| |$

$\sqrt{\hat{p}(\hat{p} - \frac{1}{2})}$

$| |$

$\sqrt{\hat{p}(\hat{p} - \frac{1}{2})}$

$Z =$  ,

$Z = 4.59$ .

The table value of  $Z$  at for a two-tailed test is 1.96. Since the calculated  $Z$  is greater than the tabulated  $Z$ , we reject  $H_0$  and conclude that the proportions of TV owners in City A and City B are not the same.

Assignment 8.1:

The following figures give the forage yield (q/ha) of ten randomly selected plots.

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21.8, 24.8, 23.3, 29.3, 30.8, 31.8, 32.4, 32.5, 32.1, 31.3.

(a) on the basis of the data, test the claim that the mean yield of forage is 30 q/ha (i) at (ii)

(b) Determine the ( ) confidence interval and limits for both a(i) and a(ii).

SOLUTION

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Assignment 8.2:

The yields from two strains of a crop was found to be as given below:

Strain 1: 15.4, 20.5, 15.9, 38.3, 8.7, 37.0, 39.2, 26.4

Strain 2: 33.6, 10.7, 6.5, 24.0, 42.5, 22.9

Test whether the mean yields of the two strains are equal at

SOLUTION

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Assignment 8.3:

Given that  $\bar{x}$ ,  $\bar{y}$ , and  $s$ .

Test the hypothesis  $H_0$  : against (i) . (ii) (iii) .

SOLUTION

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Assignment 8.4: A random sample of 1000 workers from South India shows that their mean wages are

\$47 per week with a standard deviation of \$28. A random sample of 1500 workers from North India shows that their mean wages are \$49 per week with a standard deviation of \$40. Is there any significant

difference between the mean wages in the two cities? (take ).

SOLUTION

162

Assignment 8.5:

A random sample of 100 articles from a selected batch of 2000 articles shows that the average diameter of

the articles is 0.354mm with a standard deviation of 0.048mm. Find the 95% confidence interval for the

average diameter of this batch of 2000 articles.

SOLUTION

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Assignment 8.6:

Is it likely that a sample of size 300 whose mean is 12 is a random sample from a large population with

mean 12.5 and a standard deviation 5.2?

SOLUTION

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Assignment 8.7:

To test the claim of the management that 75% employees support a new policy, a sample of 300 employees was drawn. It was discovered that only 90 of the employees support the new policy. Is the

management right in their claim at

SOLUTION

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Assignment 8.8:

A sample of 150 families in City A is randomly selected, and another sample of 250 families is selected in

City B. The number of car owners in City A and City B are 27 and 80, respectively. Test the hypothesis

that the number of car owners in both cities are the same at .

SOLUTION

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### 8.6 Nonparametric Tests

Nonparametric tests are tests that do not require knowledge about the form of the parent distribution.

Nonparametric statistics is a type of statistical analysis that makes minimal assumptions about the underlying distribution of the data being studied.

The reasons for the application of nonparametric tests include the following:

1. When the underlying data do not meet the assumptions about the population sample. Generally, the

application of parametric tests requires various assumptions to be satisfied. For example, the data follows

a normal distribution and the population variance is homogeneous.

2. When the population sample size is too small. The sample size is an important assumption in selecting

the appropriate statistical method. If a sample size is reasonably large, the applicable parametric test can

be used. However, if a sample size is too small, it is possible that you may not be able to validate the

distribution of the data. Thus, the application of nonparametric tests is the only suitable option.

3. The analyzed data is ordinal or nominal. Unlike parametric tests that can work only with continuous

data, nonparametric tests can be applied to other data types such as ordinal or nominal data. For such

types of variables, the nonparametric tests are the only appropriate solution.

The chi-square test is taken as parametric as well as nonparametric. Other examples of nonparametric

tests include the sign test, the rank test, etc.

### 8.6.1 The Chi Square Distribution:

The chi square distribution was first discovered by Helmer in 1876 and later by Karl in Pearson in the year 1900. In probability theory and statistics, the chi-squared distribution (also chi-square or -distribution) with  $k$  degrees of freedom is the distribution of a sum of the squares of independent standard normal random variables. The chi-squared distribution is a special case of the gamma distribution and is one of the most widely used probability distributions in inferential statistics, notably in hypothesis testing and in construction of confidence intervals. The chi-squared distribution is used in the chi-squared tests for goodness of fit of an observed distribution to a theoretical one, the independence of two criteria of classification of qualitative data, and in finding the confidence interval for estimating the population standard deviation of a normal distribution from a sample standard deviation, etc.

### 8.6.2 Goodness-of-Fit Test:

When the data are given in  $k$  mutually exclusive categories or classes, the goodness-of-fit test is regarded

as one-way classification test. Here, we test the null hypothesis that the probability of an item falling in

the class is or simply that the frequencies in  $k$  classes occur in the ratio .

Suppose a random sample of  $n$  units falls in  $k$  classes or categories as follows:

Categories ... ..

Observed Frequencies ... ..

The hypothesis  $H_0$ : against is not true

Can be tested by chi square where the test statistic is given by:

$$\sum \left( \frac{O_i - E_i}{E_i} \right)^2$$

$$, (8.21)$$

where the suffix  $(k - 1)$  is the d.f for , and are the observed and expected frequencies.

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Reject  $H_0$  at a given if the calculated chi square is greater or equal to the table value of chi square given

and  $(k - 1)$  d.f. Rejection of  $H_0$  means that the observed data are not in agreement with the expected

ratio.

Example 8.8: Mendel conducted a classic experiment on peas to know the genetic effect of colour and

shape in the first generation, after taking four crosses, namely, Round and Yellow (RY), Round and Green (RG), Wrinkled and Yellow (WY), and Wrinkled and Green (WG). According to his theory, the frequencies of these four classes should be in the ratio 9:3:3:1. However, the observed frequencies he

found is shown in the table below:

Table 8.4: Observed Frequencies for four Crosses of Peas

Classes RY RG WY WG Total

Observed Freq. 315 108 101 32 556

Does the observed frequencies support Mendel's theory at ?

Solution

We want to test the hypothesis that:

$H_0$ : against is not true

Sum of ratios, .

The expected frequencies ( ) based on Mendel's theory as calculated as follows:

Therefore,

, ,

, ,

The expected frequencies are rounded in such a way that the sum of observed frequencies is equal to the

sum of the expected frequencies.

The test statistic is given by:

$\sum ( )$

,  $k = 4$ ,

( ) ( ) ( ) ( )

At the table value of which is greater than the calculated value of chi square.

Hence,  $H_0$  is not rejected. We conclude that the data support Mendel's theory.

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### 8.7 Contingency Table for Independence of Factors

A contingency table is a rectangular array of order where  $m$  denotes the number of rows which are equal to the categories of an attribute or criterion A, and  $p$  denotes the number of columns which are

equal to the categories of an attribute or criterion B. An example of a contingency table is shown below:

Table 8.5: Contingency table

Attribute B

Attribute A Total

... ..

... ..

... ..

... ..

... ..

Total ... ..

In Table 8.5, cell frequency in the ( ) cell represents the number of items or individuals possessing the characteristics of and Any of the row total, and column total is called the marginal total, and is the grand total.

Contingency table simplifies the process of computations of the expected frequencies. It is very handy

when carrying out the test of independence of factors as the following example would show. The test

statistic for independence of factors is the same as in goodness-of-fit, the degrees of freedom is equal to

( ) ( ) The test is always right-tailed.

We test the hypothesis that:

$H_0$ : two factors are independent Vs : Two factors are dependent.

The expected frequency, , for item is calculated using the formula:

( )

, (8.22)

Example 8.9:

In a volunteer group, adults 18 years and older volunteer to spend one to nine hours with children in an

orphanage home. The consultant employs persons among NYSC members, University students, and nonstudents. The table below shows the types of volunteers and the number of hours they volunteer per

week.

Table 8.6: data for example 8.9

Type of volunteers Number of hours worked per week Row Total

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1-3 4-6 7-9

Community college students 111 96 48 255

Four-year college students 96 133 61 290

Non-Students 91 150 53 294

Column Total 298 379 162 839

Is the number of hours volunteered independent of the type of volunteer at ?

Solution

The two factors are number of hours volunteered and type of volunteer. This test is always right-tailed.

: The number of hours volunteered is independent of the type of volunteer.

: The number of hours volunteered is dependent on the type of volunteer.

Using (8.22) to calculate the respective frequencies, we have for instance,

since and

Similarly,

since and

since and

The complete expected frequencies is presented in the table below:

Table 8.7: Expected frequencies for problem in Example 8.9

Number of hours worked per week

Type of volunteers

1-3 4-6 7-9

Community college students 90.57 115.19 49.24

Four-year college students 103.00 131.00 56.00

Non-Students 104.42 132.81 56.77

The test statistic is given by:

$$\sum \sum ( )$$

$$( ) ( ) ,$$

$$( ) ( ) ( ) ( ) ( )$$

$$( ) ( ) ( ) ( )$$

$$170$$

At the table value of . Since , we reject  $H_0$  and conclude that the

two attributes are not independent (that is type of volunteer and the number of hours volunteered are

dependent on each other).

#### Assignment 8.9

In a recent diet survey as regards the consumption of tea in two communities in a certain city, the following results were obtained:

Community A Community B

No. of families that take tea 1236 164

No. of families that do not take tea 564 36

Test at whether there is significance difference between the two communities with regards to tea consumption

SOLUTION

$$171$$

#### Chapter Nine

##### 9.1 Introduction.

In this chapter, we shall be looking at the dependence of factors or variables which takes only numerical

values. Quite often, the goal is to establish the actual or quantitative relationship between two or more

variables. This aspect is studied under regression which is the second part of the chapter. We commence

the chapter with correlation analysis which is used in situations where the researcher is only concerned

with the quality of the relationship and not the actual relationship between the variables.

## 9.2 Correlation.

Correlation is a measure of the degree of relationship existing between two variables. It could be defined

as the degree to which variables are related hence. Unlike regression which indicates the quantitative

relationship between independent variable and one or more dependent variables, correlation gives an

estimate the quality of the relationship between these variables.

Similar to what we have in regression analysis, when the relationship is between two variables it is termed

a simple correlation. On the other hand, if the relationship is between more than two variables, it is called

a multiple correlation.

There are mathematical equations for determining whether or not correlation exists between variables.

However, graphical method called the scatter diagram can be used to check for correlation.

## 9.3 Types of Correlation

Broadly, correlation is divided in two, namely linear correlation and non-linear correlation.

### 9.3.1 Linear correlation

Correlation is said to be linear when all points on a scattered diagram appear to cluster around a straight

line. There are two types of linear correlation, namely, positive or negative linear correlation. Two variables are said to be linearly and positively correlated if they tend to move in the same direction. In

other words, an increase in the value of one variable is associated with an increase in the value of the

other variable, and vice versa.

Diagrammatically, the positive linear correlation is represented by the Fig. 9.1 below.

Y

X

Fig. 9.1: Positive linear correlation between X and Y

In Fig. 9.1 above, it could be seen that not all the points on the scatter diagram fall on the straight line.

This shows that the correlation between X and Y is not perfect. If all the points fell on the straight line,

the correlation is said to be perfect positive.

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Two variables are said to be negatively and linearly correlated if the two variables tend to move in the

opposite direction. An increase in the value of one of the variable is associated with a decrease in the

value of the other variable and vice versa.

The diagram of two variables that are negatively and linearly correlated is shown in Fig. 9.2.

Y

X

Fig. 9.2: Negative Linear Correlation between X and Y

In situations where all the points in Fig 9.2 above fall on the straight line, the correlation is said to be

perfect negative.

### 9.3.2 Non-linear Correlation

Correlation between two variables is said to be non-linear when all points appear to form a curve. As in

linear correlation, non-linear correlation could be positive or negative. The correlation between two variables, X and Y is said to be positive non-linear if their value change in the same direction described

by a curve.

The diagram of this type of correlation is shown in the Fig. 9.3.

Y

X

Fig.9.3: Positive non-linear Correlation between X and Y

Two variables are negatively non-linearly correlated if their respective values change in opposite directions, forming an inverse relationship along a curve in the XY plane. The diagram is shown in Fig.

9.4.

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Y

Fig. 9.4: Negative non-linear Correlation

### 9.3.3 Zero Correlation

This type of correlation exists in situations where two variables are uncorrelated with each other. In other

words, there is no linear relationship between the two variables. When the values of the two variables are

plotted in a scattered diagram, the points are seen to be dispersed all over the surface of the XY plane

with no suitable line to join them.

Y

X

Fig. 9.5: Zero Correlation

## 9.4 Methods of Computing Correlation Coefficient

Correlation coefficient, denoted as or simply  $r$ , between two variables X and Y, is defined as the ratio

of the covariance of X and Y to the square root of product of the variance of X and that of Y.

Mathematically,

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( )

(9.1)

$\sqrt{(\ )(\ )}$

If we are given pairs of sample observations  $(\ )(\ )$ , then two computational methods are available for determining the quality strength of the correlation between X and Y, namely the

Karl Pearson coefficient of correlation and Spearman Ranks coefficient of correlation.

The Karl Pearson coefficient of correlation is given as:

$\sum(\ )(\ )$

$\sum(\ ) \sum(\ )$

, (9.2)

$\sum \sum \sum$

(9.3)

$(\sum)(\sum)$

$\sqrt{\{\sum\}\{\sum\}}$

$\sum \sum \sum$

for (9.4)

$\sqrt{\{\sum(\sum)\}\{\sum(\sum)\}}$

The Spearman ranks coefficient of correlation, denoted by the symbol  $r_s$ , can also be used to determine

the quality of the correlation between X and Y but by first ranking the n observations for X and then Y.

Once this is done, we can make use the Spearman ranks coefficient of correlation given by the formula:

( )

(9.5)

where  $\sum d^2$  is the squares of the differences in the ranks of the corresponding paired observations.

The Spearman ranks correlation is specifically used when the observations are qualitative.

The correlation coefficient ( $r$  or  $r_s$ ) is a pure number, it is independent of the units in which X and Y are

measured. Furthermore, and  $-1 \leq r \leq 1$ .

When  $r = 1$  it means that there exists a perfect positive correlation between X and Y. When

$r = -1$  it means that there exists a perfect negative correlation between X and Y. When two variables are independent, the correlation between them is zero. However, the converse is not true because  $r = 0$  does not necessarily means that X and Y are independent. It could be that there are other forms of relationship (e.g, quadratic) between X and Y other than a linear relationship.

Between -1

and 0, and 0 and 1, we have weak correlation ((-0.1,-0.39) and(0.1, 0.39)), fair correlation ((-0.4, -0.69)

and (0.4,0.69)) and strong correlation ((-0.7,-0.99) and (0.7,0.99)).

Assumptions about Correlation Coefficient:

1. The random variables X and Y are distributed normally;
2. X and Y are linearly related;
3. There is a cause and effect relationship between the factors affecting the values of X and Y in the series of data.

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#### 9.4.1 Test of Significance of Correlation Coefficient

The random samples ( ) ( ) ( ), are drawn from the population under consideration.

Whatever conclusions are deduced from the sample or are meant to be used to make inferences about

the parent population. For us to be sure of the validity of such inferences on the population, we need a sort

of confirmation about the correctness of these sample statistics ( or ) by way of test of significance.

Assuming R is the population parameter for , then the test of significance of correlation coefficient means to test the hypotheses whether or not the correlation coefficient is zero in the population, that is, we

test

The test statistic for testing is

(9.6)

where is the estimated value of R based on the n paired observations, is the standard error of given as:

$\sqrt{(9.7)}$

Therefore,

(9.8)

If the calculated value of t is greater than the table value of t for a given

Level of significance and ( ) degree of freedom, reject , otherwise accept . Rejecting leads to the conclusion that the two variables X and Y are dependent (i.e, not independent). This means that the

correlation between them is worth considering. If is accepted, it means that the value of r is due to sampling whereas in reality, the two variables are uncorrelated in the population.

Example 9.1: The table below shows the 9 paired observations for the variables X and Y

Table 9.1: Paired observations for X and Y

X 400 440 480 550 620 650 660 740 760

Y 50 60 70 85 95 100 105 115 120

(a). Calculate the value of (b) Test the significance of r at 5% level of significance

Solution

We make use of the formula:

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$$\Sigma \Sigma \Sigma$$

$$\sqrt{\{\Sigma(\Sigma)\}\{\Sigma(\Sigma)\}}$$

Table 9.2: Data for solution to Example 9.1

S/N

1 400 50 20000 160000 2500

2 440 60 26400 193600 3600

3 480 70 33600 230400 4900

4 550 85 46750 302500 7225

5 620 95 58900 384400 9025

6 650 100 65000 422500 10000

7 660 105 69300 435600 11025

8 740 115 85100 547600 13225

9 760 120 91200 577600 14400

Total 5300 800 496250 3254200 75900

(a). Using the respective values from the table above, we have

$$()()$$

$$\sqrt{((())((())())}$$

We conclude that there is a strong positive linear correlation between the variables X and Y.

(b). To test the significance of the population correlation coefficient, we test

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where the test statistic is given as:

For we have

$$\sqrt{()}$$

The table value of t 5% and 7 d.f is 2.365. We reject , implying that there is significant correlation between X and Y.

### Example 9.2

The table below shows the 9 paired observations for the variables X and Y

Table 9.3: Data for Example 9.2

X 20 14 18 25 21 35 26 37 29

Y 25 23 20 15 25 18 17 31 32

Calculate the Spearman rank correlation between X and Y

Solution

In order to calculate Spearman ranks correlation, , we rank the values of x and that of y in ascending

order to get the respective values from the table below:

Table 9.4: Data for estimating for Example 9.2.

S/N rank(x) Rank(y)

1 20 24 3 6 -3 9  
 2 14 23 1 5 -4 16  
 3 18 20 2 4 -2 4  
 4 25 15 5 1 4 16  
 5 21 25 4 7 -3 9  
 6 35 18 8 3 5 25  
 7 26 17 6 2 4 16  
 178  
 8 37 31 9 8 1 1  
 9 29 32 7 9 -2 4

( )

( )

We conclude that there is a weak positive linear correlation between X and Y.

#### Assignment 9.1

The table below shows 14 paired observations on X and Y

X 21 25 26 24 22 30 19 24 28 32 31 29 21 18

Y 19 20 24 21 21 24 18 22 19 30 27 26 19 18

(i). Calculate (ii) Test the significance of r at 5% level of significance

SOLUTION

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SOLUTION

180

#### Assignment 2

The table below shows 12 paired observations on X and Y

X 11 15 27 20 22 35 19 24 28 32 31 29

Y 13 10 20 31 21 17 28 22 19 30 27 36

(i). Calculate

SOLUTION

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SOLUTION

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#### 9.5 Meaning of Regression Analysis

Regression is an important tool applied by researchers in order to understand the relationship between two

or more variables. It describes in mathematical form the relationship between variables in a given study.

In other words, regression analysis presents an equation for estimating the amount of change in the value

of one variable associated with a unit change in the value of another variable. In expressing any relationship in mathematical form, two types of variables can be identified, namely, the independent

variable (also called the explanatory variables, the input variable or factor, etc) and the dependent variable

(output variable, the response variable, etc) which depends on the independent or explanatory variables.

For instance, given the statistical model:  $Y = \beta_0 + \beta_1 X + e$ .  $Y$  is referred to as the dependent variable,  $X$  is

the explanatory or independent variable,  $\beta_0$  and  $\beta_1$  are the model coefficients, the variable  $e$  is called the

error term or disturbance in the relationship. Specifically, for a linear regression,  $\beta_0$  and  $\beta_1$  represent the

intercept and the slope, respectively. It represents factors other than  $X$  that has influence on  $Y$ . It is implicitly assumed that the explanatory variable  $X$  has a causal effect on the dependent variable  $Y$ , and

the coefficient  $\beta_1$  measures this causal effect or influence of  $X$  on  $Y$ .

## 9.6 Types of Regression Analysis

Regression analysis could be simple or multiple.

### 9.6.1 Simple Regression

This is a regression analysis which describes the relationship between two variables. It is also called the

two-variable linear regression model or bivariate linear regression model because it relates to just two

variables say  $X$  and  $Y$ .

### 9.6.2 Multiple Regression analysis

This is an extension of the simple regression analysis. It is a regression analysis that involves three or

more variables. Hence, any model with a minimum of two independent variables requires multiple regression technique for its analysis. For instance, is a statistical model that

involves three variables, namely one dependent variable ( $Y$ ) and two independent variables ( ).

This focus of this chapter is on simple linear regression where the coefficients and are both linear.

## 9.7 Least Squares Method For Estimating Parameter Estimates Of Regression Models

To estimate the magnitude of the parameters of a regression model or equation, there are several techniques that can be used including the Ordinary Least Squares (OLS) method, the matrix method,

maximum likelihood method, etc.

There are distinctive properties associated with OLS method and these include:

- a. the parameter estimates obtained by OLS method have some optimal properties like unbiasedness, least variance, efficiency, best-linear unbiasedness (BLU), least mean square-error (MSE) and sufficiency;

- b. its computational procedure is fairly simple as compared with other econometrics techniques and data requirement are not excessive;
- c. the mechanics of OLS are simple to understand.

It is important to note that any estimation procedure using OLS method is based on certain assumptions.

It is on these assumptions that the parameter estimates of any regression model could be accepted as

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having a dependable prediction power. Some of these assumptions include the fact that the error term is

normality distributed with a mean of zero and constant variance.

Recall that the statistical model for the linear regression line of Y on X for the population is given as:

, (9.9)

Suppose the regression line given by (9.9) is to be fitted on the basis of pairs of sample observations

( ), ( ), ..., ( ). Each pair of ( ) for will satisfy the regression line (9.9).

Therefore,

, (9.10)

( ) (9.11)

$\sum \sum ( )$ , (9.12)

To get the least squares estimates of and such that the sum of errors is minimized, we differentiate (9.12) partially with respect to and respectively to get two equations called normal equations: We also replace and with their estimated values say a and b, respectively.

( $\sum$ )

$\sum ( )$  (9.13)

( $\sum$ )

$\sum ( )$  (9.14)

From (9.13), we have:

$\sum ( )$  (9.15)

$\sum \sum \sum$ , (9.16)

Since the summation is from 1 to n, we get:

$\sum \sum$ , (9.17)

$\sum \sum$

, (9.18)

Therefore,

—, (9.20)

From (9.14), we have:

$\sum ( )$ , (9.21)

$\sum \sum \sum$ , (9.22)

Substituting the value of a from (9.20) in (9.22), we get:

$$\sum \sum$$

$$(\sum \sum \sum), (9.23)$$

$$(\sum)(\sum)(\sum)$$

$$(\sum) \sum, (9.24)$$

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Therefore,

$$\sum (\sum)(\sum)$$

$$, (9.25)$$

$$\sum (\sum)$$

$$\sum (\sum)(\sum)$$

$$\sum (\sum)$$

$$, (9.26)$$

Equation (9.25) can also be written as:

$$\sum \Gamma)(\Gamma)$$

$$, (9.27)$$

$$\sum \Gamma)$$

Therefore, the estimated or fitted line of regression of Y on X is given as:

$$\sum \Gamma)(\Gamma)$$

$$\hat{\Gamma})(\Gamma), (9.28)$$

$$\sum \Gamma)$$

For

### 9.8 Causes of Deviation of the Fitted Value from the Observed Value

In reality,  $\hat{\Gamma}$ , in (9.28) will always show deviations due to some reasons that are briefly explained.

#### 1. Omission of relevant variables from the model

It is always difficult to include all the explanatory variables that affect or influence the response variable

in a single model. This is because of the complexity of the real-life situations as well as the need to keep

the model as simple as possible. Thus, several explanatory variables that affect a given phenomenon in

one way or the other may not be recognized and included in the model.

#### 2. Error of specification

The deviation of a fitted value from the observed value could also occur due to imperfect specification of

a relationship. Quite often, a non-linear relationship is represented in a linear form. Again, some phenomena need to be studied using several equations solved simultaneously. If these phenomena are

studied with a single model, error of specification is bound to occur.

#### 3. Error of aggregation

In collecting data for statistical analysis, it is often the practice to add data from different groups with

dissimilar characteristics. For instance, in the studies that involves social behavior, since the attitudes of

an individual may differ from those of any group, lumping their data as a unit for analysis could bring

deviations of the fitted value of the response variable from the observed value.

#### 4. Error of measurement

This error arises due to the method of data collection and processing. In data collection, a wrong sampling

technique adopted could cause an error in measurement. Equally, the use in appropriate statistical process

in processing statistical information could cause deviations of observations from the fitted line.

#### 5. Inclusion of irrelevant variables without theoretical underpinning

Differences or discrepancies in the value of the fitted response and that of the raw or observed value could

arise in in situation where irrelevant variables are included in the model.

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Example 9.3 The table below gives the paired values of variables X and Y

Table 9.5: Data for Example 9.3

X 16 22 28 24 29 25 16 23 24

Y 35 42 57 40 54 51 34 47 45

(1). Find the regression line of Y on X. (2) Predict the value of y when x =31

Solution:

In this example, we wish to make use of the equations (9.28)

The values of the terms needed to compute the estimates and from equation (9.28) are obtained in the

table below:

Table 9.6: Computation of data for Example 9.3

S/N (i)  $\bar{x}$   $\bar{y}$   $(x - \bar{x})$   $(y - \bar{y})$

1 16 35 -7 -10 49 70

2 22 42 -1 -3 1 3

3 28 57 5 12 25 60

4 24 40 1 -5 1 -5

5 29 54 6 9 36 54

6 25 51 2 6 4 12

7 16 34 -7 -11 49 77

8 23 47 0 2 0 0

9 24 45 1 0 1 0

Total 207 405 0 0 166 271

$$9; \sum \sum ( ) , \bar{}$$

$$\sum \sum ( ) \bar{}$$

$$\sum ( ) ( )$$

$$(1)^{\wedge} \sum ( )$$

$$186$$

—

$$( )$$

Therefore<sup>^</sup>

(2) Predict the value of y when ,

$$^{\wedge} ( )$$

### 9.9 Standard Errors of the Estimated Parameters for Simple Linear Regression

The standard errors of the estimated parameters are given as follows:

For parameter a, we have:

$$\sqrt{(\sum ( )$$

$$) (9.29)$$

For parameter b, we have:

$$\sqrt{(9.30)}$$

$$\sum ( )$$

where the standard error of residuals is given by:

$$\{\sum ( )^{\wedge} \sum ( ) ( )\} (9.31)$$

$$( )$$

### Assignment 3

(a) Obtain the parameters of the regression of Y on X using the data in the table below:

X 25 21 19 18 20 29 32 26 42 30 22

Y 19 17 15 11 15 20 24 19 30 20 15

(b) Predict the value of y when x is 35 (c) obtain (i) (ii) (iii)

SOLUTION

$$187$$

SOLUTION

$$188$$

SOLUTION

$$189$$

### 9.10 Hypothesis Testing of the Parameters for Simple Linear Regression

A hypothesis is an assertion or conjecture about any chosen parameter (e.g. mean) of the population.

In case we are considering more than one population, a hypothesis may be about the relationship between the similar parameters of the distributions.

A hypothesis can take two forms, namely, null and alternative hypotheses.

- i. Null hypothesis: A null hypothesis is the hypothesis which is to be actually tested for acceptance or rejection. It is denoted as  $H_0$ .
- ii. Alternative hypothesis: An alternative hypothesis is a statement about the population parameter(s) which gives an alternative to the null hypothesis, within the range of pertinent values of the parameter. It is denoted by  $H_1$  or  $H_a$ .

In this Section, the three common tests shall be considered. These are:

- i. the standard error test
- ii. the t-test and
- iii. the F-test.

During hypothesis testing, we are faced with the task of finding out whether the parameter estimates

are statistically significant or not. Hence it entails determining whether the independent variable(s) significantly affect the dependent variable or not.

#### 9.10.1 Standard Error Test

In this test, a comparison is made between the parameter estimate and its standard error without reference to degree of freedom. Based on the result of the comparison, a decision is taken on the significance of the estimate.

Decision Rule

- (a). If the standard error or is greater than the half of the or , that is if , where is either or , we accept the null hypothesis and conclude that is not statistically significant.
- (b) If , we reject the null hypothesis and conclude that is statistically significant.

#### 9.10.2 The t-test

This is a test for testing the hypothesis  $\beta = 0$  vs  $\beta \neq 0$ .

The t-test is usually used when the number of observations,  $n$ , is small (say,  $n \leq 30$ ). Here, reference is made to the degree of freedom at the chosen level of significance. The calculated t-value is compared with the theoretical or table value of  $t_{\alpha/2, (n-2)}$ , at a particular level of significance with  $(n-2)$  degree of freedom.

The calculated  $t$  is given by:

$$t = \frac{\hat{\beta} - \beta_0}{\text{SE}(\hat{\beta})} \quad (9.32)$$

and

$$t = \frac{\hat{\beta} - \beta_0}{\text{SE}(\hat{\beta})} \quad (9.33)$$

where  $\hat{\beta}$  and  $\text{SE}(\hat{\beta})$  are obtained from equations (9.29) and (9.30), respectively.

Decision Rule

- (1). If  $|t| > t_{\alpha/2, (n-2)}$  we reject  $H_0$  and conclude that  $\beta$  is statistically significant, that is  $\beta$  is different from zero.
- (2) If  $|t| \leq t_{\alpha/2, (n-2)}$ , we accept the null hypothesis ( $H_0$ ) and conclude that  $\beta$  is not statistically significant, that is  $\beta$  is not different from zero.

Accepting for parameter  $\beta$  means that the regression parameter (coefficient) of  $Y$  on  $X$  has no

practical significance, that is the change in Y corresponding to a unit change in X is practically meaningless.

For parameter a, accepting means that the regression line passes through the origin, that is the regression line does not cut the Y axis.

### 9.10.3 The F-test

The hypothesis vs ,

can also be tested by the F-test using the analysis of variance technique. For the F-test, the ANOVA table is as given below:

Table 9.7: ANOVA table for F-test

| Source of                                    | DF | Sum of Squares | Mean of Squares | F-value |
|----------------------------------------------|----|----------------|-----------------|---------|
| Deviation                                    |    |                |                 |         |
| $\sum (Y - \bar{Y})^2$                       |    |                |                 |         |
| Due to $\sum (\sum Y_i - n\bar{Y})^2$        |    |                |                 |         |
| $\sum (Y_i - \bar{Y})^2$                     |    |                |                 |         |
| Regression                                   |    |                |                 |         |
| $\sum (\hat{Y}_i - \bar{Y})^2$               |    |                |                 |         |
| Dev. from (n- $\sum$ )                       |    |                |                 |         |
| ( )                                          |    |                |                 |         |
| Regression 2) $\sum (\hat{Y}_i - \bar{Y})^2$ |    |                |                 |         |
| (n-                                          |    |                |                 |         |
| Total $\sum (Y_i - \bar{Y})^2$               |    |                |                 |         |
| 1)                                           |    |                |                 |         |

If ( ), reject otherwise accept . The physical interpretation for rejection or acceptance of remains the same as given under t-test for

### 3.4.3 Confidence Limits for Regression Parameters

The ( ) confidence limits (CL) for and are given as follows:

$$( ) \quad (9.34)$$

$$CL = ( ) \quad (9.35)$$

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where ( ) is the table value for a two-tailed t-test at level of significance for ( ) degree of freedom.

Example 9.4: Given the table below:

Table 9.8: Data for Example 9.4

S/N X Y

1 36.6 54.8

2 39.5 57.6

3 43.4 58.1

|    |       |       |
|----|-------|-------|
| 4  | 47.6  | 63.4  |
| 5  | 53.4  | 72.5  |
| 6  | 58.5  | 78.4  |
| 7  | 66.1  | 82.7  |
| 8  | 74.9  | 84.4  |
| 9  | 87.1  | 90.3  |
| 10 | 100.0 | 100.0 |
| 11 | 115.1 | 109.2 |
| 12 | 131.7 | 119.8 |
| 13 | 150.0 | 129.7 |
| 14 | 162.6 | 140.8 |
| 15 | 176.3 | 153.8 |
| 16 | 190.4 | 152.6 |
| 17 | 209.4 | 153.2 |
| 18 | 233.6 | 163.0 |
| 19 | 255.7 | 175.3 |
| 20 | 271.4 | 184.3 |

- (a) Determine the regression line of Y on X using  
 (b) Calculate (i) (ii) (iii)  
 (c) Determine the significance of and using the t-test at level of significance  
 (d) Determine the ( ) confidence intervals for and .

Solution

$$\sum ( ) ,$$

$$\sum ( )$$

$$192$$

—

Table 9.9: Data for computing values for Example 9.4

S/N X Y  $(\bar{X})$   $(\bar{Y})$   $(X - \bar{X})$   $(Y - \bar{Y})$   $(X - \bar{X})^2$   $(Y - \bar{Y})^2$   $(X - \bar{X})(Y - \bar{Y})$

|   |      |      |          |          |      |        |      |  |
|---|------|------|----------|----------|------|--------|------|--|
| 1 | 36.6 | 54.8 | -88.5650 | -56.3750 | 7844 | 3.1781 | 4993 |  |
| 2 | 39.5 | 57.6 | -85.6650 | -53.5750 | 7338 | 2.8703 | 4590 |  |
| 3 | 43.4 | 58.1 | -81.7650 | -53.0750 | 6686 | 2.8170 | 4340 |  |
| 4 | 47.6 | 63.4 | -77.5650 | -47.7750 | 6016 | 2.2825 | 3706 |  |
| 5 | 53.4 | 72.5 | -71.7650 | -38.6750 | 5150 | 1.4958 | 2776 |  |
| 6 | 58.5 | 78.4 | -66.6650 | -32.7750 | 4444 | 1.0742 | 2185 |  |
| 7 | 66.1 | 82.7 | -59.0650 | -28.4750 | 3489 | 0.8108 | 1682 |  |
| 8 | 74.9 | 84.4 | -50.2650 | -26.7750 | 2527 | 0.7169 | 1346 |  |

9 87.1 90.3 -38.0650 -20.8750 1449 0.4358 795  
 10 100.0 100.0 -25.1650 -11.1750 633 0.1249 281  
 11 115.1 109.2 -10.0650 -1.9750 101 0.0039 20  
 12 131.7 119.8 6.5350 8.6250 43 0.0744 56  
 13 150.0 129.7 24.8350 18.5250 617 0.3432 460  
 14 162.6 140.8 37.4350 29.6250 1401 0.8776 1109  
 15 176.3 153.8 51.1350 42.6250 2615 1.8169 2180  
 16 190.4 152.6 65.2350 41.4250 4256 1.7160 2702  
 17 209.4 153.2 84.2350 42.0250 7096 1.7661 3540  
 18 233.6 163.0 108.4350 51.8250 11758 2.6858 5620  
 19 255.7 175.3 130.5350 64.1250 17039 4.1120 8371  
 20 271.4 184.3 146.2350 73.1250 21385 5.3473 1.0693  
 Total 2503.3 2223.5 0.0000 0.4000 111890 34549 61443

$\sum (Y_i - \hat{Y}_i)^2$

(a)  $\sum (Y_i - \hat{Y}_i)^2$

$n=20$ .

—

( )

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Therefore^

(b)

$\sqrt{\sum (Y_i - \hat{Y}_i)^2}$

$\sum (Y_i - \hat{Y}_i)^2$

where the standard error of residuals is given by:

$\sqrt{\sum (Y_i - \hat{Y}_i)^2 / (n - 2)}$

( )

(i)  $\sqrt{\sum (Y_i - \hat{Y}_i)^2 / (n - 2)}$

( )

(ii).  $\sqrt{\sum (Y_i - \hat{Y}_i)^2 / (n - 2)}$

( )

$\sqrt{\sum (Y_i - \hat{Y}_i)^2 / (n - 2)}$

$\sqrt{\sum (Y_i - \hat{Y}_i)^2 / (n - 2)}$

(iii).  $\sqrt{\sum (Y_i - \hat{Y}_i)^2 / (n - 2)}$

$\sum (Y_i - \hat{Y}_i)^2$

To determine the significance of and using the t-test at level of significance, we proceed as follow:

For the parameter estimate , we have:

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The calculated value of  $t$  is greater than the tabulated value of at 5% level of significance at 18 degree of freedom which is given as .

Hence, we reject  $H_0$ , meaning that is significant. The practical implication of this conclusion is that the regression line does not pass through the origin.

For the parameter estimate, we have:

The calculated value of  $t$  is greater than the tabulated value of at 5% level of significance at 18 degree of freedom which is given as .

Hence, we reject  $H_0$ , meaning that is significant. The practical implication of the conclusion is that plays a significant role in determining  $Y$  through  $X$ .

(d). To determine the ( ) confidence intervals for  $\alpha$  and  $\beta$ , we proceed as follows:

For  $\alpha$ , the 95% confidence limits for  $\alpha$  is given as:

( )

CL=41.96

CL=41.96 6.01

This implies that the upper limit is 47.97 and lower limit is 35.96

For  $\beta$ , the 95% confidence limits for  $\beta$  is given as:

( )

CL=0.55

CL=0.55 0.0408

This implies that the upper limit is 0.5908 and lower limit is 0.5092

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### 9.11 Curvilinear Regression

When we have nonlinear relations, we often assume an intrinsically linear model and then we fit data

to the model using polynomial regression. That is, we employ some models that use regression to fit

curves instead of straight lines. The technique is known as curvilinear regression analysis. To use curvilinear regression analysis, we test several polynomial regression equations. Polynomial equations

are formed by taking our independent variable to successive powers. For example, we could have , Quadratic Model

, Cubic Model

In general, the polynomial equation is referred to by its degree, which is the number of the largest exponent. In the above, the quadratic is of the second degree, the cubic is of the third degree, and so

on.

The function of the power terms is to introduce bends into the regression line. With simple linear

regression, the regression line is straight. With the addition of the quadratic term, we can introduce or model one bend. With the addition of the cubic term, we can model two bends, and so forth. The Figure below shows an example of a quadratic regression which is plotted for the model for the values of  $x$  from 0 to 5:

Fig. 9.6: Graph of a quadratic model

The plot of the cubic model given by is shown in the Figure below:

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Fig. 9.7: Graph of a cubic model

Notice that there is a single bend in the quadratic curve as against two for the cubic curve because of the effect of the term for the quadratic model and the terms and for the cubic model.

#### Assignment 4

Given the table below:

X 19 11 10 14 22 9 13 23 11

Y 10 12 8 15 20 7 10 17 12

- Determine the regression line of Y on X using
- Calculate (i) (ii) (iii)
- Determine the significance of and using the t-test at level of significance
- Determine the ( ) confidence intervals for and .

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#### Chapter Ten

Research is an important part of academic, scientific, social, medical, economic, industrial and management studies. The objectives of a research include to find out the cause and effect among variables and to establish their relationship between them. One of the means or tools of achieving these objectives is design of experiments.

##### 10.1 Introduction to Design of Experiment

Broadly speaking, design of experiment is a plan to collect data from individuals or units which are subjected to certain conditions or treatments in such a way that the individuals or units are free from

the influences or effects of all extraneous factors (nuisance variables) and have full freedom to show

the effect of the conditions or treatments to which the individuals or units are exposed to. Analysis of

data and interpretation of results are carried out to conclude a research design. We give an illustration

of how these are done using a category of experimental design referred to a completely randomized

design (CRD) in the concluding Section of the Chapter.

We give brief discussions of some important terms associated with design of experiments in the Subsections that follow.

#### 10.1.1 Data

Data are numerical values of a variable collected on individuals or unit in the course of carrying out experiments

#### 10.1.2 Factors or Independent Variable

This is the variable that is manipulated by a researcher in order to see elicit a reaction from the experimental units. Independent variable could be a condition or a substance that is capable of prompting a reaction (an effect) on the experimental units. Independent variable is used interchangeably with the term treatment. For instance, different doses of fertilizers applied to different

plots. Students exposed to different teaching methods. The doses of fertilizer are called treatments while the different teaching methods are called conditions.

#### 10.1.3 Factor Levels

Factor levels are groups within each factor. They are sources of a particular factor. For example, in the study of the effect of carbohydrate on the weight gain in birds, the factor levels are the different

sources of the factor (carbohydrate) under study. These sources may include wheat, dried cassava peels, maize and others. Each of these sources is a factor level.

#### 10.1.4 Treatment

A treatment is equivalent to a factor level in a single-factor analysis. However, in multi-factor analysis, a treatment is equivalent to a combination of factor levels. For instance, in the study of the

effect of carbohydrate on the weight gain in birds, each of the factor levels (that is wheat, dried cassava peels and maize) is a treatment under a single-factor analysis.

In considering treatment as a combination of factor levels, assuming we wish to study the effect of different breeds of chicken and different sources of carbohydrates on the weight gain in chicken.

We

may decide to use three different breeds of chicken and three sources of carbohydrates, and this will

give a total of nine ( $3 \times 3 = 9$ ) treatments. This is so because each of the sources of carbohydrates will

combine with each breed of chicken to form a treatment. This combination of factor levels can be illustrated as follows:

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Sources of carbohydrate (SC):

1. Dried Cassava peels (C)
2. Wheat (W)

### 3. Maize (M)

Breeds of Chicken (BC):

1. Plymouth (P)
2. Sicelica (S)
3. Borea (B)

Table 10.1: Treatment Combination

SC C W M

BC

P PC PW PM

S SC SW SM

B BC BW BM

The treatments in this experiment are: PC, PW, PM, SC, SW, SM, BC, BW and BM

#### 10.1.5 Dependent Variable

A dependent variable is one which is used to measure the effects of the various treatments/conditions

on the individuals or experimental units. Measurements of yields of crops in fertilizer experiments and

volume of milk produced of cows in feeding experiments are examples of dependent variables.

#### 10.1.6 Nuisance Variables

Nuisance variables are extraneous variables which influence the dependent variables and cause undesired variation in the measurements taken on the dependent variables. For example, the inherent

knowledge of a person in teaching method experiment affect the measurements taken the dependent

variable which is the learning outcome of the students or learners. This is so because a teacher of higher level of intelligence quotient will likely get better results (learning outcome on the part of the

students) than a teacher of lower intelligence quotient. Nuisance variables interact with the dependent

variables in some way that weakens and sometimes invalidates the results of the experiment.

Hence,

they should be controlled as much as possible or, if possible, completely eliminated.

#### 10.1.7 Experimental Unit

This is an individual or group of individuals to which a single condition or treatment is applied independently of other individual or group. It is the unit whose response to the treatment is being examined or observed and this observation is taken on each unit as a separate entity. For instance, a

patient is the experimental unit in drug experiment while a plot is the experimental unit in fertilizer experiment.

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#### 10.1.8 Experimental Error

These are error caused by extraneous factors in an experiment which are beyond the control or not controlled by the researcher. For example, if on two or more plots, seeds are sown on the same day and are treated similarly in all respects, their yields will still not be the same. This difference in yields

forms part of the experimental error for the particular study.

#### 10.1.9 Control Experiments

A control experiment is an experiment in which all variable factors have been kept constant and which is used as a standard of comparison to the experimental component in a controlled experiment.

It is also defined as an experiment designed to check the results of another experiment by removing

the variable or variables operating in that other experiment. The comparison obtained is an indication

or measurement of the effect of the variables concerned. Control experiments are used to minimize the influence of the extraneous variables.

#### 10.1.10 Randomization

This is the process in which each and every experimental unit has the same chance of being allocated

to treatments. This is best performed with the help of a random number table. Randomization eliminates researcher's bias. It also ensures that no treatment will continually be favoured or handicapped by extraneous sources of variation which are out of the control of the researcher. With randomization, statistical inferences drawn are based on the assumption of that errors are independently and normally distributed.

#### 10.1.11 Replication

This is the repetition of each treatment on several experimental units. Replication provides an estimate

of the experimental error variance which is the basic requirement to test the significance of the treatment differences and finding the estimate of the standard error of mean given by  $\sqrt{\frac{MSE}{n}}$ , where  $MSE$  is the mean square error of variance and  $n$  is the number of replications of the treatment.

#### 10.1.12 Local Control

This is a technique for controlling a known extraneous variable. In field experiment, we know that adjacent plots are more alike in respect to fertility as compared to those which lie far apart. So, the plots of land with the same fertility are kept in one block. The purpose of local control is to reduce the

error variance and make the test of significance more sensitive and powerful thereby creating a more

efficient design.

### 10.2. Validity of Experimentation

The aims of an experiment are two-folds: One, to draw valid inferences about the effects of the treatments (independent variables) on the individual or groups (dependent variable) under study. To

generalize our results to a larger population of interest, there are several threats to validity and some

of them are briefly discussed below.

(i). History: In many experimental designs, measurement (say  $m_1$ ) about a criterion variable is taken

on the test units prior to the manipulation (treatment)  $X$ . When the manipulation is over, an after measurement (say  $m_2$ ) is taken on the same criterion variable. The difference ( $m_2 - m_1$ ) measures the

effect of  $X$  on the variable. But there is a possibility that in the mean time between  $m_1$  and  $m_2$ , many

events might have occurred which could have given rise to  $m_2$  besides  $X$ . Such events are called history. They are threats to the validity of the experiment because the inferences drawn about ( $m_2 - m_1$ )

regarding the outcome of the study are actually incorrect and should be taken with a pinch of salt.

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(ii) Testing effects: The effects which are caused due to taking measurement on the dependent variable before and after the application of the treatment on the test units are called testing effects which are further classified into two types of effects, namely, main testing effects (MT) and interactive testing effects (IT). The effect that occurs when a prior observation ( $m_1$ ) after a latter observation ( $m_2$ ), is the main testing effect. An effect in which a prior measurement ( $m_1$ ) has an influence on the independent variable and hence, affecting the test unit response is called interactive

testing effect.

(iii) Instrumentation (I): Any changes made in the calibration of the measuring instrument during interviews cause threats to validity of the experiment.

(iv) Selection Bias: The differential selection of subjects (individuals or group of individuals) to experimental and control groups is a major threat to internal validity. Selection bias is almost eliminated if the subjects are assigned to control and experimental groups randomly or matching the

members of both groups on key factors.

(v) Experimental mortality: This term refers to both the death of the experimental units and changes in

the composition of the study groups during experimentation. Such changes often occur due to deaths

of subjects or some subjects refuse to continue in the experiment. This alters the distribution of the subjects and one cannot ensure that the units lost would have responded to the treatments in the same

manner as the remaining ones.

(vi) Resentful Demoralization: If the subjects come to know that the treatments levels assigned to them are inferior in terms of benefits, goods or services, they may feel demoralized and hence show

their resentment. They are very likely to perform very low. This will create larger difference than the

real one between desirable and undesirable treatment levels. Confidentiality of treatment levels be maintained to reduce this type of threat to the validity of the experiment.

(vii) Reactive Factors: If the subjects know that they are being observed, they will act or respond differently than those who are not being observed. So, the results from the experiment cannot be generalized.

(viii) Low statistical Power: If the power of a test is low, then a research will very likely fail to reject a false null hypothesis. The reason for this could be inadequate sample size, inability to control nuisance variables, the use of inappropriate test.

(ix) Violation of Assumptions: All tests are based on certain assumptions. If such assumptions are violated, then there is every likelihood that incorrect inferences would be drawn.

(x) Reliability of Measures: If the reliability of measurements on the dependent variables is low, then

there is bound to be increase in error variance which may lead to the acceptance of a false null hypothesis.

### 10.3. Statistical Designs

In the category of true designs, there are a number of statistical designs that allow the control of extraneous variables to a large degree or extent. Such designs are randomized and enjoy some advantages enumerated below:

(i) The effects of more than one treatments or conditions can be estimated and compared from one experiment.

(ii) Interaction effects among the various treatments/conditions on the subjects can be measured.

(iii) Many extraneous factors can be statistically controlled.

(iv) The designs are generally economical and convenient.

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(v) Each design has a fixed procedure of analysis; hence, the chances of researcher's bias are negligible.

(vi) Threats to validity of the experiments are very less compared to other types of designs.

### 10.4 Requirements for a good Experimental Design

The following are requirements of a good design

(i) A good design should have no researcher's bias.

(ii) It should involve the use of appropriate statistical tests in analyzing the data.

(iii) It must make use of randomization and replication.

(iv) All extraneous factors should be controlled through local control as much as possible

### 10.5 Analysis of Variance (ANOVA)

Analysis of variance (ANOVA) is a powerful tool for data analysis when a number of populations are under study and from each population a random sample or a group of units is selected.

ANOVA is used to test simultaneously the equality of two or more population means when there is one parametric dependent variable and one or more independent variables. For instance, through ANOVA, one may compare the average yield of several varieties of a crop.

It is a statistical technique used for decomposing the variation in an observed data into its different

sources. In other words, it is a technique employed to break down the total variation in an experiment

to its additive components. With this technique therefore, we can break down the total variation, occurring in a dependent variable, into various separate factors causing the variation.

It is important to note that ANOVA does not stipulate any functional relationship between the dependent and the independent variable but it enables us to break down the total variation into the different sources or causes of the variation.

The assumptions underlying the application of ANOVA are as follows:

- (i) The observations follow normal distribution.
- (ii) Experimental units are assigned to treatments randomly and vice-versa.
- (iii) Appropriate statistical model is adopted for the experimental design.
- (iv) All the components involved in the model are additive and independent.
- (v) Experimental errors are independently and normally distributed with mean 0 and a constant variance.

### 10.6 Types of Analysis of Variance (ANOVA)

Before applying ANOVA, we first consider the number of factors to be studied. Based on this, ANOVA has two basic classifications or types, which are the one-way or one-factor ANOVA and two-way or two-factor ANOVA.

#### 10.6.1 One- Factor ANOVA

In one-factor analysis of variance, there are only two variables: one dependent variable and one independent variable. It is used to find out the effect of the independent variable on the dependent variable. For instance, in the study of the level of maize yield using different sources of Sulphur, where we are interested in the effect of Sulphur on the yield of maize. Therefore, Sulphur is the independent variable (factor), while the yield of maize is dependent on it. Based on this, we adopt the

one-factor analysis of variance to determine if the variation in maize yield is due to the application

of Sulphur (treatment) or due to chance.

#### 10.6.2 Two-Factor ANOVA

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The two-factor classification contains more than two variables. It is made up of one dependent variable and two independent variables (factors). It is applied to determine the effects of the two independent variables (factors) on the dependent variable. This technique, therefore, enables us to estimate not only the separate effects of the factors (independent variables) but also the joint (interaction) effect of these factors on the dependent variable.

Consider the study of the effects of fertilizer and soil type on the yield of rice. There are two independent variables or factors (fertilizer and soil type) and one dependent variable (rice yield). Therefore, the effects of these two factors (fertilizer and soil type) can be analyzed using the two factor analysis of variance technique.

### 10.7 ANOVA Table

ANOVA table is a table that shows, in summarized form, the computations for analysis of variance. This table enables us to have a quick and convenient assessment of the sources of variation, their respective Sum of Squares (SS) and the degree of freedom (df), Mean of the Sum of Squares (MS) and the F-value.

The skeleton of the ANOVA table is given below.

Table 10.2: ANOVA Table

| Source of variation | Degree of freedom (df) | Sum of Squares (SS) | Mean Sum of Squares (MS) | F-value |
|---------------------|------------------------|---------------------|--------------------------|---------|
| Due to A            |                        |                     |                          |         |
| Due to B            |                        |                     |                          |         |
| Error               |                        |                     |                          |         |

In Table 10.2, error degrees of freedom is obtained by subtracting the components df from the total df.

Similarly, the error SS is obtained by subtracting the components SS from the total SS, whereas the total SS is calculated by taking the total of the square of each individual value and subtracting from it

a factor which is known as the correction for mean or correction factor (CF). Hence, the sum of squares for testing the equality of population means are as follows:

Suppose the observations in k random samples of sizes  $n_1, n_2, \dots, n_k$ , from k normal populations

( ) for .

$(\sum \sum)$

, (10.1)

, (10.2)

Total SS  $\sum \sum$ , (10.3)

Between samples SS  $\sum$ , (10.4)

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Error SS  $\sum \sum \sum$  (10.5)

, (10.6)

ANOVA table with full details is presented in the table below.

Table 10.3: ANOVA Table with full Details

| Source of variation       | df          | SS | MS | F-value |
|---------------------------|-------------|----|----|---------|
| Between samples ( )       | $\sum$      |    |    |         |
| Within sample (Error) ( ) |             |    |    |         |
| Total ( )                 | $\sum \sum$ |    |    |         |

$\sum \sum \sum$  .

Total ( )  $\sum \sum$  ,

In ANOVA, an experimenter tests the null hypothesis of equality of k treatment mean effects. That is

vs at least two of them are not equal, i.e., for some

The test employed for is F-test. If calculated F is not significant, then it is inferred that all treatments are equally effective and no further test is required. But if the calculated F for treatment is

significant, then is rejected, implying that all the treatment mean effects are not equal. Some of them may be effective and some are not. So there is a need to perform a test for testing the significance of all pairs of treatment means. One good example of such test is called the Dunn's Multiple Comparison Test.

The above ANOVA table is for a one way classification. It may be extended for two or more way classification. In that situation, the component factors will increase in the ANOVA table accordingly.

Example 10.1: The table below shows the gain in body weight (kg) per cow during four grazing treatments. Test the hypothesis at 5% level of significance that the mean gains in weight of cow under four treatments are equal.

Table 10.4: Data for Example 10.1

Cow No. Treatments

T1 T2 T3 T4

Total

Gain in body weight(kg)

1 67.3 74.2 63.1 48.7

2 36.9 42.2 32.9 49.0

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3 63.2 58.6 59.2 52.0

4 26.8 36.6 42.4 38.8

5 54.8 54.6 34.0 48.2

6 64.2 81.8 65.6

7 81.4

Total 394.6 348.0 297.2 245.7 1286.5

Solution

We can test the hypothesis by F-test using ANOVA.

vs at least two means are different

For the given data,

$G=1286.5$ , ;

$\sum \sum ( )$ ,

$( )$

Total SS =  $74357.57 - 68961.76 = 5395.81$ .

$( ) ( ) ( ) ( )$

$\sum$ .

Between treatment SS  $\sum$

Error SS = Total SS - Between SS =  $5395.81 - 359.89 = 5035.92$ .

Table 10.5: ANOVA table for Example 10.1

| Source     | df | SS      | MS     | F-value                 |
|------------|----|---------|--------|-------------------------|
| Treatments | 3  | 359.89  | 119.96 | $119.96/251.80 = 0.794$ |
| Error      | 20 | 5035.92 | 251.80 |                         |
| Total      | 23 | 5395.81 |        |                         |

The table value of ( ) .

Since the calculated value of F is less than its table value, is not rejected. We therefore conclude that, under the grazing treatments, the mean increase in weight of cows is not significantly different at

5% level of significance.

## Assignment 10.1

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The table below shows the gain in body weight (kg) per cow during four grazing treatments. Test the hypothesis at 5% level of significance that the mean gains in weight of cow under four treatments are equal.

Table 10.4: Data for Example 10.1

| Cow No.                 | Treatments          |
|-------------------------|---------------------|
|                         | T1 T2 T3 T4         |
| Total                   |                     |
| Gain in body weight(kg) |                     |
| 1                       | 25.1 44.8 41.0 38.7 |
| 2                       | 22.0 22.2 39.1 39.0 |
| 3                       | 23.1 48.7 41.9 42.1 |
| 4                       | 16.4 38.3 50.4 28.6 |
| 5                       | 24.0 44.6 44.0 38.9 |
| 6                       | 31.8 41.8 45.6      |
| 7                       | 21.9                |

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